

Fin 210 The Morality and Practice of Finance

Numbered Transcript

FIN 210 • Dr. Yaron Brook

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Lecture 1: Why Finance Matters

At a glance — Why finance matters.

- 1.1 So, finance, it's everywhere. You just can't avoid it. I think I took an Uber here. Maybe it was a Lyft, I don't know, one of those services over here. I mean, that couldn't happen without finance, without my credit card being somewhere in the company's database and a transaction happening in the background that facilitated the transfer of my money into the account of the Uber company, right? And they have a bank account, and I have a bank account, finance. The credit transaction, finance.
- 1.2 Almost everything we do, everything we do, touches the world of finance. You really can't avoid it in your day-to-day life, whether you use a credit card, a bank, maybe some of you remember checks. It's been a long time, right? Hopefully, you all got a little bit of money put away saving. Maybe you're saving for retirement, maybe you're saving to buy a car, maybe you're saving to buy a house. That saving could not happen without finance. Otherwise, you'd be tucking it into the mattress, and it would be a long, long time before you got that car or before you got anything. There would be no return on that money. It would be just stuff in the mattress.
- 1.3 You buy a house or a car, you'll probably take out a loan, finance. You maybe have a stock portfolio, finance, everything. There's just no transaction that we engage in in the world in which we live today, there's no trade in which we really do that does not involve a financial institution, a financial market, or a financial product. It is ubiquitous in our modern world. The modern world could not exist the way it exists today without it.
- 1.4 So we're going to talk over the next eight classes about breaking this down, trying to understand the different components, trying to figure out what this thing is, what is a finance. We're going to talk about its history, where the different products, the different institutions, the different markets have come from, how they've evolved. And in the background of all of this, we're going to talk about the fact that, in spite of the fact that finance is ubiquitous, it's everywhere, that it seems like we resent it, we hate it, we don't like it.
- 1.5 We'll talk about this, but when was the last movie you saw where the financier was the hero? Almost always he's the villain. Or novel you might have read or any depiction in popular culture, popular press, or anywhere where finance is presented as a positive versus as a negative. So we have this thing, it's ubiquitous. You have this thing that the modern world cannot survive without. And yet there is this underlying negativity about it all. Civilization couldn't exist without it, and yet we hate it, we despise it. And we'll talk about the historical causes.
- 1.6 So let's just start with what finance is, what finance does, what is involved. And if you think about different financial markets, different financial institutions, maybe we

should talk about the primary product of money. What's the primary product of money? Sorry. Primary product of finance. I gave it away. Money. The primary product of finance is money. So we're going to be talking about money, what it is, where it comes from, what kind of money we've used in history, what kind of money we use today, what kind of money we might be using in the future. Money is a human creation.

- 1.7** And as such, it's something that can change over time, and has changed over time, and might, I suspect, will change over time. So money is obviously the product of finance. What else? We've got banks, right? We've got investment banks and commercial banks and mortgage banks. And we used to have saving and loans. And we've got insurance companies, mutual funds, hedge funds, private equity funds. God, a lot of these, right? And then you've got bond markets and money markets and stock markets and all kinds of financial institutions, all kinds of financial markets. There's a huge variety and a huge array of these things.
- 1.8** We won't have time to talk about every single one of them, although we will in principle. We'll focus primarily on two. We'll focus primarily on banking, on really understanding banking. Banking has been historically at the core of what finance is and at the core of how finance functions. We'll talk about how banks work, how they function, what are the issues with them, what is the future of banking; and then we'll spend some time on the stock market as representative of markets. And we'll mention some of the other financial institutions, financial markets along the way. But those are going to be two that we really spend some time focusing on.
- 1.9** So what is common to all these institutions and all these markets? What is the same about all of them? What can we say is common about them all? Well, they all are there in some way to facilitate savings. All of us engage with financial markets because we might have excess money, we might have excesses, and we'd like to save it for the future. We'd like to use it in order to plan for the future. So all these financial markets and institutions facilitate ability to save—not only to put the money aside and to protect it. You can put it in the mattress. Risky. Somebody might come and take the mattress, might get burned down or whatever.
- 1.10** So one of the reasons we want to put it in some safe box or in a bank or somewhere like that is safety. But also because savings, for a variety of reasons we'll talk about, actually generate a return. We actually get something more. Call it interest, call it profit. Depends on the form in which the saving takes. But there is a return that is, the saving actually grows. Over time, we have more. And that's part of what financial institutions facilitate, is this ability of our savings to grow.
- 1.11** And in that sense, in the sense of saving, we have the ability here to facilitate investment. All the saving goes where? The only way in which it can grow is if it's invested in something, mostly in something that is productive, mostly in something that has generated something out there in the real world that is actually creating real wealth, that is actually ultimately creating real returns, which generate the growth of

our savings. So by aggregating all these savings, we can, as financial institutions, financial markets, we can make significant investments.

- 1.12** Think about any startup or business that wants to expand. It requires capital. That capital often comes, that capital is going to come from people out there who are trying to save their money and want it to be deployed for productive purposes. We'd like to see it grow. And what we're seeing here is really the function of financial markets and institutions is to match my desire to save with productive activity that's actually going on. And the primary, of course, has to be productive activity because that's where the return comes from. That's where the profit comes from. That's where the wealth creation comes from.
- 1.13** And that's, of course, me producing is where I get the money that I can then save. I can't save money unless I produced it first. One of the great fallacies in economics. One of the great fallacies in economics is this idea of the primacy of the consumer. This idea is that consumption is what's really, really important. Consumption is what drives an economy. It goes back to Keynes. But this idea that we can evaluate everything in economics based on whether people are consuming or not consuming, how much they're consuming, and so on.
- 1.14** But if you think about it—maybe at the very, very primitive level, consumption, you can pick berries and you can collect nuts. But even that requires some production, i.e., collection, i.e., picking, some action that you have to take. We don't just get our food and our stuff as manna from heaven. It doesn't just appear. The primary act that human beings must engage in is the act of production. Only once we produce can we consume. And indeed, most of our consumption is not just available out there in nature like berries and nuts. Most of our consumption needs to be what? It itself needs to be produced.
- 1.15** So I go to work, I produce, and then I take what I've earned from that production—we'll get back to how that all works later— and then I go and buy something from somebody. That's my consumption. But he had to produce it. So there's a sense in which, for every act of consumption, there were two acts of production, at least, if not many more, because if you look at the supply chain, there are many more. So what financial markets are doing is matching the producer's need for money and capital to build stuff, to make stuff, with somebody else who has surplus capital, surplus money, and can deploy that money towards that production.
- 1.16** So it matches savings and production. Financial markets match savings and production. And by doing so, they provide a return on your savings. They allow you to grow those savings. They expand the number of investors, if you will, of savers. There's a way in which labor now becomes capitalist. So the labor that is producing, that is creating stuff, if it has that excess, again, it's not just hoarded. That excess is now deployed as capital for other investments. And in that sense, everybody becomes a capitalist to the

extent that everybody has a 401k, the extent that everyone is saving, everyone is now an investor as well.

- 1.17** The saving provides capital to producers. It makes it possible for managers and entrepreneurs to get access to capital that they need in order to expand their businesses or to start businesses. We'll talk about this in a later class, but one of the other things financial markets and institutions allow us to do is control for the inherent risk in the kind of transaction in production in our economic activity. It could be the risk of weather going bad. We have financial instruments to protect us from the consequence of bad weather here in Florida. You buy what? Hurricane insurance. Farmers can buy options of futures on crops.
- 1.18** So there's a variety of ways in which our financial markets allow us to control the consequence of risk. I can't control the weather. Markets don't allow us to do that, but they can protect us from the consequence of weather. So again, they facilitate production and consumption. At the end of the day, they facilitate trade, and always have facilitated trade. Some of the most historical origins of finance are in letters of credit that allowed people to trade across vast geographies without having to lug money around, back and forth. Money, remember, in those days, was gold, so it was heavy to lug it around.
- 1.19** So you could write a letter of credit saying, "Yes, he gave me the money, give him the goods," and then they would trade back and forth. Again, we'll talk about this in more detail. Financial markets also provide us with information. And we'll see that, particularly when we talk about stock markets. We get real information from financial markets. Information about what people think is important, what is not, what is the future, what is not. Imagine what was happening to the stocks of companies that produced buggies— you know, buggies attached to horses—as Ford is ramping up production of the Model T. What do you think happens to buggy stocks?
- 1.20** Well, if you don't know exactly what's going on, if you're not into transportation, if that's not your expertise, just watching the price of the stock of these buggy companies go down, gives you information about, "Oh, something is happening here. Something is changing. I've got some additional information I didn't have before." So financial markets provide us with important information about what is going on in the world or what is rising and what is falling, what is of interest, what is not, what is dying, what is being born.
- 1.21** Think about the fact that AI was something none of us had ever heard of. That vaguely had some idea that there was something called AI, and now everybody's in AI. And part of that is the hype around AI, but part of it is the fact that a huge amount of capital is now being invested in AI, which is generating huge amount of changes, new innovations, new products that are coming to all of our attention, through the fact that markets have made these companies significant in our lives. We've got that signal, that

information. This is important. A lot of people are putting a lot of money into this. Must be important.

- 1.22** And we could decide, "Huge mistake, they're all wrong, they're all stupid, I know better," and that's always possible. But the fact that what's happening, all this investment is going on, is really, really important— and a crucial piece in, overall, the information that we need in order to assess the future of technology, let's say. So markets provide us with information, information for businessmen, for entrepreneurs, for capitalists, for speculators, for savers. And again, as we talk about the stock market, we'll talk more about how all these interact and how this really gets laid out and what it means and why this information is so crucial.
- 1.23** So if we have to summarize what is going on here, financial market institutions are markets and institutions whose primary function, not only function, primary function, is to allow the efficient conversion of savings into capital in the hands of producers, while allowing us to control for risk. So the transition of savings into capital in the hands of producers used to create, build wealth, while controlling for risk. They encourage saving. They encourage us to save because they offer us a return. So they increase the pool of capital that's available. And of course, that saving are invested in productive activity, which is what drives our economy.
- 1.24** Our economy is production. It's not consumption, as Keynes would tell you. It is production. That's what drives an economy. You want to get an economy going, get a bunch of businesses started, right? Get entrepreneurs. Entrepreneurs creating businesses is what creates jobs, which is what spurs economic growth. So it is production at the heart of it all. And of course, by doing this, there are multiple benefits to the economy. We grow, we increase productivity, and we raise the standard of living for everybody across the board.
- 1.25** Indeed, our modern world could not exist without them. We take for granted new innovations constantly arriving. Those innovations are mostly a product of startups, of new companies, that are getting money from venture capitalists, that are getting money from financial markets. But that's at one level. And another level, you go out to your strip mall and you see these restaurants and little stores. They exist because they're getting bank loans, and they probably had some, not venture capital as we know it, but somebody provided it with capital in order to get it all started; bank probably was part of that.
- 1.26** Everything we have is a product of this. If you don't have the markets and institutions, then everything you raise, you have to raise from your own resources, the money you've saved as an individual, in the mattress, I guess, because there are no banks, no stock markets, no place to put them. It's all in your mattress. Or maybe it's some friends and family, which a lot of businesses use that. A lot of businesses start out with friends or family, but that's it. There's no more. There's nothing beyond that. Now you

can only grow based on how much profits you generate. You can plow the profits back into the business and grow the business.

- 1.27** But it's only markets, it's only finance, that allows you to grow this in any substantial way. So you're stuck with doing it all yourself. And if you wanted to grow your savings, if you wanted to invest your money, you would have to go out there and do your own research and figure out who is trustworthy and who is not, what a good risk is and what is not, what a technology is worth investing in and what technology is not worth investing in. You would have to become an expert in every one of these things.
- 1.28** I mean, you could go today and buy an index fund that basically invests in the world. And you have exposure to pretty much every major, at least, every tradable business in the world. You can buy a piece of real estate with diversification across the entire United States, potentially even globally. Imagine if you had to have the expertise to make every single one of those investments; you couldn't, and it would all break down. So the idea of a world without financial institutions and other financial markets as we know them today is really unimaginable. It really is not possible.
- 1.29** At the same time, people don't really understand these markets. They don't understand how they function. I mean, very few people understand even how something as simple as a bank works, and how it works, and why it works, and what value it provides. There's been hostility, really, from the beginning of time, as we'll talk about, towards even the idea of a bank charging me interest on a loan. I mean, most people out there resent that fact, and there's been resentment towards the idea of paying interest to somebody, again, from almost the beginning of time, if not the beginning of time.
- 1.30** There's this perception the financial markets and institutions are only interested in this short run. You see this all the time. You see it in the press constantly. Stock markets, all companies care about is quarterly, and all the stock market cares about is the quarterly performance. Long-term, who cares? Which is really interesting given that we have this amazingly rich, prosperous world with a bunch of companies that, if you really only thought about it in terms of short-term, would never exist. Think about a biotech company, a startup that doesn't really produce anything that generates income for maybe 10, 15, 20 years, and yet it receives capital.
- 1.31** Or think about the fact that we have businesses in this country that if they really only thought about next quarter, how many of them would really exist? Think about the planning that goes into managing a significant business. If you run a small business or if you're involved in a business or just involved in life, think about what it means in life to just plan for the next three months and ignore the distant future. You wouldn't survive for very long. The fact that we have and live in an economy that is growing, maybe not as fast as we would like, certainly not as fast as I would like, but it's growing and it's producing and it's succeeding and it's innovating.

- 1.32** That in and of itself suggests that there's a lot of thinking going on, and there's a lot of long-term thinking. And yet we're stuck in this mentality of, no, markets are very short-term. We have generally a zero-sum mentality. People do. We generally think that the world is zero-sum. My gain is your loss. If I make a profit, you must have lost. If the bank's making money, then its customers must have lost. In every realm, I think the most illustrative example of this is, I don't know if any of you have ever watched the movie Wall Street. The movie Wall Street is, I believe, a must-watch. It's a brilliantly made movie.
- 1.33** It's very much from a particular philosophical perspective. The director of this movie has an ideology. He believes in something, and he's made a movie to portray it. He is an unapologetic Marxist. And the movie projects an unapologetically Marxist view of finance. And what does that mean? That means everything is zero-sum. It means that anybody getting rich is getting rich at people's expense. It means that financiers, exploiters, they're generally bad guys, and they need to be constrained. They need to be controlled.
- 1.34** Just some quotes from Wall Street, from the movie, quote, "Wall Streeters get wealthy not by creating anything of value, but by buying and selling at the expense of others." Now, that's a pretty conventional view. They're just paper shufflers, and they're exploiting people. So when we talk about the stock market, we'll talk about whether that's true or not. And what is, because it's not, what is the productive value? What are they creating? What do people in Wall Street actually create? I mean, that's abstract. It's not obvious. So we'll talk about that when we talk about stock markets.
- 1.35** Or another quote, "The rich get rich not by hard work and intelligence, but by exploiting the poor." I mean, that's a line out of the movie. He's not hiding anything in terms of what he believes, right? And then there's a number of references to the fact that money corrupts, always corrupts, it's corrupting. Making money is a corrupting activity. And again, very explicit. But he's a Marxist, the movie director is. But this was an incredibly popular movie. And it reflects, I think, many of the ideas, whether I don't think most people are Marxists, but people have internalized these ideas.
- 1.36** And again, these ideas about money, these ideas about wealth creation, these ideas about stock markets and speculating and banking are all deeply ingrained within us. They're all deeply ingrained in our culture. And we'll talk about whether these things are true. So you can look at movies, TV shows across the board. In Billions, I don't know if you guys have seen Billions. Billions, I think it was a show on Showtime. Six, seven seasons, very entertaining, worth watching. And in many respects, it's one of the better shows on finance.
- 1.37** But the underlying assumption is that the financier, the billionaire at the center of this, cannot be successful, cannot achieve anything unless he cheats, lies, and steals. That even though he's kind of the good guy in the show—there are no real good guys in the show—he's the central focus. He's doing insider trading constantly in illegal ways, even

in immoral ways, and clearly is deceiving. And that's just, and it's presented as, well, how else could you make money? I mean, he actually says, "This is how you make money. This is how the markets function. There is no alternative." Again, ingrained in our culture, and there's a consequence to this. There's a consequence to this.

- 1.38** And fundamentally, the consequence to this is that because we distrust markets so much, financial markets, because we distrust financiers so much, in every aspect of their activities, we, therefore, have to control and regulate them. Every aspect of the financial markets or financial institutions of money, every aspect of money is controlled and regulated. Where does the money today come from? It doesn't come from the market. It comes from a government institution, from the Federal Reserve, because we don't trust the market with money.
- 1.39** We've been, again, told and taught over and over and over again that financial markets, financial institutions, if left alone, not only are corrupt, not only are zero-sum, not only don't they produce anything, create anything, even though, again, if you go back to the beginning, everything we touch involves these financial markets and institutions, but somehow they don't create and produce anything, even convenience, it turns out. But they're also responsible for every economic crisis we've ever had. Who gets blamed for the Great Depression? It's the money lenders. It's the financiers. It's the Wall Street entrepreneurs. Those are the people blamed. It's speculation in the stock market. That's always the reason why bad stuff happens.
- 1.40** Why do we have, why did we have a financial crisis in 2008? Greedy Wall Street. It's always shocking to me, this idea that in 2007, leading up to the crisis in 2008, greed showed up in Wall Street. They weren't greedy before. They've never been greedy before. But suddenly in 2007, greed showed up in Wall Street and caused the 2008 financial crisis. It's greed that made it happen. Wall Street, if you understand, if greed is the constant pursuit of profit, the single-minded focus on making money, greed is always on Wall Street. Always has been, always will be. That's their business, making money.
- 1.41** Now, whether that's productive or not, different question. But that's the focus. So it can't be that greed suddenly caused the financial crisis because it's always part of what it means to be on Wall Street. And yet Wall Street gets blamed, not just for the Great Depression, but the financial crisis. And over and over again, you see this coming. We'll talk about the more recent bank crises, the Silicon Valley bank and others like that. But I will guarantee, I can guarantee now that the next recession, the next downturn, the next crisis, will be blamed on bankers, will be blamed on Wall Street, will be blamed on finances.
- 1.42** It's one of these great mysteries of, we rely on an industry. We see how important it is to our lives. We experience it every single day, but partially because we don't conceptualize it, partially because we don't bring it to awareness, we don't know or we don't appreciate, we don't acknowledge, what it's actually doing to our lives. So one of

the purposes of this course is to bring it to awareness. It's to bring the contribution to awareness.

- 1.43** And the second, there are deep historical reasons, ideological, philosophical, deep historical reasons why we distrust bankers, why we distrust financiers. And again, we'll look at that history. We'll evaluate that history and see where all this hostility bubbles up to. And in doing so, I think we'll gain a better understanding of the actual functioning of these institutions and what finance actually means and how it functions.
- 1.44** I'll give you a quote from President Roosevelt, FDR, who's trying to, again, during the Great Depression, and telling you who this is. And he's very clear. It's unequivocal. And I mean, just the language is interesting because you don't hear presidents talk like this, but that's a separate issue. Here's the quote. He says, "The money changers have fled from their high seats in the temple of our civilization." Think about all the connections there. Money changers, which is, remember what Jesus did, the money changers? Threw them out of the temple. "Have fled from the high seats of the temple." There's a temple, Jesus' temple, "of our civilization." That's where they belong and that's where they come from.
- 1.45** And they were corrupt in the biblical story, and they're corrupt now, and they've fled. In a sense, we've kicked them out of there. "We may now restore the temple to the ancient truths. The measure of restoration lies in the extent to which we apply social values more noble than mere monetary profit." Monetary profit, somehow this is bad. Somehow this is corrupt. There's something really evil going on here. "The rulers of the exchange of mankind's goods have failed, through their own stubbornness and their own incompetence, have admitted failure and have abdicated. Practices of the unscrupulous money changers stand indicted in the court of public opinion, rejected by the hearts and minds of men."
- 1.46** Now, we will talk about, at least somewhat, I mean, we don't have time to really get into it in great detail—we will talk somewhat about the causes of the Great Depression and whether it's the money changers in Wall Street that really caused the Depression and whether they fled from the temple or whether they were kicked out. And whether that is even true, because to what extent does FDR, in recovery from the Great Depression, to the extent that we recovered from it, to what extent does FDR really not use financiers, abolish financiers? Is Wall Street dead after 1933 when FDR says this, or is it just changed?
- 1.47** One last on this. Isn't it interesting that almost every single conspiracy theory out there ultimately lands up on a banker? Who's pulling the strings of the world? At the end of the day, it's financiers, it's bankers, often Jewish, and we'll get to why Jews later. But often Jews, but bankers, they're the ones pulling the strings. If we're trying to explain what's going wrong in the world from whatever perspective and whatever the crisis happens to be and whatever's going on, blaming the banker seems to be a safe way to do it.

- 1.48** We've got this juxtaposition of finance being crucial, something we use all the time, hated, despised, distrusted, and as a consequence, heavily, heavily regulated. And the regulations have consequences. The regulations have consequences. All this control stops the market from working the way it would otherwise work. It takes away the thing that makes the market work most effectively; and the market works most effectively when people are pursuing profit, profit which we don't like, but profit. And the regulations cause, and again, we'll get into details on this, misallocations of capital, which results in misallocations of human resources.
- 1.49** Financial institutions spend a huge amount of money and brainpower, huge amount of money and brainpower, trying to figure out how to get around regulations. Many of the markets we have today, or have had over the last, let's say, 50 years, are often responses to regulations and ways around them. I mean, the most obvious of that is the money market, which we all, I think, all know about. Many of us might have money in the money market, and money markets were created in order to get around a regulation. We'll talk about which one and how they do that.
- 1.50** But a lot of energy, a lot of the complicated derivatives we hear about, they don't have any necessarily any economic purpose, other than to get around things that the government will not allow bankers to do, and yet they need to get done. Somebody has to do it. So we find ways to get around the constraints that are being put on. And the amount of resources, brainpower, doing that rather than doing something productive, if these barriers were not existing, is truly astounding.
- 1.51** I think there would be a lot more entrepreneurial activity, if not for all these regulations and controls. I think that we'd have a much higher saving rate. We'd have more efficient markets. We'd have fewer booms and busts, fewer crises, fewer crashes, fewer panics, fewer panics. And indeed, horizons in terms of investing would be longer term. We'd think longer in terms of planning. I don't think the markets do it quarterly now, but they certainly could be even more long-term in terms of their thinking.
- 1.52** And the consequence of all of that is lower productivity of labor and a lower quality and standard of life that we have. We give up significant economic growth. We give up significant standard of living as individuals because we have decided this needs to be controlled, this needs to be regulated, and we don't trust financiers, and we think they're going to do bad things.
- 1.53** All right, let me tell you about the next few classes and what we're going to cover in those classes. Next class, we're going to talk about money. Just focus on money. What is it and what is its history? And it's not just money, because we're going to also talk about credit and interest rates, because you can't separate money in terms of finance without talking about interest, or another term for interest that has been used for most of human history is usury. We'll talk about the concept of usury, where it comes from, and how it's articulated.

- 1.54** The class after that, or an additional class, we'll talk about banking history. We'll talk about banks, how they function. We all use them. We all have checking accounts. We have saving accounts, maybe. Maybe we've even taken a loan from a bank. But how does it actually work? How do they make money? What do they actually do? And again, what's the history here? Do we need them? In what sense? So we'll talk about banking history, how it works, and what its future is.
- 1.55** Then we'll talk about the stock market. How does the stock market work? Which I think is, particularly its productive function is something people really don't understand. And there's a lot of interesting that we can talk about, what is in the stock market, how it works and how it functions. In both, in the context of both banking and stocks, we'll talk a little bit about valuation. How do you value a financial asset? How do you value a stock? How do you value anything, really, a commercial real estate project or a bond or a loan? How do you put a number to it? How do you get a value? What are the principles?
- 1.56** We're not going to get into the details of valuation. That's way beyond what we're trying to do here. But we'll talk about some of the principles around, how do we approach valuing any one of these financial assets, financial instruments? We'll talk about risk and how financial markets control risk. We'll talk about how fragile the financial system is, why it seems like every 10 years we have a crisis, and whether that's going to change, and what causes these crises. Also, we'll talk a little bit about bubbles. Are there such things? And if they are, what causes them? And where do they come from? How do you get a financial bubble? In a sense, what do we do about financial bubbles?
- 1.57** And in that context, we'll talk a lot about regulation because regulation plays a big role in cycles, in crises, in panics, in bubbles, in all of these things. We'll talk about regulation, and always in the background is money. Remember, everything finance does is money. And so it's crucial we have a good understanding of what that is. And then we'll end with a discussion of the future. I think the future of banking or the future of finance is potentially super exciting, super interesting. We have technologies now that, just a few years ago, were unimaginable.
- 1.58** And it's, I think, really interesting to project into the future and think about, what is money in the future? What is financial markets and institutions in the future? How do we raise capital in the future? All of these things are actually in flux. All of these things are actually changing. Part of that is because of crypto. A lot of that is fintech. And just the ability to transact instantaneously in so many different ways. And of course, that is a potential future, but will we allow it to happen? Do we trust the markets enough to allow them to evolve and to allow it to happen? All right, so we'll end here for this class, and we will pick it up with money and its history next class. Questions?
- 1.59** First of all, thank you for all of this. We talked about the human-made cultural ways that we decide we will restrict such thing, we won't do such thing in a capitalist free

market, whatever. In regards to the Near East, they don't have so much banking because usury is not halal in this part of the world. Do you think that this will change, that this is dynamic, because a lot of people want in?

- 1.60** The reality is that, in the Muslim world, charging interest, the way you get interest in the bank and the bank gives you a loan and you pay interest, is against the Quran. Indeed, as we'll talk about next class, it's against the Bible or it's against certain interpretations of the biblical writing. That's where the Near East gets it from. They get it from the Old Testament. They didn't invent anything here. Usury, which is interest, is against Islamic law.
- 1.61** But, and we'll see this a little bit when we talk about the history of money, Christians found lots of ways to get around this. There was still a Medici Bank. Genoa was this thriving finance center during the late Middle Ages or during the Renaissance, in spite of the fact that usury was a mortal sin. You went to hell for charging interest. So there are a lot of ways to get around this. Indeed, there are banks in the Middle East, and they get around it. They pay you interest. They just don't call it that. They call it something else, and they find different loopholes and different ways in which to restructure this. And I think that's only going to increase.
- 1.62** In addition, I think a lot of the Near East institutions are going to use Western financial institutions. They already do. Because in the West, we're allowed, and they'll let us in. So I don't think that'll be a barrier. In addition, there's no prohibition in Islam against stock markets or against trading in stocks, because that's profit. The prohibition is not about profit; the prohibition is just about interest. We'll talk about the difference between those two, but it's just about interest. So in the Muslim world, they can still have public companies. The Saudi oil company is publicly traded. It's majority-owned by the monarchy, but it is publicly traded out there.
- 1.63** So put it this way, I don't think that the attitude towards finance in the Arab world or in the Middle East is what is going to constrain them. I mean, there are lots of other things that I think will constrain them, primarily culture, lack of individualism, authoritarianism, these all sheikhs and kings, and there's not exactly individual liberty, even in a place like Dubai or in Saudi Arabia or any of these places. So, that is the constraint on them. The constraint on them is not finance. The constraint, I think, on the Near East today is culture.
- 1.64** Saudi Arabia wants to have an innovation center. They want to have high tech. You're not going to get high tech unless you allow some freedom. And are they willing to allow freedom? Are they willing to allow people to think outside the box? Are they willing to allow companies to rise and fail? Are they willing to lose money? We'll talk about the fact that failure is really, really important for making money. Because some companies have to fail when—the buggy industry has to fail, and that capital gets reallocated to places like the automobile industry which is rising.

- 1.65 So are they going to allow what Schumpeter called creative destruction and the dynamism of marketplaces and the dynamism of individual entrepreneurs? That I don't know. That is the challenge that they face. I don't think the challenge they face is necessarily the Middle Ages attitude towards money. That, I think, they will ultimately overcome.
- 1.66 Hey, Yaron. I'm excited to be here. I'm a big fan of your show. I suggest anybody who hasn't seen it, who's watching this, go to the Yaron Brook show on YouTube.com. Anyway, I was just in New York. I flew in, and while I was going through JFK, I noticed they're putting a new \$19-billion investment. I presume this is going to be Port Authority or whatever agency manages JFK. I guess my question is, why is it that financial markets have to match savings with productive activities, rather than the government matching taxes, let's say, with investment?
- 1.67 Well, the government is supposed to match taxes with investment. So in this case, the way they're going to fund the expansion of JFK, the way government usually funds infrastructure, is they issue bonds. They issue bonds to bond holders, and you'll buy the bond. That'll provide them with money. They will promise you a certain interest rate return on that bond. So you will collect your coupons, you will collect your interest payments on the bond. And then in order to pay you those interest payments, they will take tax money and use it in order to pay you back. Ultimately, they'll pay you the principal. All of that, supposedly, is coming from tax money. And that is how government presents itself as funding projects, whether it's local or federal.
- 1.68 Now, on a local level, to a large extent, they're going to have to match expenditures with taxes. Why? Because on a local level, they cannot print money. They cannot print money. So the New York Authority cannot print money. So the only way it can pay you back the bond is through taxes. The real problem you get with government is when you get to the federal level, where they can print money, and now they can avoid taxes, raising your taxes in order to provide all the spending that they do. They don't raise taxes, and instead, they keep refinancing.
- 1.69 So how do I pay back your loan? By taking a loan from him. And then how do I pay you back? I mean, it's a kind of a pyramid scheme. It's a Ponzi scheme. But it's a Ponzi scheme with a printing press in the background, where they can always, if they want, just print more money in order to pay everybody back. Again, they don't admit that, but it's obvious, and they don't say that. And the markets think of it this way. The markets think, look, the government has to pay this money back. It will pay this money back. We're assuming they're not going to default on it. So there are only two ways it can pay it back.
- 1.70 It can either raise taxes or, one day, when they decide to pay it back, because our debt is only increasing, or they can inflate it. They can print the money. And part of what financial markets are doing constantly is trying to figure out how the government is going to handle all this debt. One of the reasons we actually get inflation—I think we

have inflation right now—is because the markets are looking at the government's debt go like this and going, "There's no way they can pay this back through taxes. They don't have the balls to raise everybody's taxes to pay this back." We would probably rebel against that anyway.

1.71 So the only way they're going to do that is by inflating in the future. And if they're going to inflate in the future, I might as well raise prices right now. Might as well anticipate that inflation and drive prices up right now. So in that sense, the federal government doesn't have to match taxes. They can inflate. Now, you could argue inflation is a tax. It certainly lowers our standard of living. It lowers our purchasing power. It lowers our quality of life. But nobody thinks of it as a tax. But that is the reality. But the federal government could do both those things. Local governments can only raise taxes in order to pay for it.

1.72 So we're going to get a beautiful JFK. It looks gorgeous, the renovations that they're doing there, and taxes in New York, or maybe the airlines pay it, which means you pay it through higher tickets. One way or the other, we all are going to pay for that. But at least in this case, we will get a nicer airport. Now, there are lots of other government spendings where you're not even going to get something concrete that you can appreciate. And of course, you could argue that if the airport was private, maybe its funding would be even more effective and more efficient, maybe it would be even more beautiful. But that's not the world in which we live.

Lecture 2: The Origins of Money

At a glance — The origins of money.

- 2.1** What do we need this for? Where does it come from? I mean, and why is it so useful? It's just a piece of paper with a five written on it. It's a five-dollar bill. It's got a bunch of pictures on it. What makes this valuable? And how did we get to this? Is this what they were using 2,000 years ago? Where does the dollar come from? What is it? How would you identify this? Money, but what is money? Basically, that's what we're going to talk about today. And add to that my ability to take this \$5, invest it, and come back with a lot more. How does that work?
- 2.2** So today we're talking about money and interest. And to understand the role money plays, one has to recognize that the important function of trade in human society. Trade is essential for success, for growth, for prosperity. We trade with one another constantly. None of us can produce everything we need. Now, it's true to some extent that, thousands of years ago, there was, mostly people did produce what they needed, or at least groups of people produced what they need.
- 2.3** But even when you go back 10, 15, 20,000 years ago, when all of humanity were hunter-gatherers, you can still, in the archeological record, find evidence of trade. One tribe specialized in making these special cool-looking bead things, necklaces. And another tribe maybe specialized in a particular type of hunting weapon. And they discovered that they could trade those things among each other and that both tribes were better off as a consequence. This is the beauty of trade, right? The beauty of trade is that it's win-win.
- 2.4** I don't engage in it unless I'm going to be better. "I'm willing to give up the weapon because I can make more, and I really want the beads. Right now, these beads are worth more to me than this piece of weapon. So I'm willing to trade." And the other side say, "Whoa, weapons. Never seen anything like it. Maybe we can copy them. These beads, yeah, we really like them. We really value them. But that weapon, more valuable to me than the beads." That's how trade happens. So both people are better off.
- 2.5** So, basically, the fundamental way in which people interact with one another, at least economically, and I would argue maybe even more than economically, the fundamental way in which people engage with one another is through trade, is through trying to establish relationships that are win-win, trying to get to the point where both parties benefit from exchanging. And ultimately, what happens economically is we tend to specialize. We tend to specialize in making really good weapons, really good beads, really good chickens, really good wheat, whatever it happens to be. And then we want to exchange.

- 2.6** It's much more efficient. It's much more productive, this idea of division of labor. Now, 5,000 years ago, division of labor was very primitive. Today, of course, we excel at division of labor. How much of what you consume do you produce directly? Zero? I don't watch my own videos. I don't actually consume what I produce. But by producing it, I can then consume other people's videos or other people's stuff. And we all really, really specialize. And I get my food from making videos in a sense.
- 2.7** So think about a very primitive society where there's limited, but still there exists some level of specialization, some level of division of labor. And we've got these weapons, bow and arrow, let's say, and some beads, and that seems pretty straightforward. But now let's say you also have somebody with chickens and somebody with a cow. And now I want to get some chickens, and I have weapons, but the guy who has the chickens doesn't really want weapons. Doesn't need them. He's got enough whatever. So how am I going to get the chickens if all I've got to offer him is this weapon?
- 2.8** Well, maybe this guy with the chickens want beads, maybe. And the bead guy wants the weapons, so I could trade the weapons for the beads and then use the beads to get the chickens. That was already complicated, and we only had three things. Now, add the cow, and the cow is problematic because, god, it's a whole cow, and that's a lot. And maybe I have enough for half a cow. But what is half a cow worth? Nothing because it's not alive anymore. Let's say I wanted it for the milk. If it was beef, then it would be easy. We could slice it up. But if it's for milk, I have to have the whole cow. How do I arrange things?
- 2.9** Because let's say the guy is willing, he wants chickens, but he doesn't want 100 chickens, which is what the cow is probably worth. He just wants five. But he would take a few beads and he would take a few weapons. And you can see how this gets unbelievably complicated very, very quickly. And the more products we add, the level of complexity increases exponentially. It doesn't just increase flat. It increases exponentially because the number of combinations of things increases dramatically.
- 2.10** A barter economy is a very, very difficult economy to sustain and requires a lot of manipulation. And what actually happens is, it cannot grow. It cannot gain sophistication. You cannot have, I don't know, millions of people specializing. You cannot have dozens of people specializing. You can do it with 5, 10, 20 goods, maybe, maybe. And even then, think of the accounting that has to happen, keeping track of it all. And think of societies where math and writing are still fairly primitive. This is hard stuff, very hard stuff.
- 2.11** So somebody comes up with an innovation and says, "Look, everybody seems to like beads." Beads are pretty cool. And beads are hard to make. So there's not going to be a huge number of beads. There's going to be a constant increase in beads, but they're not easy to make, particularly these special beads. They're unique beads. So what we're going to do is, we're going to use beads in order to facilitate all the trade. So we're going to say, how many chickens are the beads worth? So we're going to denominate

the chickens in beads. And then how many are the weapons worth? And how much is the cow worth? And how much are the hundred other goods that are out there worth in terms of beads?

- 2.12** And beads are what circulate. And people can now buy different things using the beads. So the first guy, he takes the chickens and he gives them to the beads and he gets some beads for the chickens and maybe sells extra. And he gets the beads. And now he can use the beads to buy maybe a weapon and maybe buy some beef and maybe buy some other things. And everybody is now accessing the beads in order to transact, not because they want the beads, because they're using the beads as a medium of exchange. It's the way in which we exchange goods. It facilitates trade. It makes it possible to denominate every good out there in terms of beads. So beads become money.
- 2.13** And indeed, money early on was all kinds of stuff. Shells, pretty shells, beads, certain crops, even. All of them had certain limitations. Beads are maybe difficult to produce, but once there's an incentive because you can use them to buy pretty much everything else, somebody's going to figure out how to make more beads, and suddenly you get lots of beads out there. And what happens to the prices of all the things out there? It goes up. And what happens to the value people place on the beads? It goes down because it's now common. "I don't want to be like everybody else. It's just common."
- 2.14** And people try to use crops. What happens if you use a crop? Well, it's fine if the crop is limited and not a lot of people are growing it. But if I hold on to the crop too long, what happens to it? It's perishable. It's gone. So it loses its value. It goes literally to zero, from having a high value to zero. That's not good. Crops are supposed to be used and quickly. What if I use something like a cow for money? Well, again, perishable. Longer lifespan, though. Very hard to divide it. Not divisible. There's no meaning to a quarter of a cow.
- 2.15** So out of this, out of this experimentation, because it was experimentation, nobody had a theory of economics. Nobody said, "Oh, we need a good money." I mean, there was no concept of money. There's no concept of what that would look like. They tried different things. And through this trial and error, they pretty much figured out the kind of characteristics that would make a good money. And money here is, it's exactly that. It's a facilitator of exchange. It's a facilitator of trade. It's what smooths out trading. Money is its primary function. The thing that defines it as money is that it's a medium of exchange. It's the thing that facilitates our exchanging things.
- 2.16** So what did they figure out? They figured out this money should be durable. It has to last. So it should maintain the way it looks and its value. It can't deteriorate, it can't die, it can't go rotten. So living things, not good money. It needs to be portable. We need to be able to carry it around because I don't know, well, an account or something that is something that I want to purchase. So it needs to be portable. We need to be able to move it and carry it around.

- 2.17** And I think part of that portability is it needs to be divisible. I need to be able to have a \$1 bill, a \$5 bill, a \$10 bill, \$100 bill. I need to be able to divide it. And of course, we have coins, so we can give even change to the \$1 bill. So it has to be durable, portable, divisible. It has to be scarce. If there's a lot of it, then it's not that valuable to people. We'll get to valuable to people in a minute. And I can make more of it. So we don't want to be able to make more of it. If I can make more of it very, very quickly, then we call that in our modern world the inflation.
- 2.18** But I think everybody realized this in primitive society. Again, they tried it out. They saw what happened. Prices went up, and it lost its value. So we want it to be scarce. So durable, portable, divisible, scarce. And then finally, it needs to be widely desirable. It needs to be something that people want, like the beads. Everybody likes beads, right? They're cool. If you don't like the beads, your wife likes the beads. There's a use for the beads beyond the fact that it's used as a medium of exchange. I want it for some other reason than just as a medium of exchange.
- 2.19** So money started out as just this medium of exchange, playing with different things, experimenting, trying different things out. No guiding, no central authority. And you see this all over primitive tribes. You see them, you can go to islands, to different places. They all are looking. They're all experimenting with things as medium of exchange. And some of them come up with something really good at the end, depending on what's available in their environment. And some of them never do.
- 2.20** So what was this island? There's an island somewhere where they have these big rocks shaped in a certain way. They use that as money, and they just keep track of who owns what rock, where they never move the rock. The rocks are just there. And they just keep track of who owns what, and they exchange the stuff constantly. Not a very useful money. It's not portable. It's not divisible. But it's all they came up with, and they never had exposure to the rest of the world, and they have no other real resources there.
- 2.21** But everybody needs, unless you live in a very isolated tribe, and even then, everybody really needs some medium of exchange because, again, barter becomes too damn difficult. In most of the world, in most of the world that developed civilizations, starting with Mesopotamia, Egypt, the whole Mediterranean, really, basically, the money evolved to being some precious metal, usually gold and silver. And the reason gold and silver is it's durable. Gold doesn't change into something else. It's always gold. It is portable. You can carry it around. It's a little heavy, but not too bad. Given the small quantity of gold that existed, it was very valuable. It's divisible. You can splice it up in different quantities.
- 2.22** You remember in the westerns, they come into the bank with a little bag of gold and they weigh it because it's the weight that tells you how much it's worth. Everything is denominated in weight of gold. So it's durable, portable, divisible. It's scarce. It's not impossible to find your gold reserves, but it's hard. It's difficult to get the gold. So the

supply of gold grows very slowly, again, particularly in primitive societies that didn't have access to sophisticated mining equipment and everything.

- 2.23** And it's widely desirable. Why is it widely desirable? For the same reason the beads are. It's shiny and pretty. You can make jewelry out of it. Again, you go to society after society, different places around the world, different places all over the globe, people like gold. They like the shininess of it. They like gold jewelry everywhere. So it is widely desirable. And indeed, it became the money. Money, the facilitator of trade, the thing we use to facilitate exchange, was gold and silver. Some societies, it was silver. Sometimes it was a combination of both, which created a lot of havoc. But it was both.
- 2.24** Silver was always priced less than gold because it was, I guess, less scarce. And so, one of the ways in which you could have divisibility is the smaller denominations would be silver and the higher denominations would be gold in a sense. The other thing that's nice about gold and silver is you could turn them into coins. So you can actually print them and identify them at particular weight so that not everybody has to have a scale, because that's the big problem with gold or commodity money, is how do you weigh it? How do you know how much it is worth?
- 2.25** And it's hard exactly to tell what came first, but it's likely that what came first is money, gold as money, being transacted. And then some autocrat, some king somewhere said, "Well, we need to create coins. I'll put my portrait on the coin, and I'll guarantee that it is of a particular weight." And he then probably said something like, "And nobody else can create coins. So if you want to create a coin, you can only trust me. Can't trust anybody else. Those private people, private enterprise, we never trusted them. So you have to bring your gold to me. I will stamp it into coins and give it back to you, and maybe take a fee for it."
- 2.26** And of course, ultimately, gold mining was nationalized. So the gold finds were perceived to be owned by the king, and all money originated in that sense from the state. And it all came in little gold coins or silver coins. And of course, there's a temptation to cheat. You might say it weighs a certain amount, but it's just a little bit less, but not enough for people to usually perceive how little it is. Again, these are different ways that gold and silver were devalued by the people who printed them or created them.
- 2.27** But pretty much for most of human history, since the beginning of the civilized world until about 100 years ago, but really, if you really think it through, maybe until about 40 years ago, 50 years ago, gold was money. That was the money we used. Everything was gold. We'll get into paper money in a little bit. But basically, so when you think about money, money is a facilitative trade. It's a medium of exchange, and one of the characteristics, one of the reasons we want it to be durable is it maintains its value.
- 2.28** So it's also a store of value. It's a medium of exchange. That's its primary function, but also serves as a store of value. I can keep all my gold in mattress. It won't lose value. It

will always be worth what X number of ounces of gold are worth. It's fixed. Unless there's a big gold find and gold floods the market and it's devalued a little bit. But usually the gold finds are small. It's interesting today, if you think about it, we basically have today access to all the gold human beings have ever mined. It's all accessible. It's either on what people are wearing and the jewelry people are wearing or in gold bars in Swiss and Singapore vaults or in Fort Knox.

2.29 That's all the gold. We know where all the gold ever produced by human history is. And the most we can mine on an annual basis is, maybe we can increase that supply by maybe one, two, maybe if there's a gold rush a little bit more than 2%. That's it. Because we've got such a huge supply. So the base is so big. So the supply of gold never increases so much as to cause any real inflation. And we'll talk about inflation later. All right. So that is an introduction to money. We'll get into more depth into modern money in a little bit.

2.30 I want to take a slight deviation a little bit to talk about interest. Because it is interesting, it appears that interest actually comes before money. That the idea of charging interest comes before money. In Mesopotamia, people used to borrow and lend from each other stuff and demand interest payments, that is, that more be returned to them than was lent of that stuff.

2.31 So for example, I want to start a farm, and you have a farm and you have a lot of seeds. This is the beginning of the agricultural revolution, right? Then I say, "Will you give me 100 seeds? And I'll give 110 back or 120 back." It turns out interest rates were closer to 30, 40%. So maybe 130, 140 seeds back. And I would plant my 100 seeds, and it would grow, and of course, from that, you get a lot more seeds, because every seed produces more than one seed. So I would take a portion of that and plus the 100 and return to you and use the other seeds to continue my business.

2.32 And you can imagine most of this lending occurring with regard to agriculture, because that is the business. That's the entrepreneurial business of the time. There's no manufacturing, there's no tech, there's no biotech. All there is is agriculture, and that's how they transact. And it's pretty interesting. The archeological record seems to suggest that they're real. If you go out there, you can find documents all engraved in stone because there's no paper, there's no papyrus, there's no writing in the sense of a paper, but there is on stone. You find these stone tablets of all the loans people are giving and how much it was paid back and how much they owe still.

2.33 And what's interesting is—I mean, again, I think this is somewhat speculative, but it makes sense—that if I returned all the money to you, if I returned all the seed plus the interest to you, then you destroyed the tablet. If I didn't return it all, you kept the tablet. So what we actually have records for are the bad loans. All the loans that weren't repaid, because all the loans that were paid were destroyed. It's a tablet. So you just crush it and on you go. Who wants to preserve? But you want to preserve the bad debts because I want to have a claim on you guys; if you haven't paid, pay me back.

- 2.34** So this idea of using one person's, call it excess, one person's supply, to provide productive capital for somebody else so that they can start a business or maybe they can grow a business, has ancient roots 5 to 10,000 years ago in the first civilizations that we know of, the Mesopotamian, in the Mesopotamia, of course. If you go back, that's also the first period where you get laws, you get Hammurabi's laws, you get those on tablets. So this is the first time we're organizing in cities and communities because of agriculture.
- 2.35** We're coming in from the wilderness, from the hunter-gatherers. And now we need to be able to govern ourselves. We need to figure out how we behave towards one another. And now there's this impetus, not only—there's even more trade going on. There's more specialization going on. And out of these cultures came money, came gold, ultimately, as money. So there's a lot of thinking going on. Later, what is this interest? What am I being compensated for? Why am I charging? I've given you seeds. Why are you giving me back more? I mean, I get that you should give me back what I gave you, but why are you actually giving back more? But what is the justification for demanding more seed from you?
- 2.36** Or if you think in terms of money, if I give you a loan of \$100, you should give it back to me. It's mine. But why are you giving me more than that? Why are you giving me \$105, \$110, \$120 back, let's say a year from now? What is it I did that makes me, makes me deserve extra, the interest? Now, this is probably one of the questions that has baffled humanity for a very long time. Certainly in the ancient world, they had no idea. So when it comes to grain, when it comes to things like, if I give seed and you give it back, they would think, "Yeah, well, you used my grain and you produced more than what I gave you. Then it makes sense for you to share some of that with me because I was the one who facilitated this creation, this something out of nothing."
- 2.37** There were 100 seeds. Now there's 500 seeds. I don't know how many seeds a plant has, but let's just say it's some multiple. It makes sense for you to share some of that with me. There's a certain sense of justice in that because I facilitated more. But we still don't have a real understanding of, OK, but what is it that I did, other than give it to you? You could also argue that what I forgo using those seeds. So if you think about it, I could have planted those 100 seeds, and I could have got all the bounty. So again, it makes sense that if you're doing it—now, I'm not doing the labor and all of that, so I shouldn't get all the seeds that you produce, but I should get a certain percentage of it.
- 2.38** But then when it comes to money, if I give you X amount of gold, you're not planting that gold. The gold doesn't grow more gold. Why am I demanding more gold from you? How does that even make sense? So if I give you the gold, you should give it back to me, at some point. But it didn't grow into more. The argument here is that money is barren. As compared to the seeds, it's barren. It doesn't grow. It doesn't do anything. It just sits there. It's just money.

- 2.39** Now, think about this. This really means that there's no concept of wealth creation. We might today understand that if I give you the money and you invest that money in a business, and the business grows profits, well, then it's not so barren. But that's a modern idea. There really are no businesses, really. There are some. But that understanding of how business works and the importance of capital and the importance of money, god, that's 1,500 years away. So Aristotle, the great philosopher in Greece, basically says, "Look, money is barren. It makes no sense to pay interest. Interest is, you're getting something for nothing."
- 2.40** By the way, let's do a little bit of terminology cleanup here. When they're talking about interest in those days, granted, they're doing it in Greek, but there's a different term that's being used, and the term is not interest. The term being used is usury. Usury is the term for what we today call interest. Any return on a loan is usury. Now, today, it has somewhat of a negative connotation. It did back then, too. It's always had a negative connotation. Here's the Oxford English Dictionary on usury. "Usury is a premium on money given on a loan. Or the fact, the practice of lending money at interest."
- 2.41** So it's not special interest like we perceive it today. It's just getting a return on my money. It's what we consider today as taking interest. It's only in modern usage do we have this difference between usury, high interest rates, exorbitantly high, usually, and interest, which is normal. We'll talk about why that separation happened. It goes back to how Christianity got around the limitations in usury. OK. So Aristotle comes out clearly against usury, against interest. And his argument is, there's no value created. There's no plus.
- 2.42** Now, remember, this is an era in which trade is limited. There's no economic theory. There's no even theory that sees the world as growing, as wealth being created. There's no such thing as wealth creation in their mind. There is, but there's no such thing. Everything in their mind is zero-sum. I buy an iPhone from Apple for \$1,000, zero-sum. It was worth \$1,000 to me. I didn't gain. Apple didn't gain. The world is the same as it was before. Now, when we think about it, we go, "Wait a minute. Apple made a profit. And hey, you're on, you probably value this more than \$1,000. That's why you're willing to give up the \$1,000. Yes, I value it a lot more than \$1,000."
- 2.43** But that's the essence of trade, this win-win. They didn't have that conception, intellectually. They didn't think of it. They traded, of course. Greece, during this period, was as global as a trader as you could be in those days. They were trading all over the Mediterranean. But they didn't have an intellectual conception of the benefits of trade. And they didn't have an intellectual conception of the win-win nature, and that wealth, in a sense, is created every time we trade. Both parties are better off. And in that sense, wealth is created.
- 2.44** And as a consequence, they couldn't see the value being created through lending. Lending was happening. Transactions were going on, but they were viewed as,

"Something's wrong here. This can't be right. Why are you charging interest?" It was tainted from the beginning, even in Greece. Now, on top of that, the Bible is pretty clear about usury. It's pretty clear about, this is not good. So this is what God tells the Jews. "Thou shalt not lend upon usury to thy brother. Usury of money, usury of victuals." And later on in the passage, it says, "Unto a stranger, thou mayest lend upon usury."

- 2.45** So the Jews are told in the Old Testament that, to their brother, they cannot charge interest. No usury when we trade with our brothers. But to strangers, to foreigners, you can charge them interest. Now, Jews, of course, only view their brothers as other Jews. So the idea was, coming out of the Old Testament, you cannot charge interest to other Jews. But to the non-Jews, God is OK with that.
- 2.46** Now, in the New Testament, there's no direct prohibition on usury, but it makes it clear that your moral duty is to help people. It's to help people who need help. And the idea of charging them interest, that's exploitation and that is, that is sinful. As Luke states, "Lend, hoping for nothing." Again, so lend out of generosity. We'll get back to this with an example from Shakespeare later. Jesus expels the money changers from the temple. It is deeply ingrained in our attitude towards money changers and towards the activities that they engage in.
- 2.47** Now, the idea is, they were cheating in the temple, but it's much deeper than that. There's something about this idea of money changers. Money changers would, they would exchange currencies because there were different gold coins in the realm. This is the Roman Empire. Different gold coins being issued from different places. And they were—what the money changers were was a, you could think of it as a currency exchange. And they were taking a profit. And that profit, doing the exchange, was deemed usury, and therefore, sinful, and therefore, kick take them out of the temple. Certainly, you shouldn't be doing it at the temple, right? But the temple was where everybody aggregated, where everybody came from all these different provinces and exchanged.
- 2.48** So Christianity was opposed to usury from the beginning. And think about it. Think about the difference between Christianity and Judaism. One of the crucial differences between the two religions is that Judaism is very, if you will, sectarian. There are only a few Jews. It doesn't claim to be universal. Jews are chosen people. You can't just be chosen like that. So it's a particular people. And it clearly is an in and an out. There's Jews and then there's everybody else.
- 2.49** Christianity is much more ambitious. Christianity is a universalist religion. Everybody is my brother. Everybody is either a Christian or potentially a Christian. Therefore, when God says you cannot charge interest, cannot charge usury from your brother, now it applies to the whole world. It applies to everybody. It's not just this one tribe, it's all tribes. So there are legal restrictions right at the beginning of Christianity against any

usury. In the Council of Nicaea, it banned usury among clerks, so the clerks couldn't participate in usurious activity. They left the rest.

- 2.50** But then, later on, under Charlemagne, the church expanded the prohibition to everybody. And they defined usury simply as "when more is asked than is given." When more is asked than is given, can't do that. So, no interest is allowed, according to Christian dogma. In 1139, the Second Lateran Council in Rome denounced usury as a form of theft, required restitution from those who practiced the sinful act. In the 12th and 13th century, they denounce usury all they want, but it was going on. It was happening all the time.
- 2.51** But bankers like the Medicis and others developed really, really sophisticated ways in order to provide loans, receive interest, and pretend it wasn't. They called it all kinds of other things. So for example, I lent you money. I'm not really lending you money. I'm taking an ownership stake in your company and you're paying me dividends. It's not interest. And when you repay the loan, you're buying my share back. It's one way, it's very popular in Islamic banking to do exactly the same thing today, because the Islamic banking never evolved out of this, whereas we in the West have.
- 2.52** So there was this clear prohibition on usury. Indeed, in the Council of Vienna in 1311, they declared that any person who did claim that there was no sin in the practice of usury was going to be punished as a heretic. So this is serious. This is damnation. But there was one group of people who could practice usury. And it was somehow OK for them. And those were the Jews. So it's no accident the Jews became moneylenders. There's a lot of reasons they did that, one of them being that they couldn't own property. They couldn't do—most professions were closed to them. So they became merchants and usurers and moneylenders because that's what was available.
- 2.53** But also because usury for them was not a sin. It was OK. And Jews during the Dark Ages, Middle Ages, into the Renaissance, and of course, since then, became moneylenders, moneylenders to kings and princes, moneylenders to businesses. Pretty much, they became the, some of the most important moneylenders in Europe. They never built big—they didn't build big banks until much later, partially because they always wanted to keep a low profile. I mean, think about the Christians' attitude towards the Jews just during this period.
- 2.54** "God, I need a loan. I need to borrow money. I know it's sinful to pay interest on that money, but I need the money. I need the money to grow my business. I need the money to sail a ship off to the New World or whatever it happens to be. So I'm going to have to go to the Jew who's going to lend me money. He's going to commit a mortal sin, according to my religion, but not according to his." That's weird. That's wrong. "And I'm going to pay take him back more than what he gave me, which I shouldn't be doing, but I have to." I mean, you're not going to get a lot of love by being a Jewish moneylender. Moneylending is bad anyway. Now, on top of that, it's Jews.

- 2.55 But you can see, "He's doing something that is a mortal sin. I have to pay him back." You can see how anti-Semitism, which already existed, made, this made it much worse. And there's no question this made it much worse. And of course, one of the ways in which you can get rid of your loans, one of the ways, let's say I borrowed a lot of money and I owe a lot of interest on it. I don't want to pay it. I don't want to pay my loans back. What's one way in which I can get rid of my loans? I could kill the moneylender. But that's a reality that happened all over Europe.
- 2.56 The local aristocrat owed a lot of money to the Jews, would arrange for a pogrom, arrange for the murdering of Jews, usually riling up people, "they killed Jesus," whatever. They'd go, they'd burn him, and he'd make sure that the first thing burnt was the book where the moneylender kept all the records of the loans. In a famous case in England, in 1190, the Jews of York were massacred. And basically this was an attack orchestrated by the nobility who owed money to Jews. By the way, not long after that, Jews were kicked out of England completely. For 200 years, there were no Jews in England.
- 2.57 So there is real hostility in Christianity towards usury and interest, towards this productive activity. Remember, in those days, there were no stock markets. This is finance. Finance is the issuance of coins. OK, that's what government does, the king does, right? Not a government, really, but the authoritarian king does. And then there's moneylending. That's it. Those are the transactions, basically. Different lines of credit, credit for trade, but fundamentally, that's it. And yet that one activity that everybody needs and everybody wants and everybody's engaged in is sinful. It'll get you into hell.
- 2.58 Here's a prominent French bishop arguing, "God created three types of men—peasants and other laborers, to assure the subsistence of the others; knights to defend them; and clerics to govern them. But the devil created a fourth group, the usurers. They do not participate in men's labors." Remember, there's no productive activity; they're just giving you money. "And they will not be punished with men, but with the demons, for the amount of money they receive from usury corresponds to the amount of wood sent to hell to burn them."
- 2.59 Now, that's pretty typical of the rhetoric of the time. It's not unusual. I could have given you 100 quotes like that. So usurers are evil, says the church, because their profits come from, we don't know. We don't know where it comes from. We don't understand. It doesn't come from any productive activity, and it's clearly exploiting others, and particularly if they're charging interest from poor people, that is the worst. And by the way, most of them are Jews. So another reason they hate the Jews. So, again, I think this is an important part of the roots of anti-Semitism. So they become rich. They don't produce anything. It's all zero-sum.
- 2.60 Now, this is the understanding of economics. This is the understanding of trade that they have. It gets to the point where the Christians are willing to lend money to the Saracens, the Muslim Empire, because they believe that when you lend somebody

money, you do them harm. So in a sense, the Christians were funding some of the Saracens' economy, some of the Saracens' weapons, production or whatever, by giving them loans because they thought that because they gave them money back with interest, somehow the Arabs were losing out somehow. That is the zero-sum mentality that they really had.

- 2.61** Now, this starts changing in terms of the economic understanding, not the moral understanding, but the economic understanding. It starts changes in the Middle Ages and the later Middle Ages, moving into the Renaissance. Certainly, Thomas Aquinas has a much more sophisticated understanding of usury. But he thinks that, "Yeah, it's understandable that I require some return on the money I've lent you because time has passed and there's some value to the fact that I can't use that money over this time." And he's the first one to start conceptualizing interest as, which is what it is, I think, the price of time. The price of time.
- 2.62** But, say many of the medievals of the period, "But wait a minute, time doesn't belong to you. Time belongs to God. And you want to benefit from something that belongs to God." So again, they are really fighting this. They really don't want this to happen. So in a sense, usurers are attempting to defraud God, taking stuff that belongs to God, from Him. All this time, of course, as we said, bankers are cheating. They're finding ways to give out usury. They're finding ways to make these loans in spite of everything. And somehow the economy grows. During the Middle Ages, you get small economic growth. More trade is happening.
- 2.63** Also, the Christian world is exposed to the Arab world, and the Arab world is just—this is a little post its golden era, although in Spain, the golden era is still going on. They know a lot more than the Christians do about science, about a lot of things. And one of the things that the Arabs have that Christians don't have is an actual numbering system that is useful. Do you know, I mean, this is mind-boggling when you think about it, that all math in the Western world, all math in the Western world, was done using Roman numerals? Try to add, subtract, and multiply with Roman numerals. Try to keep track of stuff with Roman numerals. It's impossible to do.
- 2.64** You'd never have calculus. You don't even have algebra. You can't have algebra. I mean, it's just impossible to do the math that we take for granted with Roman numerals. But as they're trading with the Arabs, they discovered that the Arabs have a different numbering system, a numbering system the Arabs have learned from the Indians and have brought from the Indians. And that numbering system is the numbering system we use today. It has the concept of zero, which is a huge conceptual achievement, which the Indians in India have. And they have 10 numbers. Now you can add, subtract, multiply, easy.
- 2.65** And suddenly, math takes off. Math can't develop as long as it's bounded by—and math is much more developed in the Arab world than it is. They have algebra in the Arab world at this point. They don't have it in the Christian world. And then an

innovation out of this is double-entry bookkeeping, which is going to be really, really important when you start doing banking, liabilities, assets, figuring out, matching them, who I owe what, keeping track of who owes what to whom. All of that is made possible because of double-entry bookkeeping, and all of that is during this late Middle Ages period.

- 2.66** As I said, bankers are cheating constantly. The Church is also cheating. So this is a period where the Catholic Church is doing all kinds of things on the side. And it often is the lender. Sometimes when the moneylenders are charging lower interest than the Church is, they prosecute the moneylenders for moneylending, for usury, so they can get rid of the competition, so they can gain the profits. Then, on the other hand, some moneylenders are treated differently, like the Medicis. One of the things that the Medicis had going for them is they were very in with the popes. One of the reasons they were very in with the popes is at some point they became the popes. The Medicis were the popes. Those popes are not going to go after the Medici bank. They might go to the other banks.
- 2.67** At Genoa, during the Renaissance, developed into this real finance center. It was the place, and basically everybody ignored usury laws, and the city officials got it. They understood that their wealth was dependent on ignoring these things, so they just allowed it to happen, and they completely ignored it. This was also the period, as early as 1220, but it doesn't really catch on until much later. But 1220 is the first time you see a new concept come about, and that is the concept of interest. It comes about in this funny way.
- 2.68** Interest derives from the Latin verb "intereo." And intereo means to be lost. So interest payments are a loss. But now there's starting to be a separation. So think about it this way. Compensation for a loan is illegal, right, if it's a gain or if it's a profit. But if it's a loss, interest, then it's OK. So conceptually, they're trying to wrap their heads around, trying to come up with excuses, trying to come up with ways where we can continue the behavior that we're engaged in, but for it not to be perceived as sinful and immoral.
- 2.69** Over time, they gain an understanding that if the lender is, for example, putting some effort and time into evaluating the loan, into choosing who to give money to and who not, then that might be deserving of something in return, some compensation. So it's OK to charge interest in those cases. If the lender can show, in a sense, that they could have used the money for something else that was profitable, then maybe it's OK for them to charge interest. So now you're getting the beginnings of the concept of opportunity cost.
- 2.70** What else could I have done with the money? What else could I have done with the money? And if I can do that and I'm giving it to you, should you be compensating me for not being able to do that? So you're starting, during the late Middle Ages into the Renaissance, starting to see this grappling going on among the scholastics. These are the, these are the Catholic thinkers inspired by Aristotle, inspired by Aquinas. And so

there's a little bit of a loosening up. But by the way, the Catholic Church does not eliminate the prohibition on usury until the late 19th century.

- 2.71** So during all this period that exists, again, I want you to think about how this weighs on people and how this weighs on our attitude towards banking, towards interest, towards finance more broadly. These cultural elements don't just go away. They really do stay with us. I mean, anybody reading Dante, and we still read Dante because it's amazing literature, but in Dante's *Inferno*, guess who's in the seventh rung of hell with a bag of money around their neck? Moneylenders, the usurers.
- 2.72** They've got a big bag of gold, and it's pulling them towards a fire. And that image, that image and that literature, stays with us and stays with us and impacts, I think, much of, of our perception of finance. All right, let's wrap it there. We'll get back to this issue next class.
- 2.73** Hello. When you talked about the lowest level of hell reserved for the moneylender—
- 2.74** Well, seventh one, I think there are nine rungs of hell. So it's not the lowest, it's like almost the lowest.
- 2.75** Do you think it's because profiting on money is like, it's a purification or just looking at profit very abstractly? I mean, there's nothing else except for profit. The guy who grows corn, he can feed people. This person is benefiting off of just pure profit.
- 2.76** Absolutely. There's no question. I mean, the profit of a financier is the purest of profit in the sense that it's not connected directly to any product. So think about it this way. If I'm Steve Jobs and I'm making a lot of money at Apple, I'm still not a great guy because I'm making a lot of money and nobody likes that. But every time somebody comes at me and says, "Why are you making so much money?" I pull out an iPhone or a Mac or something and say, "Look, this is what I gave you."
- 2.77** But now I'm Mike Milken. I don't know if you guys know who Mike Milken was, but a famous financier, made a lot of money. I think he, one year, had a salary, this is in the '80s, of \$200 million. It was a lot of money back then. \$200 million. OK, explain this to me, Mike Milken. There's no product. I can't show you a product. Now, I can, because without the Mike Milken, there is no iPhone, there is no Mac, none of these products exist. But that's one step removed. And it's a huge abstract step. To show that the activities of the financier lead to the production of all these goods is difficult to show.
- 2.78** So in that sense, you're right. It's the purest form of profit. It's, in its origin, requires abstraction, requires thinking, requires figuring out what's going on. Most people don't do that work. They see a moneylender shuffling papers and making a lot of money. Where's the product? Where's the service? What have you provided me? And financiers are very bad, very, very, very bad, at explaining what they do. Part of what I'm trying to do in this course is trying to explain what they do, but they are very bad at doing it themselves.

Lecture 3: The Birth of Banking

At a glance — The birth of banking.

- 3.1 So in the last class, we discussed the history of money and the history of usury, of interest, of the prime mechanism by which finance works, uses money, and then a return on that money in the form, primarily, of interest, a reward for lending out the money. And we got to the point where it's clear that Christianity and, therefore, the Western world of the time were really antagonistic towards this innovation, which was, which was interest and the banking that started to develop throughout Europe.
- 3.2 And the banking that started to develop was a direct response to trade, the need for trade, the need to plan for trade, the need to think about long term, and also think about geographic distances. Europe was becoming more integrated, more trade was happening. And this really brings us to the beginning of changing attitudes. I mean, up until now, usury, up until, I'd say, the Renaissance, the Reformation, usury is viewed as a sin, a mortal sin, a really, really bad thing. People are doing it on the side, but there's no question about how bad it is, and there's very little understanding about its economic function, why it's any good.
- 3.3 And that starts to change in the Renaissance and the Reformation. The Reformation in particular, both Luther and even more so Calvin, but Luther has this alternative view. Luther basically comes to the conclusion that, look, usury is bad. It's sinful. It's horrible. But we live in a sinful world. And it looks like it's needed. We don't quite understand why, but everybody does it. So fine, let them do it. So for Calvin, sorry, for Luther, civil authority should allow usury. And because it's inevitable, because men are sinful and it just is, this is the way it is. But it should be condemned in the harshest terms in terms of morality.
- 3.4 Late in his life, Luther writes, "There is on Earth no greater enemy of man after the devil than a gripe-money and usurer, for he wants to be God over all men. And since we break on the wheel and behead highwaymen, murderers, and housebreakers, how much more ought we to break on the wheel and kill, hunt down, curse, and behead all usurers?" But we're not going to do it because it's convenient to have them do what they do for the economy and for things to happen. And again, everybody's sinful, so what can you do?
- 3.5 Calvin was a little bit, I think, more understanding. He believed, Calvin really believed that, and this is the first time you get this really in Christianity, Calvin really believed usury did not conflict with God's law. He was like, it can't be that God allows the Jews to do this with people outside when it's sinful. God would have never permitted it for the Jews to do with foreigners if it was truly as sinful as Luther argues. So it can't be that bad. So we need to now fine-tune what we mean by usury. If usury is done in the course of business, then it's OK. But to be a usurer, to be in the profession

of moneylending, that's still not good. That's still too pure—what was the term we used before? Too purely motivated by profit, not enough of the public good.

- 3.6** But Calvin is already reflecting the fact that people are starting to gain a better understanding of what's going on, of how this process work. Again, we all take it for granted. You borrow money, you have to pay it back with interest. This was not obvious in those days because they wanted a justification, and they didn't yet have it in terms of a totality. So usury is not Christian. That's agreed upon. But it serves an economic purpose, so we're mostly going to ignore it.
- 3.7** Indeed, in England, in 1545, for the first time, Parliament passes a law that allows for the charging of interest and sets a boundary on it. It was 10% and it was lower to 5%. And then they said, "No, this is wrong. We need to do away with interest." So they kept changing their mind. But for the first time, there was a political recognition. "No, interest is something that happens. Maybe because it's sinful, what we should do is, rather than stop it from happening, we should regulate and control it. Regulate and control it and put limits on it." And that's what the bank, that's what England does, is it put—there's no Bank of England at this point. It puts a limit on it.
- 3.8** For example, Francis Bacon believed that the ideal would be lending at zero interest. That's a Christian ideal, of course. And it's what the Bible required, he believed. But he thinks that's utopian. And he says, "It's better to mitigate usury by declaration than suffer it by rage, by connivance." So he's saying it's better for the government to regulate it than to let it just be out there, even though the utopian ideal is zero. So he proposed two rates of interest, one 10%, one 5%, depending on now the risk involved. And this is the first time that this concept of risk gets introduced. "Maybe I should be compensated if I'm taking on more risk."
- 3.9** But maybe the thing that is lasting and most remembered about the attitude towards usury in this period comes to us from Shakespeare. Shakespeare, of course, is a genius, and he has a way in terms of how he expresses these things that is quite amazing. And he wrote a play about usury. I highly recommend it. It's one of his great plays, *The Merchant of Venice*. There's actually a movie of it, I think, with Al Pacino playing Shylock, which I highly recommend to watch if you can follow Shakespeare, but it's worth it.
- 3.10** In this play, he actually articulates. All the debates that are going on about usury are in the play, in the language of the play. For example, Shylock, who is, of course, a Jewish moneylender, is confronted by his nemesis, who is a Christian moneylender. But the Christian moneylender doesn't charge interest. And that is the conflict in the play, is between the two. In the story, this Christian moneylender needs cash. The boat with his merchandise is late. It hasn't come back to port. He's short on cash, and he can't find cash anyway. And so he goes to Shylock. He goes to Shylock, the Jewish moneylender.

- 3.11** Shylock hates the Christian. He hates Bassanio. And Bassanio has been very rude to him and has been bad for business. If you're trying to make money off of lending money and this guy is charging zero interest, not good for business. So he doesn't like, I mean, he hates this guy. So he says, "OK, I'm willing to lend you the money, but instead of—and I'm not going to charge you anything for it. But," he says, "if you cannot pay me back, I want you, if you cannot pay me back in money by the time we approved, you have to pay me back with a pound of flesh, cut out from the body, a pound of flesh."
- 3.12** So here's the Jewish moneylender. Again, think of all the Jewish stereotypes. Think of anti-Semitism, Shakespeare's reinforcing them all in this play. It is an unbelievably anti-Semitic play, but a work of genius, nonetheless. So you get here a situation where he's basically, if the guy can't pay him back, he's asking for his life. This is Shylock explaining his hatred of Bassanio.
- 3.13** He says, "How like a fawning publican he looks," because he dresses nicely. "I hate him for he is a Christian, but more for that, in low simplicity, he lends out money gratis and brings down the rate of usance," usury, "here with us in Venice. If I can catch him once upon the hip, I will feed fat the ancient grudge I bear him. He hates our sacred nation, and he rails, even there where merchants most do congregate, on me, my bargains, and my well-won thrift, which he calls interest. Cursed be my tribe if I forgive him."
- 3.14** So you see here, there's competition, and he hates him. He's putting down Jews, and he rails against the fact that these Jews are taking interest on their loans when he is a good Christian. And generally in the play, Shylock is portrayed as angry, vengeful, greedy, not a good guy. When his daughter elopes, his daughter elopes and takes with her the father's money; she runs off with the money. He cries, "My daughter, oh my ducats, oh my daughter." Like, not clear who he cares more about. Does he care more about the ducats or does he care more about his daughter?
- 3.15** Here's Massimo. Massimo, he says, "If thou wilt lend this money, lend it not. As to thy friends, for when did friendship take a breed for barren metal of his friend? But lend it rather to thine enemy, who, if he break, thou mayest with better face exact the penalty." So again, Massimo is reflecting this idea of money being barren. Money cannot produce money. So don't do it to friends, only do it to enemies. Only lend money to enemies. So he represents the traditional Christian view of the time regarding usury.
- 3.16** And I'm not going to tell you how the play ends or how it develops. You're going to have to go watch it. You should. But you get all the stereotypes, all the attitudes towards usury in this play, of course, embellished with the genius of Shakespeare. If we go historically, after the Reformation and the Renaissance, we get the Enlightenment. And here, likemost of our modern world, we see real knowledge. We see a real change, a real shift. During the Enlightenment, you really get people starting to understand what is going on here.

- 3.17** What is interest about? Why is this projection right economically? And they start hinting at the fact that maybe it's also OK morally. Although, I think even to this day, we still haven't come to grips with that idea. Even to this day, we're still in Shakespeare's world and Dante's world, at least to some extent. So you get a French scholar teaching in Holland who basically makes a thorough argument about why it's not barren, right? Because the money that could be used for a business who generates profits, those profits are the fruit of that initial investment. So it's not barren at all. It actually produces stuff.
- 3.18** Turgot is also an 18th-century French economist. He identifies usury's connection to property rights. Well, it's mine. If I'm going to give it to you, well, I'm forgoing use of it. And you should compensate me for using my property. He says, just like we pay rent when we use somebody's house or we use somebody's land, we should pay rent for using somebody's money. So again, another argument for why this is.
- 3.19** He also says, look, it's also true that time changes the value of the money. He says we perceive the value of money differently if we're going to get it five years from now than if we have it today. What would you prefer? You could ask anybody. Do you prefer \$100 today or \$100 in five years? And even put aside inflation, put aside risk, because you're certain to get it. I got to have it today. So there's a time value for money. This is a massive innovation that Turgot is doing.
- 3.20** And he says, "Time doesn't belong to God. Time belongs to the individual." So a recognition of property rights over yourself, in a sense that it's your time. And since it's your time and you're giving some of it up through money, you should be compensated for it. So now you're getting this idea again, but in a fuller sense of the time value of money, the fact that money is ultimately compensation for time. Bentham, the famous English philosopher, wrote a treatise in the late 18th century called In Defense of Usury. Bentham argued that any restrictions on interest rates where you economically harmful.
- 3.21** The government shouldn't control it. Nobody should control it. Because it basically reduced innovation and it reduced the ability of the innovator to raise capital. And it's not an accident. This is the industrial revolution. People are now raising capital. They're starting businesses. They're starting to use the money that they can borrow to actually build industrial projects. In the Middle Ages, even in the Renaissance, economic activity is very minimal. There's very little going on in comparison to anything we know today. The industrial evolution becomes vibrant, energized. And it's vibrant and energized off of the ideas coming out of the Enlightenment, so this all goes together.
- 3.22** The source of this is the Enlightenment ideas. So strict usury laws are harmful to borrowers. Credit markets will shrink if you have restrictions on them, but demand will still be there. And how do you satisfy that demand? Well, it goes into the black market. There's an illegal market, and in illegal market, interest rates are going to be much higher. So Bentham is saying, if you put in caps, if you put in restrictions on interest, it's

not going to take away the demand. They'll just pay higher interest rates, and that's just unjust, and that's just wrong. Get rid of the demand, get rid of the restrictions, let the market function, and demand and supply will match up on the different rates.

- 3.23** And he even makes a moral argument. He says, "My neighbors, being at liberty," so a liberty argument, "have happened to concur among themselves in dealing at a certain rate of interest. I, who have money to lend, and Titus, who wants to borrow it of me, would be glad, the one of us to accept, and the other to give, an interest somewhat higher than theirs. Why is the liberty they exercise to be made a pretense for depriving me and Titus of ours?"
- 3.24** The idea is, if people come to an arrangement, if people voluntarily agree to pay a certain level of interest, who is the government to step in and intervene and try to control and try to say, "No, that's not acceptable. That interest rate is OK, but this interest rate is not"? So here's an argument for economic liberty, economic freedom, at least from a practical perspective. Of course, you get support from Adam Smith, who also says, "As something can everywhere be made by the use of money," so money can be—you can use it for anything. You can make anything for money, right? "Something ought everywhere to be paid for the use of it."
- 3.25** So again, it's not barren. It's the opposite. It's unbelievably fruitful. It creates whole industries. There is no industry without money. There is no industry without capital. There's no industry without an investment. Well, of course, then you should pay for it. Of course, the lender should be awarded for it. So by the end of the 18th century, into the early 19th century, there is a pretty much complete understanding of the role of interest, where it comes from, what it is, and why it's as high as it is. So there is a basic interest rate that is just to compensate me for the fact that you are using my money for a period of time. I can't use it.
- 3.26** The time value of money, the fact that I will only get my money in the future. I'm going to ask for some compensation for that. So that is clearly recognized, and it's viewed as this idea of some future. So this is a compensation for time. There's also a recognition that risk matters. That if I have two alternative people who want to borrow money from me, and one, I'm more confident will repay it, and the other, I'm less confident they'll repay it, I should charge a higher interest rate from that who I'm less confident with. Because there is a chance, particularly if I'm making a lot of loans, there's a chance that I won't get my money back. I need to be compensated with the interest for that risk.
- 3.27** So there's a recognition of risk, a recognition of time value of money. And maybe a little later in the story, there will be a recognition of inflation. So the fact is that I can buy less with the money, particularly in our modern times, buy less with my money in the future. I need to be compensated for the loss in purchasing power. So if you think about what the components of interest today, what goes into an interest, it's time-

value money, which is often called the real rate of return, the risk-free rate of return. It's just what people will, to delay receiving the money.

- 3.28** And that delay comes from, I could use it to buy ice cream. I'm not going to buy ice cream. I'm not going to consume it now. There's a price to that. I have to delay my consumption until later, and I want some compensation. So that's the base interest. And on top of that, you overload risk. And on top of that, you overload a compensation for the loss in the value of the money. Those are the three components of an interest rate—a real rate, compensation for time, a risk, compensation for risk, and inflation, compensation for loss in purchasing power.
- 3.29** But of course, in spite of this, there are plenty of thinkers who are not ready to give up on the fact that finance, broadly, even usury interest rates are still bad. Certainly, Marx does not accept this. According to Marx, moneylending, other financial activity are not productive. They're exploitative. All value comes from manual labor. All value comes from the workers. The capitalist is just exploiting. He hasn't done anything. He hasn't produced anything. He hasn't picked up a hammer. He hasn't used his muscle, which is what matters to the labor theory of value, which is Marx's theory. And therefore, any return he gets is exploitation, because that return comes at the expense of the worker who's actually done the work.
- 3.30** As a 20th century Marxist has put it, "The major argument against usury is that labor constitutes the true source of wealth." So you're just exploiting people. Even Dostoevsky has a negative, generally a negative view of finance and speculation and money. In *The Brothers Karamazov*, he writes, "It was known, too, that the young person had been given to what is called speculation, and that she had shown marked abilities in the direction, so that many people began to say that she was no better than a Jew. It was not that she lent money on interest, but it was known, for instance, that she had, for some time past, in partnership with the old Karamazov, actually invested in the purchase of bad debts for a trifle, a tenth of their nominal value, and afterwards had made out of them ten times their value."
- 3.31** Today, we call that vulture investing. Vultures, right, picking on the dead? You find companies that the world thinks are terrible and failing and everything. You can buy their shares or you can buy their bonds for very, very cheap. But you've identified the fact that they probably will survive. They'll probably do fine. And then you wait until they do fine and you sell and you make a lot of money. Clearly, the perception is that's somehow bad, right? Whereas you could argue that's an incredibly productive activity. By buying that stuff, you're letting other people exit who don't believe in it, so they have a way out; they can get their money back.
- 3.32** And you're also expressing your belief that this will do better. And that could, in and of itself, lead others to provide capital. And we'll talk about this more when we talk about stock markets and how they work. So during this period, you get a real shift. While the old attitudes toward usury sustain themselves, the world is moving on. The world has

given up on those attitudes, and it is driving forward with one big major exception, that there's almost universal agreement that we need to watch those usurers. We need to watch those bankers. We need to regulate them. We need to control them. We need to monitor what's going on. We can't just let the market go.

- 3.33** And you get massive innovations during this period. You get the Amsterdam Stock Market, created in 1602. So the first stock market where people can trade shares in companies. US Stock Market, 1792. Again, industrial revolution, the beginning of it. You get Lloyd's Insurance, the first insurance company in the world. There's always been insurance, always been insurance markets, but Lloyd's is the first organized place. Anybody know where Lloyd's come from, the name Lloyd's? It's a coffee shop where they would all meet and get together. Most of these things are founded in coffee shops. I think that's true of the US stock market as well.
- 3.34** In England, coffee became a big deal during the Enlightenment; that's where people would meet, that's where they'd do business. And Lloyd's was founded in 1689. You'd see derivative contracts, options, futures. We'll talk a little bit about them in a future class. They become popular in Northern Europe. Again, they've always existed to some extent, but they became real popular and became traded during the 17th and 18th and into the 19th century. So finance is innovating, finance is growing. These activities are massively increasing during this period as the attitude becomes more and more relaxed.
- 3.35** It's not clear what comes first, but certainly without this relaxation of attitude, the authorities would have crushed whatever financial innovation was going on. So ideas usually come a little before the actual execution. But here there's constantly this struggle between people doing it in spite of the authorities, in spite of the ideas and the ideas catching up to it. So there's a constant tension going on.
- 3.36** So this is where we are, 19th century. The markets are allowed to function for the most part. We've got some controls, not a lot. They're going to get much worse. We'll get to that. There's suspicion of usury, but it's allowed. It's allowed. Let's talk a little bit about banks and how banks fit in to all of this. Now, to some extent, there'd always been banks, the Medicis and Jewish moneylenders and so on, but they weren't formal institutions, particularly the Jewish moneylenders. Usually, it was one person who would lend money and collect interest or an affiliation of individuals who would do this.
- 3.37** The Medici Bank did not function like what we associate with a modern bank. Most of what the Medici Bank did was letters, in a sense, letters of credit that allowed for trade over large distances. So you would put the gold—gold was money during this period, right?—you would put the gold at the Medici Bank. But the purpose of putting the gold was to pay a merchant. Let's say you did it in Florence and you were going to pay a merchant in Paris, but you didn't want to have to carry that gold all the way to Paris. Very dangerous in those days. So you put it in the Medici Bank.

- 3.38** The Medicis would give you a letter of credit, saying the gold has been deposited here. You would take that letter of credit to the merchant in Paris. Maybe a Medici affiliate in Paris would give that merchant the gold based on that. And the two affiliates, the one in Paris and the one in Florence, would settle after the fact. And if there was gold being transported, the Medicis would transport it in a heavy guard, and it would be a large quantity. It wouldn't be you trying to transport the gold by yourself across these distances. And that's how trade was facilitated through these letters of credit, and that was the main activity of bankers in those days.
- 3.39** But as people became wealthier in Europe, particularly in Northern Europe, during the late Renaissance and into the Reformation and then ultimately into the Age of Enlightenment, people became wealthier and they started to have gold. They actually possessed gold as a representation of their wealth. And these might have been merchants. They might have been a variety of different types of merchants or even professionals in Northern Europe. And they found it cumbersome and difficult to hold all this gold at home or in a hole in the backyard, or whatever it was. People would break in and steal it. And it just took up a lot of space.
- 3.40** And they went to people who deal with gold. And the obvious people in those days were the goldsmiths, who dealt with gold. They say, "Look, you already have gold. So you already have a vault. And that vault is really, really protected. And you don't let anybody break in. Please, would you be willing to store my gold in your vault?" And the man says, "Sure, I'll do that for a fee. You deposit your gold here in a fee." And then what they would do is they would give them a receipt. "Here's a receipt for two bars of gold."
- 3.41** And over time, the merchant who might have received this receipt, he's going to buy something, something that is worth two bars of gold, let's say. And he says, "I could go to the bank, to the goldsmith, and get my two bars and then take it to this other merchant to get whatever it is that I'm buying." That's cumbersome. It's a lot of work. And again, there's risk involved in taking the gold out. Or, "Maybe I can go and ask the goldsmith if he would be willing for me to transfer this receipt to this other merchant, and then he can collect the gold when he wants it." And the goldsmith says, "Sure." And they made these notes negotiable. They're transferable. Anybody could use them, as long as they were signed over.
- 3.42** And you can see how this would work, right? If the notes were of small enough denomination, now you can start using them to buy stuff, and everybody would be willing to take the note because they'd know that if they went to the goldsmith, they could get the gold. And this was a great deal. Suddenly, you had a massive efficiency in trade. No longer people lugging around gold and handing it over to people, even within the same city. But just like in the days of the certificate of credit, now you could do it even within your own city and within your own community. All you had to do was pass a piece of paper around.

- 3.43 Soon, the goldsmith made it not only negotiable, but basically anonymous. So you'd come to the bank—the goldsmith, it's not a bank yet—you'd give them the gold, and they would give you just a piece of paper saying this piece of paper is redeemable into X ounces of gold. Now everybody's using these pieces of paper as money. The real money is in the bank, in the vault. But the paper is what people are actually using as the medium of exchange. But the reason why other people are accepting it is because they know they can always convert it into gold.
- 3.44 And this is really the beginning of a gold standard, a paper money gold standard, a paper-based gold standard. Before, everything was gold coins. Now out with coins, now I can use paper. Much easier, much simpler than dragging around coins. The next step in this evolution is that the goldsmith is sitting there with a vault full of gold. And he knows people that would like gold to invest in businesses, to grow, to grow their existing business, to buy some seeds, to do whatever they want. And they would love to borrow that gold to facilitate economic activity.
- 3.45 And the goldsmith realizes that it's almost never the case that a lot of people come all at once demanding the gold in this vault. And basically, the vault is full all the time. And it's a waste because the money is just sitting there. So what the goldsmith starts doing is, starts making loans of that gold. Now, again, he might not literally take the gold bars and give it to somebody. He doesn't have to anymore. All he has to do is give them a note saying they can access it. But that goes together with the fact that they owe him this note back. So it's two directions.
- 3.46 Now, this is the beginnings of what's called fractional reserve banking. It means that X is deposited, but the goldsmith or the banker now can actually make loans of the money that was deposited. Now, the banker is saying, "Oh, the more deposits I get, the more I can loan out. So I want to encourage people to give me their gold. So if before I was charging them to store their gold, maybe I can give it to them for free, or maybe even I can pay them a small interest so they give me their money, and then I can lend it out at a higher interest and make some money."
- 3.47 Now, they can't lend out all of it because then they'd go bankrupt very quickly. People would come demand their gold, and it wouldn't be there. So they only lend out a fraction of it, but they lend up enough of a fraction of it to be able to make money off of the difference. And this is really the beginning of modern banking. And you can see, it's also the beginning of paper money. You can see suddenly paper is becoming the thing that we associate with money. That's what we use to transact.
- 3.48 But at the end of the day, it's all backed by gold. You have more paper money than gold because of the fractional reserve, because they're issuing some money based on other people's gold. But that is offset by the fact that the person owes money back to them. So there is more money circulating at any given point in time than the gold reserve, but that is capped by how much reserve they keep and by the fact that their loan has to be repaid. So economic activity is thriving or starting to thrive. It's starting to grow. It's

substantial. But the amount of money in the economy is still constrained by the amount of gold. It's not one-to-one, but it's still constrained by the amount of gold.

3.49 Now, this constraint poses a problem, a problem primarily for governments. And maybe the best example of this was the government of Louis XIV. They loved to spend money. He was prolific. He spent all over the place, I think he was the owner, built Versailles, and lavishly spent money on parties and on everything else, money he didn't have. The tax revenue was not enough to pay for the way he lived. So he borrowed. He borrowed huge amounts of money, huge amounts of money. And when he died, his young son, Louis XV, was going to become king, but he was too young. So there was a regent appointed.

3.50 And the reality was that this regent faced the situation where there wasn't any money. All these debts had to be paid. And you can't conjure up money out of nowhere. So he had to raise taxes, but taxes were very unpopular, particularly with the aristocrats. They're the only ones who had money. He didn't want to piss off the aristocrats, so he had a real problem. And this is where a mathematician, economist you could call him, by the name of John Law comes into the picture. John Law is the first one to innovate and to really innovate with something, what we call today fiat money, which is what we have today. Fiat money was actually used in China in the 13th century, so they got a head start. But in the West, John Law is the first one to really do it on scale.

3.51 Law was actually convicted of murder in England. He's a Scotsman. He was convicted of murder. He killed somebody in a duel, claimed the guy had fallen on a sword, nobody believed him, convicted to death. He was sentenced to death, but he was well connected. They tried to get him out. No, the authority said, no, he has to hang. Anyway, he had important influential buddies. They got together, they arranged an escape. He managed to get out and escaped to France. He had written a lot about, he was a mathematician. He'd written a lot about what he called central banking and about how governments could create their own currency and about how this limitation from gold was barbaric and wrong, and there were ways to do this that were much better.

3.52 And when he came to France and he met with the regent, he was a charismatic guy. He convinced this guy that, let's experiment, let's try this out. So John Law started a bank authorized by the French government and started issuing notes, bank notes, not on the basis of the gold reserves that he had, but he started issuing notes by buying government bonds, buying the government debt, buying all the loans that the king had made, and money started going out. On top of this, Law started a company called the Mississippi Company, which controlled—he got a grant from the king to control all the trade coming from the Americas.

3.53 France, if you remember, this is before the Louisiana Purchase. So France controlled much of the land that there was around the Mississippi, Louisiana, western United States. A lot of that was owned by France. And now he controlled all the trade coming

from there. He also then got the king to give him permission to basically hold a monopoly over all trade coming from the French colonies into France. So he had a monopoly over all this trade. He sold shares in this company, and he was flooding the market with these banknotes, and then encouraging people to use those banknotes to buy the shares.

- 3.54** The company then would buy more government debt, so getting the loans off the books. The company ultimately incorporated the bank into it. And it became this massive business enterprise. At the peak, John Law was probably the richest man ever. And it became this scheme where the stock price only went up. No real substantial revenues to justify all of this. But most of this money was going to bail out the government. The government was already giving them all the favors of monopoly power. Money was being printed by the bank, by this Mississippi company, really. Money was being printed. That money was being used to buy the shares. More money would be printed, used to buy the shares.
- 3.55** And what you got is, everybody in France suddenly felt rich. The French economy was doing fantastically well. People were buying everything. There was a huge boom in consumption. Everything was doing fantastic until somebody started asking questions about what underlies all this? What's at the base of all this? Of course, inflation picked up. Prices were going up, not just the price of this one share, but prices just generally in France were going up. It's inflation. So people were getting really, really upset. And a few of the smart players were like, "Something's rotten here." Whether they understood exactly what was going on or not, they started taking the gold and moving it out of France.
- 3.56** So the government said, "Can't do that anymore. Stop moving gold out of France." They literally would search, like, buggies and horsemen leaving France, search for gold and confiscate it if it was found. But soon enough, as people realized what was going on, they started selling their shares. Soon enough, there were no buyers for those shares that were being sold, and this entire pyramid scheme collapsed. John Law managed to get out of France in time and escape and make it, I think he went to Amsterdam, make it out alive.
- 3.57** John Law is considered maybe one of the founders of the idea of a central bank, one of the founders of the idea of fiat money, one of the founders. The argument today would be, he got it a little wrong, but in principle, he was OK. In principle, he was right. Indeed, this idea of a central bank, a bank controlled by the government, a bank that funded the government, was already percolating. By the way, this all happened in the early 18th century. So this is about, what, 70 years, maybe 80 years before the French Revolution.
- 3.58** But the consequence of this, by the way, was basically a depression in France. I mean, this crushed the French economy completely. It took it a long time to recover. You could maybe say that the seeds, at least economically, of the French Revolution were

being laid by John Law's scheme. I mean, it probably would have happened anyway, but differently; and clearly, France was a lot poorer for what he had done to it. Nevertheless, his ideas became prevalent. Even before Law, you'd already had the Bank of Sweden founded in 1668. The Bank of England was founded in 1694.

- 3.59** The Bank of England was the lender to the government, but it was very much constrained by the gold standard. They could only lend based on gold, initially, of course. Over time, these central banks, which now spread throughout Europe over the 18th and 19th century, over time they gained more and more and more power. They were constrained by the gold standard, but they could do away with the gold standard during war. Because what does the government need money for anyway, mostly? This is not a period of the welfare state. This is not a period where the government is building infrastructure. It's not really building anything.
- 3.60** What is the one activity the government needs a lot of money for? War. Almost all the lending that's going on is for warfare. Maybe Louis XIV needed it for lavish parties, but that was not the case in most of these European countries. The English were in a long-term war with France. They needed funding for that war, and the government would give it to it. And every time the war got heated and they needed a lot of money, they would suspend the gold standard, print a bunch of money, and then resurrect the gold standard afterwards. And that's a pattern that went through in England all the way to, not accidentally, the start of the Great Depression, because it has a lot to do with the actual Great Depression.
- 3.61** Over time, banks became more than just lenders to the government. They also became clearinghouses between other banks. They became the place where these other banks would deposit their reserves. So any gold that these other banks had, they would put at the central bank, or if they had as part of their capitalization, they had bonds, they had maybe government bonds, that was all to be deposited at the central bank in a form of reserve, reserves for these individual banks. Over time, they also became the main regulators of the banks, the central banks, started regulating the banks, determining the reserve requirements.
- 3.62** And ultimately, what they became is the land of last resort, which means they were the agency, the mechanism, by which other banks would be bailed out if they got into trouble. Or ultimately, really ultimately, they were the mechanism by which the entire economy could be bailed out if there was fear of a recession or fear of an economic downturn. So these central banks are still, again, constrained by a gold standard. You get this period in American history where everything is backed by gold in the 19th century. The United States experiments with central banks.
- 3.63** There's two somewhat weak central banks, one from 1791 after the Constitution to 1811. This is Hamilton's baby. It was how the US—you see the US, how did it fund the Revolutionary War? Basically, Congress basically just printed money. It printed money, which by the end of the war was worthless. And the only way to get, but at least

Hamilton believed, that the only way to get the US back on track was the establishment of this Bank of USA, which basically bought back all this junk currency that had been printed and issued proper currency and reestablished a banking system and got banks rolling.

3.64 So we had a mini central bank, 1791 to 1811, and then another one from 1816 to 1836. But Americans didn't like this, the centralization of economic power. And from 18, from the Civil War, really, again, what did the government do during the Civil War? How did Lincoln fund the war? Just printed money. Greenbacks. No, greenbacks were in the Revolutionary World. But just printed money. From the end of the Civil War to 1914, the establishment of the Federal Reserve, which is the US Central Bank, basically what we had in the United States is called free banking.

3.65 During a slice of this in England, in Scotland, not in England, in Scotland, they also had a free banking era where the central bank was not involved. And a free banking system, basically, actually, the Scottish one was much freer than the American one, funnily enough. Basically, under a system like that, banks issue currency. They issue currency based on the gold that they have. And the central bank doesn't play any role because in America there is no central bank from the Civil War until 1914. Banks transact with one another. And there are regulations, even during this period, particularly about how big a bank can get and where it can do business. But for the most part, they're left alone. They are free banking.

3.66 Scotland had such banking, Canada had such banking, the United States had banking, and the central bank played—there was no central bank in the US, and in other places, it played no role. The Scottish free banking era ends when the Bank of England takes control, when the Bank of England imposes itself as a central bank over these other banks that are in Scotland and unifies the United Kingdom's banking system. From 1914 to 1945, the United States has a central bank. During this period, the idea is that the central bank has gold. It basically takes all the gold reserves from all the banks, and it centralizes them. It now issues currency. The currency is that dollar bill that we have today. It's this.

3.67 But it says on it, not like this, it actually says on it, redeemable into X amount of gold. That is its gold-back paper currency, but issued by one issuer. Money before 1914 looked a little bit like this, not really, always had the name of the bank on it because every bank issued its own money. There wasn't one issue of money like there is today. The Federal Reserve issues all currency. In those days, there was every bank. You had a bank of New York, you had a bank of Connecticut, every bank, local bank, private bank. So all money, pre-Federal Reserve, was private money, was issued based on private incentives, based on private gold reserves. Once the Federal Reserve enters the picture, you get money that is uniform, issued by the government, controlled by the government.

3.68 From 1914 really until 1933, this system kind of works. Not really. We'll talk about that. I mean, ultimately, I think this system caused the Great Depression. In 1933, FDR says—he runs on a platform, by the way, he runs for President saying there are rumors that he's going to confiscate everybody's gold. And he says, "No, no way am I going to do that." First thing he does when he becomes President is confiscate everybody's gold. So I don't know how much you know about this in American history, but in 1933, gold from private US citizens is confiscated. I mean, it's redeemed for dollars, but you cannot own gold. Indeed, it is illegal in the United States to own gold except for jewelry until 1971. Illegal. You go to jail for having a gold bar in your home.

3.69 And at this point, basically, the Federal Reserve has all the gold in the world, and theoretically, it's supposed to issue currency only based on the gold reserves that it has. But Federal Reserve ultimately does what it wants. From 1945, right at the end of World War II, until 1971, we have something called the Bretton Woods gold system. So in this case, the dollar bills that you get, don't say redeemable into gold, but the Fed is not, supposedly, not supposed to print any more money than there is gold. You can redeem your dollars, but foreign governments who have dollars, they can redeem the gold at a fixed rate, \$35 per ounce. That was the rate. So if they brought \$35 to the Fed, they would get the gold. They would get an ounce of gold in return. But only if you're a government, and individuals cannot do it.

3.70 Now, what does the Fed do during this period? It cheats. It prints a lot more money than they have gold. Ultimately, you've got to fund a war. Wars, always wars. Vietnam. You've got to fund a welfare state, war on poverty in the late 1960s. So the Fed is printing money like crazy. Government is running huge government deficits. Money is being printed. Inflation, the value of the dollar in real terms has gone down because now if everybody came to the Fed with their dollar bills, everybody, there wouldn't be enough gold in the vault to pay it.

3.71 The French, of all people, the French, de Gaulle, the French President de Gaulle, realizes this, and he starts pulling gold out of the Fed. He literally sends boats to New York, carrying dollars, hands them to the Fed. "I want my gold. It's redeemable." Bretton Woods, remember. And shiploads of gold are sailing across the Atlantic back to France. And the US is losing gold at a rate that is unsustainable, given the value that they placed on it. As a consequence, Richard Nixon goes off the gold standard completely in early 1970s. Gold stops being redeemable even to foreign countries. The Fed is no longer bound to only print money based on its gold reserves. Gold becomes irrelevant. You, as individual citizens, can now own gold. That becomes legal.

3.72 And in a sense, this is the beginning of the real fiat money era, where money is completely controlled by a centralized authority. Money can be printed at its whim. Congress has assigned to the Fed two standards by which they should print money. One is no inflation, so price stability, what they call. And second, low unemployment, because they believe that there's a trade-off between the two. Questionable, but they

believe there is one. So you have to have low inflation and low unemployment. That is the goal of the Fed. And they can put as much money as they want, based on those standards.

- 3.73** And here the challenge really is, and this is the challenge that we're facing right now, I think, because we're facing inflation and we're facing a Fed that's— how do you determine how much money to print? How do you determine it? And not only is there printing money, but the Fed, really, since its beginning, since 1914, also has the power to set interest rates. It sets the interest rate in which banks lend and borrow from one another. It sets a number of interest rates. It also now pays interest on reserves the bank holds at the Fed. So it's setting the interest rate environment.
- 3.74** So what should you set interest rate at? How much money should you print? All of that now has become an issue for the Federal Reserve to determine, a governmental issue. And by what standard do you make those determinations? In much of economics, I'd say in the economics field generally, there's a huge suspicion of central planning. We generally don't trust governments to decide how much food should be produced and at what price it should be sold. Or if I ask any audience in the world, "What do you think this would look like if a government committee designed it?", everybody laughs because they all know it wouldn't look like this.
- 3.75** There's generally suspicion of government design, government control, of central planning. I think not enough suspicion, but there is. Except in one realm, which is money. When it comes to money, we completely trust the government. We completely trust the Federal Reserve. We completely trust central banks to determine what the right number is, what the right figure is. Now, the consequence of this, I think, could be pretty horrific in terms of the financial crisis. We'll talk about that in a future class. But that is the reality in which we live.
- 3.76** We've taken the ability to create money away from bankers, or at least partially away from bankers. We'll get to fractional reserve banking, creates money on top of what the Fed does. We've given that primarily to the Fed. We've taken the power to determine interest rates, or at least the base of interest rates, the starting interest rate, from bankers, and we've given it to the Federal Reserve; we've given it to the central banker. And there are massive consequences to this that we will look into in the future.
- 3.77** So overall, our modern banking system has evolved, I think, out of this suspicion of banking, out of the suspicion of finance, to centralize it around a government that we trust more somehow than private markets. It has evolved with the constant idea that there's something morally suspicious about finance, but we need it, so we need to let it go. We need to let it evolve. We need to let it work. The modern banking system is very much dependent on a central bank. And when we come back for the next class, we'll talk about how private banks function in this world, how they make profits, what they actually do. That'll be in the next class. Thank you.

- 3.78** So you mentioned that the three components of interest rates, the primary components, are the time value of money, default risk, and inflation risk. When we go back and we talk about early development of money, I guess it's in the distaste for usury. Time value of money is a bit abstract. Gold doesn't really inflate, but default risk seems pretty straightforward. Why should I get a premium? Because, well, he might not pay me back. So what is the view during this time, in your opinion, of default?
- 3.79** Well, I think they recognized that default is a risk. Default is something bad. And as I said, even in Mesopotamia, they would keep records saying who defaulted. And maybe there was certain social ostracism or there were other ways in which you would punish, penalize those who defaulted. But the connection between that and interest was not made until much later. That is, the idea of getting, in a sense, compensated in advance or getting compensated in contract for that was not really comprehended at all as an option, as an idea, until much later. Because, again, it's an abstract connection. Yes, they default, and I'll try to get as much as I can out of it, and maybe it'll take a little longer to get it back, but again, if you go back, this is even in seeds, this is not even in money. They understood.
- 3.80** And indeed, one of Hammurabi's laws from Mesopotamia was something like, if I lent you seeds, because there's no money, if I lent you seeds, and you planted them, and it was bad weather, and your crop was destroyed, and you couldn't pay me back, tough for me. That is, the loan is gone. The loan is dismissed. So I suffer the consequence, not you. So they recognized these issues of default, and they differentiated between, I guess, default from, what is it called, the acts of God, or default, willfully not paying me back. That I think might have been even criminal. But the acts of God, you just wipe out the loan.
- 3.81** You did the discourse about the Fed. Now, within the argument principle of decentralization tends to be a better decision-making. What would be the path towards Fed elimination or—
- 3.82** End the Fed.
- 3.83** Yeah, end the Fed. And why do we need it, in your opinion?
- 3.84** Well, in my opinion, we don't need the Fed, and ending the Fed would be a good idea. But it's a challenge. It would not be easy to end the Fed. I think people who advocate for ending the Fed as a gratuitous, as a slogan, don't realize how embedded the Fed is in everything that we do. So it would take a real unwinding of much of our current financial regulations and industry to get to the point. And the whole mentality around banking, the types of banks that we have, how we view money, all of that would have to change.
- 3.85** And so how would you do it technically? I mean, I think it's doable. You would have to get rid of a lot of the banking regulations. Then you would basically set a date and say, "We're going to basically connect the dollar to gold. We're going to price gold in terms

of dollars on this date," and let the markets, in a sense, trade gold and dollars to determine what that price will be. And it's frozen at that point. Let's say right now gold is at, what, 2,300, something like that? Let's say it's 3,000. So you say, OK, it's 3,000. 3,000 dollars equals one ounce of gold. So that's step one. Fed, you can only issue money based on gold.

- 3.86** Then you tell the Fed, "You have to basically take all the reserves that banks have with you, and instead of paying them back in dollars, you have to pay them back in gold." So all the banks now have their own stash of gold. And then the final step is, we're getting rid of Fed notes, dollars, and now banks, you get to print your own money. Now, of course, we'll talk about this in the last class. As we're moving into a world in which you don't need currency, printing money is just an electronic book entry, a ledger, an electronic book entry that reflects that.
- 3.87** So that would not be—you don't have to involve printing presses. You just have to involve. And then every bank would issue you their own money in terms of an electronic thing that you could then use. There'd be some issue about, is the money in the Bank of New York backed by, are they fully backed by gold versus— but all of that would have to be dealt with. You'd have to deal with all of that. There's a way in which you could create unit banks, banks that had 100% reserve for checking accounts. We can talk about that.
- 3.88** And then finally, there's a real opportunity here, I think, for some form of cryptocurrency, maybe not in the current forms, maybe not the exact, maybe not Bitcoin or whatever, ether, whatever. But an evolution to the point where, yeah, maybe it's backed, maybe it's not backed, but the market gets to decide. So the ultimate is, let the market decide what money is. The market chose gold. It doesn't mean it'll choose gold in the future. Free banking really means private money, which means, let's figure it out. We don't know. And that's where I think crypto has a place or doesn't. Competition will decide.
- 3.89** Not central planning, not me, not some economist, not some theoretician of crypto or whatever, but the market will decide. Do we really need Bitcoin now that we have free banking or don't we need Bitcoin? Or are there other things that we can't even imagine today, which I think is the case? Are there other things we can't even imagine that would devolve into money? But that's the last class, we'll talk about that.
- 3.90** How did a marginalized group, a small group of people, let's say, I don't know, the 15th century, 14th century, have access to power and enforcement when they were a pogrom away from being kicked out? When someone owed the Jews money, did they have access to power to enforce the debt, or were they just riding popular opinions at the time?
- 3.91** So how did Jewish moneylenders actually enforce getting their debt repaid? The main mechanism of enforcement was that most people who borrowed money wanted to

borrow money again. So the main mechanism was the fact that these were repeat customers. And if they didn't pay back, they would suffer the consequences. And look, the moneylenders could go to the local authority and say, "Look, these people are not paying back." And the local authority would have to evaluate, "Well, we don't want to piss off these moneylenders completely because we might have to borrow money." And maybe they would enforce it for them.

- 3.92** So they would use the rule of law to enforce it for them. Again, not because they loved the moneylenders, but because they knew that they might need access to the moneylenders. In Genoa, for example, during the Renaissance, they were funding a lot of the global trade. They were funding a lot of the exploration of the Americas and a lot of the trade that resulted from that. And it was going through Genoa. And Genoa became very, very rich as a consequence. So the city fathers of Genoa had every incentive to make sure that this system continued to function.
- 3.93** And they got in a lot of conflict with Rome, with the papacy, over this. But they basically told the papacy to take a hike, and they enforced the contract because there was— Genoa was benefiting from it, including the people in charge of the city itself. So there were built-in incentives and mechanisms to make sure that these were enforced. And sometimes they weren't. Look, as we said, in York, they were massacred. A hundred or 200 years later, all the Jews were kicked out of England. I'm sure that that made it hard to enforce contracts once they were kicked out, and now were in Spain.
- 3.94** But really, if you wanted to keep trading between the UK and Spain, you probably had to deal with Jewish bankers one way or the other, Jewish moneylenders one way or the other. Maybe now they're in Spain instead of here. They didn't come back. So it's interesting because when Shakespeare wrote Merchant of Venice, where he has Shylock as a character, he'd never met a Jew. There were no Jews in England during the time of Shakespeare. He'd never met a Jew. There wouldn't be Jews for another, I can't remember, for 50 or so years.
- 3.95** So he was getting his vision of Jews from other people's writings about Jews. And the story of the Merchant of Venice is a story that I think comes from somewhere in Europe that he, of course, embellishes, as Shakespeare always did. But he didn't have any direct contact. He had no direct experience, which is fascinating.

Lecture 4: The Business of Banking

At a glance — The business of banking.

- 4.1 All right, so we've discussed the history of usury, of interest, of banks, how we got to where we are today. Central banking, we've talked a little bit about that. Of course, there's a lot more to say about every one of those topics. What I want to do now is really talk about maybe the most central financial institution that we have and the one that we commonly all use on a regular basis, which is a bank. What does a bank actually do? How does a bank actually function? How does a bank make money? And what productive function does it have out there in our economy?
- 4.2 So, "What do banks do?" is the question on the table. And basically what a bank does is, it lends money to businesses. It lends money to consumers. It lends money to productive activity, but also for consumption. We haven't talked about the benefits that we all get from what's called consumption smoothing. Consumption smoothing is pretty cool. You want to buy a house for a million bucks, but you don't have a million bucks. But you know that over your lifetime, you will make money and you could have a million bucks through your earnings over your lifetime. But you want the house now; it would be nice to be able to live in a house now.
- 4.3 Well, consumption smoothing is the idea that you can take a loan, buy the house, and pay it off as you make the money over time, so that you can enjoy the house for all those years without having to have the full amount that you would necessary need to buy the house right in advance. You do the same thing with a car; you buy a nice car, under the assumption that you're going to keep making money and you have enough money on a monthly basis to be able to pay for the payments for the car. This is called consumption smoothing.
- 4.4 It means you're living in a sense at a higher standard of living than the cash that you have available allows you to. It raises our quality of life. It makes it possible for us to enjoy a beautiful house, even though we can't afford to pay for it. And I think almost everybody uses a mortgage to buy a home because we don't want to wait until we're 60, 70, 80, 90 years old to buy our first house, if we can afford to do it in our 30s with a loan. And think about it, it's a big loan. You're getting like 80, 90% of the value of the property, you're getting in a loan, collateralized by the house, so if you fail, the house belongs to the bank.
- 4.5 But think about the value that that provides you. A huge value living in that house for 30 years before you pay it off. That's pretty cool. That is what banks provide us. They provide us with that consumption smoothing, that ability to borrow, so we can enjoy the fruits of our future labor today. But they also, of course, fund businesses, real estate projects, infrastructure projects, a variety of different things where they make

loans to a variety of different people. And one of the important things we should think about in terms of the productive role of banks is, they don't loan money to everybody.

- 4.6 So one of the things that bankers do, and this is true of all financial markets and all financial institutions, is that they are selective. And if you will, that selectivity— who they give a loan to and who they don't give a loan to—is one of the most important functions that they serve. They keep money away from irresponsible people. They keep money away from stupid projects. They keep money away from projects that don't ever make a profit, that are destructive to the economy. And they provide money to the responsible, to the good projects, to productive projects, to profitable projects.
- 4.7 That selectivity is why they make money. That selectivity is the work that they're doing. And it's work you, as an individual, you could theoretically do on a limited scale with limited amount of resources. But it's a lot of work if you had to decide who to loan money to, who is safe, who is not, who is worthy, who is not. And then you could only loan them a little bit. If they wanted a lot, you would have to get together with your friends. I mean, think about what a world would look like if you didn't have a bank doing it, and individuals were doing it. It would be completely inefficient, completely fall apart, and loans would not be made.
- 4.8 Banks make loans. In order to make the loans, they attract deposits. They attract, they encourage people to deposit money with them. If you think about it, a bank has two types of liabilities, things they owe, money they owe to others. One type of liability are checking accounts, what we call checking accounts. This is money that you expect to have whenever you need it. You expect it to be at your beck and call. You can write a check, you know it'll clear. Or you can go to the bank, take out cash, or you can use your ATM or do electronic transfers or whatever, but the cash is there; you can use it anytime.
- 4.9 Now, note, just so you know, in the fine print of the contract that you signed with the bank when you opened that account, it does say that in certain cases they might not be able to pay you back the money immediately. They will at some point, but they might not be able to pay it immediately. We'll get to why in a minute, but note that that is in your contract. As people in Silicon Valley Bank discovered, it's in the contract. So one form is this checking account, what we call demand deposits. Whenever you demand it, you expect it to be there.
- 4.10 A second form of liability for a bank that it attracts is a saving account. Different forms of saving accounts, CDs, money markets, saving accounts. Where you agree, where you deposit money, usually there's a certain minimum requirement. Usually it's locked up in some way. You're limited in how much money you can withdraw at any given point in time from it. Sometimes it's locked up for a year or two, like a CD. Sometimes it's a saving account, so you can take some money out, but only periodically, not whenever you want it. You don't expect to just have it available whenever you need it.

- 4.11 So two types of deposits, what we call demand deposits, and the second form is time deposits. They have a longer expected— it's expected they will stay with the bank for longer. You don't have the right to just demand them whenever you want. On the asset side, they have loans. They make loans. And here is the challenge. A loan is typically not very short. There's no demand loan. I'm not going to give you a loan, which I can call anytime. You have to pay me back whenever I want you to pay me back. Usually loans have a set amount of time. A one-year loan, a five-year loan, a 10-year loan, 30-year loan, if it's a mortgage, right?
- 4.12 So they have these long-duration loans, and yet, on the other side, very short. If you think about this idea that the bank has—if the bank had to keep reserves for all of its demand deposits, so imagine demand deposits, or those that can demand immediately. You can imagine a world in which the bank says, "OK, 100% of that money, we're going to keep in the vault." When it was gold, the gold would be there waiting for the people to come and withdraw it. "The saving accounts we will lend out, and we will try to match when the saving account is due with when the loans are due so we can match the cash flows. So we're never in a position where people want to withdraw their money and it's not available."
- 4.13 So that would be what's called a 100% reserve bank, a bank that keeps 100 % of its deposits in cash or in very short-term securities. You could buy maybe money markets or you could buy treasury bills, no risk, and you could sell them immediately in order to pay the cash to whoever wanted the cash when they want it. We don't have 100% deposit, 100% reserve banks in the United States. They existed, they existed before the Great Depression, before the bank regulations, before the bank controls. There were such things.
- 4.14 But today, the amount of reserve that a bank keeps is not determined really ultimately by the bank. It's determined by the Fed. And the Fed could say, "You have to keep 10% of your total deposits. You have to keep in cash." Usually not in cash, literally, but in the deposit at the Federal Reserve itself. 90%, you can loan out again. Now, this is called fractional reserve banking. And the reason no bank says, "No, we're not going to do that. We're going to go all the way to 100%. We're going to keep 100% reserve," it's because they couldn't compete if they did that.
- 4.15 Once the Fed establishes a lower bound, competition is going to drive everybody to that lower bound. If the Fed tomorrow eliminated the lower bound and said, "Let the market work. Let the market determine what the reserve requirements is. We're not interfering," what would happen? Banks that kept it at 10%, we as customers would look at those banks and say, "That's risky. If I give them my money, I might not get it back because they only have a 10% reserve. So I'm not going to give them my money. Or if they want my money, they better pay me a higher interest rate because there's risk." Remember, risk and interest?

- 4.16 "So if they want my money, they'll pay me a higher interest rate. A bank with 100% reserve, God, I'd give them my money. I don't even need an interest rate because I know my money is 100% secure." And now you can see there'd be competition on the dimension not just of how much interest I'm getting. Of course, the bank that pays me a high interest might benefit from the fact that it can lend out 90% of the money, but also has to pay me a lot of money. So it's not clear that it would be more profitable than 100% reserve bank. And indeed, before banking regulations, before the Fed established these floors in terms of reserve requirements, banks typically had over 50% reserves.
- 4.17 I think the average was something around 60 and 70, and there was some competition between them, and there were some banks that had 100% reserves on the demand deposit. We live in a world today where all banks are somewhere around 10%. All banks. And most of us don't care. Why don't we care? Why don't we care? FDIC. We have deposit insurance. We have deposit insurance that basically says, "As long as you don't have more than \$250,000, you'll get your money back. And by the way, if you have more than \$250,000, you'll also get your money back. We just won't announce it in advance," but that's actually what happened. Again, we'll get to Silicon Valley Bank in a minute and tell that story.
- 4.18 Banks, so banks lend out the 90% of demand deposits they get. They keep 10. And if you think about it, what happens to that 90%? So that 90 %, so I lend you money. And what do you do with that money? You deposit it in a bank. And now that bank can make a loan for 90% of the money. And that loan gets deposited in a bank. They can now make a loan of 90% on the money. So every dollar, right, deposited in a bank, actually creates many, many, many more dollars. It creates \$9-plus. Not quite 10 because it's not zero reserve requirements, but it creates \$9.
- 4.19 And this is the nature of fractional reserve banking, that banks actually create money. Banks actually create money by taking a dollar that was invested and turning it into \$9 lent out. That \$1 of those dollars is invested and \$9 again, and \$8. 1 are then lent out. So it's always 10%. So \$8.1 is lent out, and then 90% of that is lent out, 90% of that is lent out, and so on. So a lot of money is created through this process. And every time a loan is made, every time a loan is made, money is created.
- 4.20 Now, you can calculate how much. So for every time the Federal Reserve, Federal Reserve thinks it controls money. But the reality is, for every dollar the Federal Reserve creates that is deposited in a bank, many, many more dollars are actually created. So that's something to remember when they talk about money supply and they talk about all this, how much control do they have? A lot of what determines how much money is actually circulating in the economy is a question of how robust loan demand is. How many loans are actually being made?
- 4.21 Banks, by the way, make fewer and fewer loans. Have been making fewer and fewer loans over time. Because the other thing that a bank could do with the deposits that it

gets is that they can buy securities. Now, they're not allowed to buy any securities. They're not allowed to go into the stock market and speculate. They're not allowed to buy derivatives. They're not allowed to buy other stuff. They're allowed to buy government bonds and government securities, and usually government treasuries of various maturities.

- 4.22 In the past, one of the reasons for requiring banks to buy government bonds, why would you want to require banks to buy government bonds if they're going to buy anything? Well, it's a built-in market for government bonds. If the government wants to finance stuff, the banks are basically financing it. And this has a long history in the United States, even before the existence of a central bank. Government securities were considered as basically like gold, as reserves. So it was, again, a way in which to get banks to fund government. Banks could buy these, they were supposedly riskless, so they were like gold.
- 4.23 How does a bank make money? It makes money by charging more on the loans or getting more on the securities than it pays on the deposits, so the difference in interest rates. So if it can get you all to deposit money in the bank at zero and it charges on the loans 5%, then it makes 5%. Now, not completely, because, again, it can only loan out 90% of what was deposited. So you'd have to do the math to figure out how much it is. But that's basically how banks make money. They make money on the difference between what they pay depositors and what they get on the loans and securities that they invest in.
- 4.24 Interesting phenomena today with mortgages, we talked about mortgages earlier, is that almost no mortgage that is issued today by a bank is actually held by the bank. Usually what the bank does is it sells them to Freddie and Fannie or to one of the investment banks in New York. They aggregate, they take a pool of all these mortgages, and they securitize it. They sell an interest in the pool of mortgages. The mortgages are generating cash flow. Anytime you have cash flow, you have a value. And so as a consequence, banks don't hold. Usually, they get a servicing fee. But the actual mortgage is held by somebody who bought this new security, collateralized debt obligation from one of these packages.
- 4.25 And that relates to the financial crisis and what happened in the financial crisis. But that hasn't changed. The reality is that mortgages are still being packaged and still being sold because it makes a lot of sense. It is a cool investment vehicle for a lot of insurance companies and pension plans and people like that. So banks make money off the spread. A lot of times that spread, the difference in interest rates, reflects— what does it reflect? Why can I charge more on the loan from a business over there than I have to pay you? Well, one is, you're willing to take a low interest rate because you get a lot of benefits from the bank. She gets a debit card, you get checks, you get services.
- 4.26 So whereas if you were just lending the money to somebody random or not random, or somebody you trusted, you would charge them 3%. You're willing to charge only 1%

because you're getting benefits. It's, the money is secure. It's in a vault somewhere. At least that's how we perceive it. It's all electronic, but it's in a vault. You're getting the debit card, you're getting the services that are needed. You get to do electronic transfers, you get all these things. You don't pay for them. So you will need to take a lower interest rate because of that. But of course, that's also a cost for the bank. It costs them something to be able to provide you with all those services.

- 4.27** So what else leads the loan to have a higher interest rate? Well, one, risk. There is risk with the loan. There's almost no risk of the bank going under, again, because of deposit insurance and a bunch of others. I'm willing to take 1%, I get all the services, and I know my money is always going to be there. I don't worry about it. Whereas the bank is actually taking a risk vis-a-vis. And again, so you can see there's a difference in risk profile, between its liabilities and its assets.
- 4.28** And then finally, there's usually also a difference in duration. The loans that it's giving out to people have longer durations, 2, 3, 5, 30 years, versus the deposits, instantaneous, or even if it's a saving account, maybe 1, 2, 3 years, but not 30. You don't have a saving account for 30 years. So money loaned, usually, not always, usually, interest rates over time are much higher for long-term than for something very short-term. So they're getting compensated for the difference in duration; they're getting compensated for risk; and they're getting compensated for the convenience that they provide you, and that's why you're willing to take a lower interest rate.
- 4.29** And basically, that's how banks make money. They make money off of this fractional reserve issue. That's a big part of it because they can, again, leverage. In a sense, they're being leveraged. They're taking on a lot of risk, and they're getting compensated for the risk that they're taking on by lending out \$9 for every \$10 that you deposit. Now, this creates a lot of risks within the system. Creates a lot of risks within the banking system. And what does that look like? So first we mentioned that you're willing to deposit your money in the bank. Almost nobody ever thinks, "I wonder if this bank is solvent. I wonder how well they're doing." Have you ever looked at your profitability of the bank you have your money in? I'm sure nobody has. Nobody has.
- 4.30** You just give it the money because, again, assuming you have less than \$250,000 in a bank account, and most people don't have more than \$250,000 in a bank account, right? Because there's deposit insurance. So the government has insured your money. Now, note that this creates a real problem. This is called moral hazard. If the banker knows that if they do something stupid and everybody comes to ask for their money and they don't have it, the government will step in, FDIC will step in and bail them out, then they don't care about doing something stupid.
- 4.31** Oh, let's change that a little bit because it's not necessarily stupid what they're doing. But let's say if the banker knows that they can take on a lot of risk. And what it means to take on a lot of risk, usually, is that if the risk works out, if things work out well, the banker makes a lot of money. The bank does well, the CEO does well, gets a nice bonus,

everybody does well. But if the risk turns bad and he can't pay his depositors, the FDIC steps in and bails them out. So he hasn't lost anything. He might lose some of his shares, but he hasn't—the depositors are fine.

4.32 Now, think about a normal business. Think about a normal business. Let's actually use the whiteboard more than I thought. So let's say a normal business is doing badly. They've taken on risk, they're not panning out, and their value is going down. And what does bankruptcy mean? When do you go bankrupt? You go bankrupt when what? When you can't pay your debts, when your liabilities exceed your assets. So let's say this is the value of your assets, that's things going badly. And let's say this is the value of your liabilities. Cross that line, you are going bankrupt.

4.33 Now, if you're here, and let's say I'm a big shareholder, I own stock in this company, and you come to me, the company comes to me and says, "Look, we want to take a big risk here. If the risk succeeds, then things are going to go like that. You'll make a lot of money. But if things fail, it'll go like that, and you'll lose everything." Then the shareholder is going to go, "No, guys, that's not a risk that I'm willing to take. Tone it down. No reason to take on, this is basically buying a lottery ticket. No reason." And you could put the odds on either one. But remember, once you cross this dollar line, what happens to shareholders? What do they get? Zero. They get wiped out. So he goes, "No. Over here, too much to lose. So I'm not interested."

4.34 But imagine you're now here, and you go to the shareholder with the same proposition. So one option is, look, we're going bankrupt. That's the trajectory, is going bankrupt. But if this risk pans out, we're not going to go bankrupt. We're going to go that way. You're going to make a lot of money. You're not going to go bankrupt, and you're going to make money. But if we lose, it goes like that, all the way to zero or whatever, but big, it's a big loss. Now the shareholder goes, "Wait a minute. I'm going to probably go bankrupt anyway. But if I go bankrupt doing this, I don't suffer this entire loss. I only lose that. I only lose that little bit. But look at my gain. Wow."

4.35 Now, that's called moral hazard. I'm strongly incentivized at this point to take the risk. What prevents me from doing that in a normal company? Who bears this loss? The debtor, the bondholders, the people who've lent money to the business. Usually bondholders, maybe a bank. But somebody's lent the business money, and they're the ones who would lose it all. If the company just goes bankrupt, and it just continues along this path, and it crosses right there, and it declares bankruptcy, what happens to debtors? They get most of their money back. What happens to shareholders? They get nothing. But the debtors get most of their money back.

4.36 So the debtors are the ones who lose if you take the big risk and it pans out, it doesn't pan out. And if the risk succeeds and it goes through the roof, what did the debtors get? Nothing. Same as they got before. This line doesn't change. So who has an incentive here to stop the company from taking the big risk? Debtors, bondholders, banks. Guess what? Bond covenants, bond agreements, always have provisions that

relate to what happens with regard to decision-making as things in the company deteriorate, as they get close to bankruptcy. Debtors have to approve them. You can't take on risks unless we give you a thumbs-up.

- 4.37 So here they might say, "Yes, but the shareholders are going to kill it." Here they say, "No." So the company can't do it. It'll just drift into bankruptcy. Creditors will get it all. Shareholders will lose. But you see the incentive that the managers and the shareholders have. Take risk. Now, imagine that these creditors have a loss guarantee. They don't lose a dime. No matter what happens to the company, they're made whole. Then what happens here? Take the risk. Nobody is there to stop you. And that's the situation in banking.
- 4.38 Because what are we as depositors to a bank? We're the creditors. And here we are, right down here. And sometimes banks have bondholders as well, and often those are bailed out as well. But certainly, deposit insurance guarantees all of these people, and therefore we don't care. Take risks, do whatever you want. If you go bust, we're taken care of. This is moral hazard, and it exists because of the government guarantee. As long as that government guarantee holds, there's moral hazard.
- 4.39 Now, what does the government say? The government says, "Don't worry. We know this. We understand moral hazard. We appreciate what's going on here. And as a consequence, we're going to hire a bunch of regulators, and they are going to supervise the bank, and they are going to make sure that when you're here, you don't make that bet." They're just going to ban it. They're just going to stop you from doing it. There's certain loans that are just not acceptable to the regulators. And the regulators will tell you, "Stop, can't do that."
- 4.40 It doesn't work too well. It doesn't work too well. Where were the regulators? Where were those regulators stopping Silicon Valley Bank, where were the regulators stopping the S&L crisis, where were the regulators stopping banks from doing what they did before the financial crisis? They were nowhere to be seen. And the Silicon Valley Bank one was easy. This was not hard to see that this could happen, and yet they couldn't see it. So the system does not work because of this moral hazard.
- 4.41 And notice that there's a lot of regulations on banks. Everything in a bank is regulated. Why? Because of this. Because regulators, and the government is so afraid of this, and they can't do away with the deposit insurance. That's unthinkable. So they load up more and more and more regulations to try to prevent that. Dodd-Frank, which was passed in the post-financial crisis in 2010, was aimed at eliminating this risk-taking as a consequence of moral hazard. And yet, we got Silicon Valley Bank 10 years later or 12 years later. Didn't stop it, and it won't stop it. It can't stop it. They just don't have the tools to do it.
- 4.42 Banks are super leveraged with massive moral hazard. And what happens is that, even though some of the debt might be what's called uninsured, not covered by FDIC, like

\$250,000 or more, or bondholders that have lent money officially through a bond to the banks, it turns out that, in almost all cases, the government will come in and bail them out as well. So there's simply no market protection against banks doing stupid, crazy, high-risk, whatever things— nothing to stop them except counting on a bunch of regulators who might be able to identify what's going on and stop it in time, which has been proved over and over and over and over again as just not the case.

4.43 Let's think about a few examples of the kind of risk that is involved here. I'm going to start with a classic case, which is the S&L crisis. I don't know, you guys are too young to remember the S&L crisis. It's a crisis that happened during the 1980s. S&Ls are saving and loans banks. Basically, the entire industry was wiped out, or almost the entire industry was wiped out in the 1980s, completely disappeared. Everybody was, as we'll see, everybody was bailed out. But basically, the industry, there's still a few S&Ls going into the financial crisis, and then they all disappeared, but there are basically no S&Ls today when there were hundreds, if not, there were thousands of them back in the 1970s.

4.44 So you have to add here the fact that the government, back in those days, regulated banks in different ways than they regulate them today. So for example, after the Great Depression, the government imposed something called Regulation Q on banks. Regulation Q basically said, you can't pay interest on either checking or saving accounts above a certain rate. Oh, no, you can't pay interest on checking. Checking account is zero interest. On saving accounts, we determine what interest is going to be. Let's say 5%, can't pay more than 5%. That's what we do. So that is capital. You can't pay on deposits, and you can only pay 5% on time deposits, on saving accounts.

4.45 Now, what did these S&Ls do? The S&Ls were regulated so that they could only make mortgages. They could only make 30-year fixed-rate mortgages. This was a way that the FDR administration wanted to guarantee that it would be funding for homeowners. So they created a whole industry called the S&L industry that was just there to make mortgages. Now, imagine you in the '50s or '60s. Again, you're all too young to imagine this, but in the '50s and '60s interest rates are very low. And these banks are making mortgages just like they were making a few years ago, at maybe 3, 4, 5% max, 3, 4, 5%. And these were 30-year fixed. So for the whole life of the loan, all the bank would get is 3, 4, 5.

4.46 In those days, banks didn't package and sell them. Most of the loans, certainly in S&Ls, were on the balance sheet. So the income was from the mortgages, somewhere between 3 to 5%, regular coming in. Nice business, particularly because regulations didn't allow you to pay any interest on checking accounts. So they all came at 0%. So you made the difference between zero and the 3, 4. Cool, all is good. And you could keep doing that for a long time. Even the saving accounts, which you had to have some, you didn't pay much. The government set it below what you were lending out.

- 4.47 And people were happy to give S&Ls money because interest rates were so low, it wasn't like they had an alternative. And when inflation is close to zero, inflation is zero, 1%, something like that, yeah, OK, I'm getting a service from the S&L. I don't need to get interest on my checking account. But then come the 1970s, and the 1970s see inflation grow dramatically, well over 3, 4, 5%. It grows into double digits. And now people are saying, "I'm not going to give the bank my money and get zero. I'm going to find something that yields more than that." And indeed, you get a lot of financial innovation during the 1970s, where people create instruments that can pay more than that.
- 4.48 Money market accounts are the primary thing that comes into existence. And money market accounts, they take in the money of depositors and they buy short-term securities that are trading regularly. But what are short-term securities trading at? They're trading at, in the market, government securities are trading at the 10, 11, 12%. They're trading above inflation because otherwise, nobody would trade. So now these people can pay you, I don't know, 7, 8% and still make a profit because they're getting 10, 11, 12%. So everybody starts withdrawing their money from the S&Ls and putting it into money market accounts.
- 4.49 And now the S&Ls, they're gaining 3, 4, 5%, which is way below inflation. So it's not worth a lot. And all their deposits are going out the other side. At some point, Congress says, "OK, well, that's not fair. Something has to give here." So they allow the S&Ls to pay whatever they want on, like, checking accounts and saving accounts. They do away with Regulation Q. Well, now the S&Ls have to pay in order to attract people's money. Let's say they have to pay less than what the money markets would sell, 5, 6, 7%. All they're getting from the mortgages is 5, less than 5. Now, they could issue new mortgages, but they're not going to be able to do that much. And people are starting to worry.
- 4.50 In those days, deposit insurance kicked in at, I think, 25,000 in the late '70s, 25,000. So people are starting to worry. And anybody rich didn't keep more than 25,000. They spread it across a lot of banks. So money is flowing out. They're paying their deposits more than they're getting from their old mortgages. They're issuing new mortgages as fast as they can, but people are not borrowing that much at 13% mortgages, 14 % interest rates. So they're having a hard time kicking in.
- 4.51 So then Congress decides to do two things. It says, "OK, there's a real problem here. So one, we're going to increase deposit insurance to 100,000. That'll keep the money from flowing out. Second, you can now, you're not limited to mortgages. You can now give loans to anybody." Now, S&Ls were basically here. You could argue even that they were even further along the line, that they were basically broke. They were basically bankrupt. Or if not immediately, they were going to be within months, years. So what did they do?

- 4.52 Now, before, they couldn't have taken the big risk because all they could do is mortgages. But now, oh god, they could give loans to oil exploration. They could give loans to commercial real estate. They could give loans to any speculative business out there that promised them a high return. They gave loans to, in mass. And of course, a lot of those loans failed, so they never got that. What did they get? They got this. So by allowing them, deposit insurance to be higher, and then on top of that, allowing for them to be able to do whatever they want, the government took a crisis that was going to happen and made it 10 times bigger and 10 times worse.
- 4.53 And all these incentives, all of them were predictable. None of this is rocket science. It's all predictable. Indeed, there was a guy, Ed Kane, an academic from Boston College, I think, who was writing a book, *The S&L Crisis*. This is what's happening, or this is what will happen, this is what's happening, told you. And by the end of this, I can't, I think the crisis cost the US taxpayers something like \$200 billion. The equivalent of the FDIC, which was guaranteeing the saving of loans, was wiped out because it didn't have enough money. So the taxpayers had to step in to guarantee all of this. And it wiped everybody out.
- 4.54 And the lesson from this was not, of course, hm, I guess this regulatory mechanism with deposit insurance doesn't work. The lesson was, we'll double up on it. We'll reinforce it. We'll have new regulations and more robust regulations. We'll spend more money on financial regulations of the Fed and the FDIC and everywhere else, in order to make sure this doesn't happen again. But it's just too obvious, right?
- 4.55 So here you had a combination of two things. You had a combination of loans going bust, speculative loans, crazy speculative loans. The mortgages were fine, but these crazy speculative loans going bust, and as a consequence, bankruptcy. A consequence, no liquidity. Silicon Valley Bank is a little different. So let me tell you the story of Silicon Valley Bank. Now, Silicon Valley Bank was an incredibly unique bank that had a very unique business model. Really, the only one of its kind. It had a few other banks that tried to copy it, were not successful. It was considered one of the best banks, one of the most profitable banks, one of the most innovative banks in the entire United States.
- 4.56 It was founded in the 1980s, basically to be a bank for startups and venture capital firms. It didn't make a lot of loans, but it often made loans, and part of the loan, they'd get an equity kicker. They'd get a piece of the business they lent to. So when that business went public, they could make a lot of money. Now, like venture capital, most businesses don't, but they always got their capital back. And once in a while, they'd get this incredible kicker on the other side. Over time, like a lot of other banks, they made fewer and fewer loans. In particular, during Covid, an enormous amount of money flowed into the venture.
- 4.57 Anyway, the bank was very successful doing this. I lived in Silicon Valley. It was like an institution there. It was a real institution, and the founders of it, they all did very well, and it was considered one of the great banks in the United States. During Covid, a lot of

money flowed into the venture capital community. One other point, before we do that — the way they got their deposits was basically startups. Let's say a startup got \$10 million of funding from the venture capitalists. It would take that \$10 million of funding and put it to Silicon Valley Bank as a deposit.

- 4.58** Now, notice that exceeds the \$250,000 of deposit insurance. But they figured Silicon Valley Bank is a great bank. I mean, it's not a problem. So they would put the money in Silicon Valley Bank. And the venture capitalists, when they raised a gazillion dollars fund, they would put the money in Silicon Valley Bank. And then there was a certain pacing to the whole thing. A startup spends the money over several years at a certain pace, and the bank would know that there would be a demand for its money like that. Most of the money that Silicon Valley took in as deposits, it bought securities with.
- 4.59** And it didn't pay any interest on the deposits, partially, for a variety of reasons, again, regulatory. It was still true until a couple of years ago that you couldn't pay deposits on corporate checking accounts. The interest, couldn't pay interest on corporate checking accounts. The interest was zero. So they paid zero. And then they could buy securities. Maybe securities pay 3%. They were very safe. They were government securities, so there was no risk. And they were always buying them short-term so they could sell them quickly in order to cover their deposits.
- 4.60** So here's the dynamic of what's happened. During Covid, venture capital raised a huge amount of money. Suddenly people, and during the period before Covid, they raised a lot of money. And why? Well, I mean, interest rates, if you remember during that period, were basically zero. The Fed had kept interest rates, real interest rates in the United States and around the world were negative. So if you wanted a return, if you were a pension plan or an insurance company and you wanted a return, you weren't going to find it in government bonds, you weren't going to find it in traditional investing. So there was a huge appetite for risky investing, venture capital-like investing, private equity investing, because they offered real returns. They offered real money.
- 4.61** And a lot of these pension plans are underfunded. A lot of the insurance companies want to juice their returns. So they poured money into venture capital. And it also seemed like there were all these innovations. If you remember during Covid, there was Zoom. There was just this attitude of, "Now that we're constrained to our homes, we're even more dependent on technology, so we need more technology." So during the period in the 20-teens, leading into Covid, venture capital accumulated huge amount of money. They funded a lot of companies at ridiculous valuations often because of these very, very low interest rates, and all their money flowed into Silicon Valley Bank.
- 4.62** Silicon Valley Bank has a pile load of money. And yes, it's paying zero on its checking, but when it looks at its securities portfolio, securities in the Covid era are paying zero. How is it going to make any money? It can't buy these short-term securities because they pay zero. So they made a decision, which pretty much every bank made, to buy

longer securities. Instead of buying one year, three months, six months bills, they started buying three, five, 10 year bonds. Still safe. The government will pay, but the interest rate is now higher because it's longer duration. But now you see there's a massive mismatch. They have these bonds that are going to pay, but they pay over a long period of time, and yet they have these liabilities that are short-term.

- 4.63 Now, after this explosion during Covid, suddenly people realized tech might be a little bubbly. Remember that there was a year where tech returns were actually negative. Maybe venture capital took in too much money. Maybe that's not such a great idea. People stopped investing in venture capital. This was in 2022. At the same time, what happens? At the same time, inflation is going up. So inflation starts going up. Now, what happens when inflation goes up? What happens to the value of those bonds that they bought? It goes down. When interest rates go up, when inflation goes up, the value of securities goes down.
- 4.64 Now, it's true, if they'd held them all the way to maturity, they would have got the promised interest rate. But if they have to sell them, they're going to take a big loss because the interest rates were zero, maybe 1%, now they're 5%, 4, 5%. So that's fine as long as you don't need to pay off these liabilities very quickly. But suddenly there's a crunch in venture capital. Venture capital is not raising as much money as it did. These startups got their money, but now they're starting to use that money. They're starting to withdraw it and withdraw it and use it. And suddenly, Silicon Valley Bank is in a position where a lot of its money is left, it's been withdrawn, to pay salaries and to pay for startup costs and all of that.
- 4.65 And yet it doesn't have any money in the vault. And what it does have in the vault are long-term government securities that are now worth a lot less than they were a year ago because of inflation. It starts selling some of those securities because it has to pay off. But every time it sells, it's taking a loss, taking an actual loss. It has to now report losses. So it realizes this is not susceptible. So in that fateful week, March 2023, it actually goes out into the marketplace, and I think it contacts Goldman, and it says, "Look, we need capital. We can't sell this portfolio. If we sell this portfolio at a loss, we're wiped out. If we pay out all these deposits, we're wiped out. So we need capital. We need cash. We'll sell shares. Will you come in?"
- 4.66 Can't remember the amount, X number of millions of dollars. "Will you come in and basically capitalize the bank?" And this happens on a Thursday night. I think they're still negotiating. Goldman is hesitant. The rumor mill starts that Silicon Valley Bank is in trouble, that they're going to Goldman for basically a bailout, and everybody starts pulling their money out of Silicon Valley Bank. And you know in the old days, I don't know if you've seen these old movies like It's a Wonderful Life, people line up in front of the bank, demanding their cash. Oh, no, not this time. This time, 24/7, I can go to my phone and I can move my money from one account to another.

- 4.67 And like that, there's just no money in Silicon Valley Bank. It's all been withdrawn. They can sell the securities, but it's not enough to cover it. Again, this is the problem with leverage. This is a problem with fractional reserve banking. And by Friday afternoon, the bank is bankrupt. Thursday afternoon to Friday afternoon, bank is bankrupt. And at the same time, because investors see this happening, they're looking at other banks. "Oh, a lot of other banks, they have a lot of uninsured deposits. We'd better take our money out of them." So it's not only Silicon Valley Bank that goes under, ultimately; two other banks go under as well, Signature and First Republic.
- 4.68 But when Silicon Valley goes under, the government then steps in and says, "Well, we're not going to bail you out in a sense of keeping you alive, but we are going to make everybody whole. If you had \$25 billion in Silicon Valley Bank, even though it's above the \$250,000 threshold, we're going to give you the whole \$25 billion." If you're anybody, except shareholders, everybody else was bailed out. And that was true of Signature, and that was true of First Republic. It wasn't— deposit insurance didn't just count for the first 250,000. It covered everything.
- 4.69 Now, think of the moral hazard there. If it's only 100,000 or 250, there's still a moral hazard, but it's limited. But once you allow it all, once you bail out everybody for everything, there's no limit. And again, it's not new for here. The first bank that was deemed too big to fail and was bailed out completely was Continental Illinois, in 1984, for the same reason as the S&Ls. It was in the same situation. It was all bailed out. Everybody was made whole, except shareholders. And note the pattern here. All this does is emboldens more and more. The moral hazard only gets worse; it does not get better.
- 4.70 Banking is an unbelievably fragile system, unbelievably fragile, highly leveraged. It's got real moral hazard problems. And we'll talk about this in our last class, but in a modern environment of fintech, not clear exactly what values it has. So we have an incredibly fragile system, which is heavily leveraged. There's a mismatch of duration, no diversification, because banks, by the way, until 1994, banks were not allowed to branch out of their main headquarters. So if you were established in South Carolina, you could have a bank in South Carolina, and maybe in different counties in South Carolina, though not always that. Sometimes you could only have one branch in the entire state, but you were not allowed to go to North Carolina or anywhere else.
- 4.71 This is a regulation that was passed at the founding of America. It's the only country in the world that has this. You're not allowed to branch out of your state or sometimes out of your county. When Continental Illinois failed, it was one of the, I think it was the fourth largest bank in the United States. It was one of the top 10 banks in the United States. It was the largest failure in American history to that point. It had one branch in downtown Chicago. That's it. Because Illinois was what's called a unit, unit banking. One unit, that's all you could do.

- 4.72 So American banking, unbelievably fragmented. So up until in the 1960s and '70s, there were something like, I don't know, there were over 20,000 banks in the United States. Since then, because of bankruptcy, because the law was changed, in 1994 the law was changed. And now you can have branches everywhere. You can see them all around. Here you have Citibank. Citibank could not have a bank in Miami in 1993. 1994, Congress passed a law that says you can branch anywhere, you can take over anybody, you can go anywhere. And that completely changed banking. But there are still 5,000 banks in the United States because there's only so much you can do, even over 30 years, in terms of consolidating banking.
- 4.73 And the regulators don't like it. And so there's only so much you can do. Today, there are 5,000 banks. If you do a calculation, if you look at other developed countries and how many banks they have per GDP, the United States should have maybe 400. So 90% of the banks in the US should disappear. If we were like anybody else, Canada has five major banks, 50 small banks, that's it, and it has a huge geographic area. So the United States has too many banks. They're fragmented. No geographic diversification. Moral hazard. It's just not a sustainable system. It's a completely fragile system.
- 4.74 And that has been proven over and over and over again. We had the S&L crisis. We had the great financial crisis in 2008. We had recently the Silicon Valley Bank crisis. I would bet anything that over the next 10 years, there will be more of these. This is not ending because nothing has been fixed. What we do is, every time there's a failure, we overlay it with regulation, without dealing with the fundamental problem. And a number of solutions, which again, we can talk about in the last class, to do it, but nobody is really talking about it. Certainly, nobody in government is talking about it.
- 4.75 The larger banks, banks like Citibank that we mentioned, J. P. Morgan and those, they're now diversified. They have geographic diversification, but they have huge moral hazard problems just like this. Citibank has repeatedly gone bankrupt and been bailed out. It was bailed out in the 1980s because it had made really dumb loans to Latin America. You remember? Well, again, you guys won't remember. But at some point, the United States bailed out Mexico, \$50 billion or something like that in the 1980s. But it wasn't bailing out Mexico. It was providing Mexico with enough money to pay its debts to Citibank and other American banks. So it was a roundabout way of bailing out the American banking system.
- 4.76 In 1992, Alan Greenspan engineered the fastest decrease in short-term interest rates in American history. Went from 6.5 to 1% in a very short period of time, or one-point-something percent, again, to bail out the American banking system because they had really a lot of bad debt on commercial real estate. And one way you can make them suddenly profitable is if you lower the short-term interest rate; that means the deposit, how much banks have to pay on deposit, goes down. And if the long-term rate stays high, then they can charge on loans a lot of money while only paying deposit, a small amount of money, you've increased their profitability.

- 4.77 So time and time and time again, and of course, we all know about the big bailout of the banking system. During 2008, Bear Stearns was bailed out, Merrill Lynch was bailed out, AIG was bailed out, an insurance company. The only one that wasn't bailed out was Lehman Brothers. Who knows why exactly, but it wasn't bailed out. But it was completely random that this one wasn't bailed out. It didn't have anything unique about it, other than it happened that the Secretary of the Treasury and a number of other people associated were all former Goldman Sachs people and not former Lehman Brothers. People would think that if it was Goldman Sachs in trouble, it probably would have been bailed out. But Lehman Brothers is not that important.
- 4.78 But anyway, Lehman Brothers was allowed to go bust, which was completely unexpected. Nobody thought it would happen, and the world went nuts over it. And then they raised \$700 billion to basically bail out the entire banking system. And the solution to that was Dodd-Frank, and that hasn't solved the problem—and it can't solve the problem, because the moral hazard is only increased every time they do this.
- 4.79 So we live in a very, very fragile world of banking. Banking generally seems to be in decline. There's so many options we have today to keep our money, where to keep our money. There's so many ways in which we can handle our money. Banks are slow, particularly the smaller banks, in dealing with the technology that is available to all of us now in terms of managing money. And these problems keep haunting banks, and they're not going to go away.
- 4.80 And I said earlier, bank lending is in decline, and they're much more inclined to buy securities. Who needs them to buy securities, to give us deposits? That seems not something that is really necessary or that you'd pay somebody a lot of money in order to do in the world in which we live right now. Short of dramatically structuring the American banking system, which is necessary, even though unlikely, some catastrophe is going to happen in the future. It might be a small catastrophe that grows. It might be a big catastrophe or something. Given the fragility of the system, something has to give.
- 4.81 All right. That is banks, a little bit about banking. Next class, we're going to talk about the stock market. And then based on how much time after that, we'll talk about maybe some other markets. But we'll devote next class, maybe a class or two, to the stock market. And then we will talk about crises, risk, fragility of the entire system. And then the final class will be about the future of finance. Any questions?
- 4.82 You gave a brief explanation, but could you explain why bank loans are decreasing?
- 4.83 I think there are so many other options for businesses to get loans that are less cumbersome and less problematic than dealing with banks that are so heavily regulated, and it's hard for banks to compete. So private equity raises large amounts of money, and they give out loans, their private loan pools, their private loan funds. And so that's part of it. A big part of it is Dodd-Frank. Dodd-Frank has encouraged banks to

go into securities. Securities are more liquid. It's easier for them to deal with crises through security. So regulation is suppressing the incentive of banks to give out loans.

- 4.84 And at the same time, they've got their real competitive pressures. Other parties are rising to deal. It's not like there are fewer loans out there. It's just fewer and fewer of them are made by banks. So in many respects, we're being debanked. The amount of business going through banks in America is decreasing, and other types of financial institutions, like these private funds, are replacing them.
- 4.85 You mentioned the burden of regulation. In your opinion, what is the built-in incentive system for the government to keep on doubling down the regulation, given its ineffective outcome for the past, I don't know how many years, 30, 40, you name it?
- 4.86 What is the incentive of the government to keep the regulations going and in a sense to double up on them? The most fundamental is control and power. The government almost never gives that up. Think about if you're working—do you know how many regulatory agencies exist to regulate banks? I think six. So the Fed regulates banks. The FDIC, which is responsible for deposit insurance, regulates banks. The OCC, the Office of the Comptroller of the Currency, which was an agency established during the Civil War to regulate the issuance of currency, of fiat money, during the Civil War, and never went away. So you just found new things to do, right?
- 4.87 If you're a public bank, then there's the CFPB, the Consumer Financial Protection Board, which is a new regulatory agency established by Dodd-Frank through the influence of Elizabeth Warren, which has just been in front of the Supreme Court that bailed out the CFPB. Anyway, that is there to relate how the bank deals with customers, with consumers and consumer products that it provides. Of course, if it's a public bank, there's the SEC. And then the sixth regulator is every single state. Every single state has bank regulators, has a bank regulatory agency responsible for the banks that are incorporated in that state. So you got six.
- 4.88 Now, you work for the OCC. What's your incentive to double up on the regulations? Your job. You don't want to lose your job. You want to keep your job. If you're in Congress, if you're Elizabeth Warren, what's the probability Elizabeth Warren would ever to say, "Too much regulations, we should do away with the CFPB." I mean, zero. It was her baby. It was her creation. What, she's going to admit she made a mistake? Have you ever heard a politician do that? No. So, the incentives are the opposite. Look, people who work at regulatory agencies are incentivized around their job. They want to keep their job. They want to grow their job. They want to become more powerful at their job.
- 4.89 They want to increase their market value. Why does your market value go up the more regulations there are? What is the job you're going to get when you leave the regulatory agency? No, what's the job you're going to get? Who's going to hire you? You're going to be hired by the bank because every bank wants a former regulator in

order to deal with regulations. Who is best to deal with regulations? A former regulator. So regulators go and work for the industry which they regulated. That's the case in pharmaceuticals. That's the case in every single field. Regulators go to work in the industry that they regulated. And the more regulations there are, their salaries are going to be higher.

- 4.90 Where do politicians go to work after they leave politics? For anybody who wants anything from Washington, because they're the best lobbyists, because they know how it works. They know their friends are all politicians. They're all going to their fancy schmancy law firms. But what are those law firms doing? They're lobbying government. Why would they hire a politician to lobby government? Because they're the best lobbyists. They know everything. So it's a self-perpetuating system, and there's no way to break out of it, short of some kind of either major philosophical change, where people just say, "Enough is enough. We're getting rid of all this," or a crisis, where such a big crisis, where you have to rethink what's going on in the world, completely rethink it.
- 4.91 And again, even then the likelihood is that you double up, you don't actually change course. But think about how many people generally in life are willing to recognize their mistakes and change course. It's an important skill and talent to have in life, but not many people have it. Admit your mistakes and do things differently next time. Most people don't do that, sadly.
- 4.92 So people throw out the phrase self-fulfilling prophecy a lot. But a true self-fulfilling prophecy is a run on the banks. What are the ingredients that make up this noxious stew? Because it happens so quickly.
- 4.93 Well, it doesn't need to take much to get a run on the bank. All you need is some kind of rumor or not rumor, just fact, that suggests to people that we're there, that the bank is in trouble, that the bank is approaching a point where it can't pay its deposits, where it's taking on too much risk, where it's not liquid enough. These are liquidity crises because, look, Silicon Valley Bank, if it could have sold, if it could have waited for all its bonds to pay back at the interest rate that was promised, they would have been fine. But once they sold the bonds, they took a hit. And that led them to bankruptcy.
- 4.94 So once you realize that these mismatches exist, I mean, there's a sense in which it's a miracle that we don't have a run on banks every single day. Every single bank in the United States fails if you run on it. There's not a single bank in the US that can survive a run, not a single one, because they're all leveraged nine to one, every single one of them, because the government has incentivized them to do that. So nobody can survive a run on the bank. There's nothing unique about Silicon Valley Bank, other than they got caught, and a rumor happened, and people were texting each other, "Hey, get your money out of Silicon Valley."

- 4.95 And I think Peter Thiel or somebody went online and said, "All my companies should get out of Silicon Valley Bank." And of course, there, every company had a big chunk of money. So it wasn't like people were drawing a hundred bucks. They were drawing millions of dollars. So every bank in the United States cannot survive. There's not a single bank that could survive a bank run. Which is, I mean, that's just factual. It's not speculative. That's factual. So it's just a question of what triggers it. And if most of the money the bank has is insured deposits, then if I'm insured, I don't worry. I don't run on the bank.
- 4.96 So one of the things that have happened since the Silicon Valley Bank crisis is that a lot of banks have gotten rid of their uninsured deposits. Over 250,000, they don't want your money because they don't want the run. Under 250 is fine because you're not going to run on the bank, because your money is secure, because FDIC is there. Although right now, one of the reasons it runs is basically the government has said, "We guarantee everything." Implicit. They haven't said it explicit. Implicit because they bailed everybody out in those three banks.
- 4.97 So there's a sense in which that's good because nobody's running on the bank. But there's a sense in which that's really, really, really bad because it creates this unbelievable moral hazard which could wipe out the FDIC and could cause a crisis that the American government cannot bail us all out of.

Lecture 5: The Business of Capital

At a glance — The business of capital.

- 5.1 1976. Steve Jobs and Wozniak working on a little project in Woz's parents garage. Project is a personal computer, logic boards. Woz views this as a cool hobby. He thinks this is really fun to do. Jobs is already starting to think maybe there is a business here. Jobs consults with somebody he knew at Atari. Atari was pretty big back then. I don't know if you guys even know what Atari is, but they used to create these game consoles for very, very primitive computer games. And he, as a consequence, builds up a business plan.
- 5.2 And one of the things they realize is they need to create a demo of what it is that they're working on. They need to build a prototype. But for this, they need money. They don't have any money, right? And they need about 1,500, 2,000 bucks, but neither of them has much. Wozniak lands up selling his HP calculator. I don't know if you guys had a calculator. \$500, he got for the HP calculator. He got \$500. Jobs sells his Volkswagen. I mean, it's the perfect car for Steve Jobs. He had a Volkswagen. He sold this Volkswagen, was probably one of those little Beetles, for \$1,500, but it broke down. So he only got \$1,300 in the end for it. And with that \$1,800, they built a demo.
- 5.3 They took that demo to a computing club, and they showed it around, and people were enthused. There seemed to be something real here. Indeed, Paul Terrell, who owned a chain of electronics stores in the Bay Area, said he would buy 50 of these if they could build them. If they could deliver 50 of them, he'd build them. Again, they needed money. How do you build 50? They asked around, they went around, and they managed to get a high school friend's father lent them \$5,000. Still short 10. They went to the bank, and the bank took one look at Steve Jobs and said no.
- 5.4 I don't know if you guys have ever seen photos of Steve Jobs in the 1970s, but long hair. He went through these phases where he didn't believe in showering, so he generally stunk. Unclean, raggedy clothes. And he was the business side of this. Wozniak was the range, was the guy who tinkered with the actual computer. Anyway, where did they get \$10,000? The bank won't give it to them. So they land up going to the supplier, the store, the electronic store that actually sold them the parts to build the thing. And they said to the guy, they said, "Would you give us the parts on loan? And once we sell it to this other guy, we'll pay you for it."
- 5.5 Now, the guy in the electronics store, again, looked at Steve Jobs. He said, "Call him. He's committed to buying 50." So he did. He called him up, got a commitment that the guy would actually buy 50 if they produced it. He said, "OK, I'll give you the parts on a loan." Lent the parts, they built the computers, they sold 50. And that was the first transaction made as Apple computers. And they paid the guy off. Notice the capital required here, right? First type of capital when you start a company is your own, friends

and family, whatever you can muster around. You sell your car, you sell whatever to start the business.

- 5.6** Second round might be a bank or it might be a vendor credit. But all of this is finance. There is no Apple if these guys don't get those first 50 logic boards produced and sold. So at this point, there's kind of a proof of concept. And Steve Jobs meets with a guy, again, goes back to Atari, and the Atari guy says, "Look, if you really want to understand how to raise real money, go speak to a guy named Don Valentine." Don Valentine was a former Fairchild Semiconductors engineer. Almost everybody in Silicon Valley during this period, important, comes from Fairchild Semiconductor. They all landed up—the guys who left Fairchild either started Intel or became venture capitalists or became other prominent people within the Valley.
- 5.7** He said, "Go speak to Don Valentine. He is a venture capitalist." And Don was the founder, one of the two founders, of Sequoia Capital, one of the legendary venture capital funds. So he goes and meets with Valentine. And Valentine describes this, describes it this way. "So we had our meeting. I went to Steve's house, and we talked, and I was convinced it was a big market." He was convinced with the market proposition. So Jobs says, "Tell me what I have to do to have you finance me," he said. And Valentine says, "You have to have someone in the company who has some sense of management and marketing and channels of distribution."
- 5.8** This is a very young Steve Jobs. Later, he is that someone. He said, "Fine, send me three people." I sent him three candidates. One he didn't like, one didn't like him, and the third one was a guy named Mike Markkula. Mike Markkula had worked at Fairchild, not surprising. And he liked Steve and Steve liked him. And Markkula says, "Look, I went there. And he was a tough sell because, basically, Steve Jobs looked like a hippie, acted like he was a hippie. And we were now going to make an investment in this guy? I mean, this is crazy." But Markkula actually stayed around, built a business plan with Jobs, and he became convinced that he thought this company could be a Fortune 500 company in less than five years, which is pretty stunning.
- 5.9** You've got Steve Jobs, Wozniak, and you've got a logic board, and you believe this is going to become a Fortune 500 company. Markkula basically set up a \$250,000, guaranteed a \$250,000 line of credit with a bank. Once they had that, once Markkula became, in a sense, the CEO of the company, now venture capital started flowing in. Sequoia invested, other venture capitalists invested. They raised the money, built the Apple One, which was the first real personal computer that people had. It sold fairly well.
- 5.10** And in December 12th, 1980, Apple IPO'd, did an initial public offering on the stock market, and share was priced at \$14. It opened at \$22, and sold on the first day of trading to \$29. And of course, as we all know, Apple today or yesterday or tomorrow, whatever, largest company in the world, most successful company in the world. It's gone through a lot of ups and downs through the many years that it's been around. But

I tell you the story to illustrate what it takes to start a company and how dependent so much of the businesses we have around us are on finance.

- 5.11** Without the vendor financing, or for other businesses, a bank loan or the friends and family, well, there is no company. Without a line of credit, there is no Apple. And of course, without venture capitalists coming in and giving the money and actually allowing for the building of the first Apple Ones and creating a market and distribution and all of that, and all of that costs money, there is no Apple. And then the initial public offering, the initial public offering raises huge quantities of money. As a consequence of that, you get the Apple II, the Macintosh, and again, the rest is history.
- 5.12** So every step of the way, in pretty much the life of every single company, is going to be dependent on their ability to attract capital. So let's talk about the particular form of capital that's involved here, and that is, this capital is almost always in the form of equity. What is equity? Equity is another word for stock. But what is a stock? What does stock represent? Well, equity and stock represent ownership. Steve Jobs and Wozniak sold basically two-thirds of the company before they went even IPO. So even though they were the founders, they owned just a third of the company.
- 5.13** Now, it turned out that when you're the biggest company in the world, even owning a tiny little fraction of it is a lot of money. But the venture capitalists, Mike Markkula, others who came in and funded this company all took equity, took a stake that is ownership. What is ownership? What constitutes ownership in the context of equity, in the context of a stock, in the context of a corporation? It means that you have control. That is, 51% of the shares control the company. You get to vote, you get to choose managers, you get to choose the board of directors, you get to choose the strategic direction of the company. So you have control over the assets, over the business.
- 5.14** And on top of that, you have what's called the residual claim. You own whatever's left over after all the debts have been paid. Debt always has a priority in terms of payment over equity. So a company will always pay its debts, legally has to, always pays its debts, its rent, any other contracts where they have to pay regular payments and it's bonds and it's bank loans and any other loans that is taken before it then pays dividends or stock buybacks, however, or if it's dissolving, just the capital back to investors.
- 5.15** So you get the residual claim. You get what's left over after everybody else gets paid off. Employees are going to get their money before owners. If you are in a business, any kind of business, first you pay your employees, and then whatever's left over, you get to spend as an owner. So this is the same here. The other aspect of this, which is really, really important, is that the corporate form allows these equity holders, allows stockholders, this ownership form, to have a limited liability. What does that mean?
- 5.16** That means that you can't lose more than what you invested. That is, let's say that the company—you remember yesterday we talked about, in a previous class, we talked

about the fact that you can go bankrupt a little bit or you can go bankrupt a lot. You can lose just a little bit or a lot. And all of that is borne by the creditors. That is, the creditors cannot go after you as an owner for whatever it is that they've lost. Limited liability means they cannot come after you for more than what you invested, which is really crucial.

- 5.17** I mean, imagine if you owned a stock portfolio, if you owned, I don't know, 100 stocks or even just 10 stocks. One of them going bankrupt could bankrupt you if you didn't have a limited liability, if you had unlimited liability. Because the one company going bankrupt, they could come after you for whatever your net worth is, to compensate them for the losses that they took. So the very existence of our stock market, the very existence of venture capital, the very existence of the equity form of financing that exists today that funds pretty much everything that we do, is this concept of limited liability.
- 5.18** And it's a concept, it's a new concept, relatively speaking. In the big picture, it's a concept that really evolved in the 19th century with the establishment of the modern corporation, what's called the C Corp, which is a, or an LLC, a limited liability corporation. But they're all basically limited liability corporations, where you cannot lose more than you invest. Really, really, really important. Because otherwise, stock market could not exist, not in its current form, not with the level of investment that occurs.
- 5.19** All right. So equity is the way in which we fund most of our businesses, ultimately, in one way or another. And any growth business, any real substantially sizable business, is going to raise capital in the form of selling ownership stakes, not just in the form of lending, not in the form of loans, bank loans, bonds, things like that, but in the form of equity. At the early stages, and today, even beyond early stages, certainly at the early stages, all of that gets done privately. That is, it doesn't involve what today we call the stock market or a marketplace.
- 5.20** For example, the first level in which, for example, Jobs and Wozniak got their money was from venture capital. What is venture capital? Venture capital is a financial institution. Like any financial institution, it has two sides. It has, in a sense, a liability side, a side that facilitates saving. So it takes people's money and promises them a return, although that promised return is not guaranteed at all with the venture capital fund. It's just telling them, "We expect to be able to return you this amount of profit."
- 5.21** It takes in those funds, those savings, if you will, and then it invests them. But it invests them not in public companies, but in private companies, typically very young companies. Typically, this is capital that is used to really grow a small company into a larger company. Venture capital has come to prominence through its investment in technology and in things like biotech, although it was used to finance all kinds of other industries in the past, but it is primarily today a form in which we invest in technology and in biotech and other things like that, the cutting-edge stuff.

- 5.22** The problem, of course, with very, very young companies, particularly companies using science or technology to push the envelope of new products and knowledge, is that the risks are massive. And what do we mean here by risks? What we mean here by risk is, there's a really high probability the company never makes it. That is that the company is actually worth zero, that they will lose all their money. So very few startups actually make it. I mean, this is true generally. How many restaurants that open actually make it to year five or the year 10? A lot less than half.
- 5.23** This is true of all businesses. But in particular, in technology, very risky, potentially highly competitive. Think of how many AI companies have just been created. How many of them are actually going to be successful? Of the 100 new AI companies that exist in the world today, and international, how many of that 100 are actually going to be successful, given, if you think about it, that Microsoft and Google and Apple and all the big guys are also investing in AI? So of the 100, if five make it big, that'll be huge. If 10 to 20 make it at all, that will be huge. But that means 80 to 90% of them are going to go bust, are going to go to zero. And that's the reality.
- 5.24** So, what does a venture fund do? How does it invest? You're going into a meeting with a Steve Jobs, a hippie, hasn't had a shower in three days. He's on a fruit diet or some crazy diet and meditating or whatever, and you're going to give him money. What's the probability you're going to get that money back? It's very low. So how do venture capitalists make it? How is this model? How do they provide a return to savers? How do they make any money?
- 5.25** A good venture capitalist basically develops the skill of being able to tell who is going to be a loser. So that's the first skill you have to have. Who doesn't have a shot? Who doesn't have a shot? You might not be able to tell who's going to be a home run, who's going to be the next Apple, but you should be able to tell who's not going to make it. Great venture capitalists are really good at screening out the garbage, or at least they're not so successful.
- 5.26** And a typical fund, and a venture capital fund is organized as a limited partnership. It's usually 8 to 12 years. They get all the money. Let's say they raise a billion dollars, and then they invest it over their first, let's say, three years, and then they sit on it, and hopefully the investment returns money to them within 8 to 10 years, and then they return the money to the investors. They keep a nice chunk of it for themselves. But the only way the model works, because of the high risk, is if they're returning a lot of money to the investors and they're making a lot of money for themselves. So first of all, screen out all the bad stuff. That's step number one.
- 5.27** Step number two is, try to figure out a model of who is going to be successful. Who do you think can be successful? And here it's a lot of their particular expertise, a lot of their particular genius, their particular knowledge of human beings. Don Valentine could see through the externality of Steve Jobs and discovered an incredibly smart, incredibly ambitious, incredibly, somebody who could really see far, who could see into

the future. Remember, Woz, during this period and I think to this day, he was a tinkerer. He was a builder, but he wasn't a business guy. He had no vision. He had no sense of what the potential was.

- 5.28** Steve Jobs saw it right from the beginning. He saw where this could take us. He saw how it could change the world. You get a sense of what kind of people they are, and then you get a sense of also, how much influence can I have on them? Because one of the things a venture capitalist brings to a business that they invest in is their own experience, their own knowledge. Usually, these are people who've done businesses in the past or have worked for businesses, know something about technology, know something about business, know something about human resources, marketing, all the things you need to know in a business.
- 5.29** And if they don't, they know people who do, and they bring in, they typically sit on the board of directors, they have a real influence. Will the founders listen? Will they pay attention? Are they coachable? So there are a lot of parameters that go into a venture capitalist, figuring out who is out and who is in. And then you make a typical venture capitalist, and I'm massively simplifying here. Let's say we make 10 investments because they've decided they like the technology. They think there's huge upside. Every single one of those investments they believe is going to be the next Apple.
- 5.30** Otherwise, they don't make the investment. Every single one of them has to be a home run. They make 10 investments in companies they believe strongly that are home runs. These are companies that are going to be the next big, today we call them unicorns, billion-dollar companies, whatever it's going to be. Of the 10, almost for sure five are going to fail. Even though the venture capitalists thought they were going to be home runs, five will fail. Four will be singles and doubles, and one will be a home run. And that's a great fund.
- 5.31** And the home run is so much, they'll make so much money off of it that it doesn't matter that five failed and that four were just OK. It's the one. I think eBay, when eBay went public, I think the venture capitalists got something like 1,000 percent on the whole fund from that one investment. So not only did they make a huge return on this one investment, but that one investment was such a huge return that the whole fund made a huge, I think eBay is one of the biggest returns ever, just in terms of the amount they invested versus the amount they pulled out in the end.
- 5.32** A fund like Sequoia, which certainly during the time of Don Valentine, but I think even till today, is legendary—they have, the reason they're so successful is they've hit home run after home run after home run after home run. But in any particular fund that they invested, only one was probably, maybe two of the 10, were home runs. So incredibly risky. As a consequence, you have to believe that every single investment is going to do phenomenally well. And then, to some extent, it's out of your control. Some of them will do phenomenally well, some of them won't. And on average, again, one in 10, two in 10, do really well, and five in 10 will probably fail.

- 5.33** And that's a life of a venture capitalist. Who are the investors in venture capital? Mostly the investors are people, insurance companies, pension plans, maybe wealthy individuals. People who have very, in case of insurance companies and pension plans, very, very large portfolios, huge diversity. We'll talk about diversification later. But as part of this diversification, they want a piece that could be a home run. Just in a sense, just like the venture capital, they want some piece that could be a home run. The rest is pretty safe. They're going to make regular returns. They want a piece that could be massive.
- 5.34** Now, a home run here is not thousands of percent, because by the time it gets to them, it might only be 20% or 30% or something like that. It's still good on an annual basis. We're talking about annual returns. But that is most of the money in venture capital funds today is pension plans, insurance companies, big endowments; university endowments give a lot of money. Any big pool of capital that's widely diversified in terms of wants a piece in the venture capital community, particularly given that since the 1980s, since Apple went IPO in 1980, this industry has done very, very well.
- 5.35** I think just in the '80s, you've got Apple. Intel was earlier, but Intel stocked it very well afterwards. You've got Oracle, you've got company after company after company, Sun Microcomputers, Compaq, Dell, Michael Dell, and Dell. Dell started out building computers in his dorm room at the University of Texas. Raised some capital, got some VC money, and today, Dell is massive and huge, and in all kinds of businesses. So venture capital is the way businesses get started when you think about equity capital, and this is, of course, a big part of finance, is equity capital. We'll talk about what happens once they go public and in the stock market in a minute.
- 5.36** But there are other forms of private equity. That is equity in the private market. You can think of venture capital as a form of private equity. It's a particular type of private equity that invests in startups, that invest in very high-risk. But there are other companies that invest, other finance companies that invest in the private markets, not in the stock market, not where it's publicly traded the securities. They're typically called private equity funds. And these private equity funds have a lot of different strategies that they can engage in. Almost all of them, not all of them, almost all of them, involve equity ownership stakes in companies.
- 5.37** But for example, you could have a nice manufacturing business that's family-run, family-owned. You've built it up. You've done really well. But your kids are not interested. They don't want to run the company in the future. So at some point, you're going to sell the business, and you could sell it to a competitor. But there might be opportunities where somebody could come in and invest a significant amount, keep you on maybe as a CEO for a while, keep you with some minor interest in it, or maybe your kids still maintain some equity, maintain some ownership of the business.
- 5.38** And so you have these funds who specialize, for example, with family businesses. They come in, they buy a significant stake, they provide capital to grow the business. They

maybe make it more efficient. It was run for the family, so maybe it's a little loose in terms of efficiency. Maybe there are too many people employed. Maybe there are too many family members employed who are not very good, and they make it more efficient, and they make a profit off of it. And how do the private equity guys get a return on their capital? Because usually private equity is also 8 to 12 years.

- 5.39** They make it efficient, and then they either sell it to somebody else, maybe a competitor, maybe another private equity fund, or they take it public. They put it on the stock market. Either way, they make here, you don't expect five to go bust. Here, you want all of them to return something, typically. Some will return very low rates of return, but a few will return very high rates of return. These are much easier to model. The risks are much lower. There's already a cash flow. Now you're just modeling, "OK, how can I improve the cash flow? How can I make it more efficient?" And hopefully you can.
- 5.40** And some of these deals will give you a huge return. Some of the deals will give you a very small return. Some will go bankrupt. That's always a possibility. But it's not something like a venture capital you take for granted. That is private equity. Family business is just one way to do it. There's a variety of different reasons. You might own a company and you want to grow, but you don't have the capital to grow. So you bring up a partner. The partner could be a private equity fund.
- 5.41** They don't have to take a majority stake even, but they provide the capital, again, and the expertise— the management expertise, the marketing expertise, whatever expertise you need to make the company more efficient, more productive, and therefore increases valuation, and then sell it and make a profit. And again, there are a variety of different private equity funds. They specialize, some specialize in a particular industry. Some specialize in a particular structure of purchasing the firm. Some use a lot of leverage, which means a lot of debt. Some don't. Some do family businesses. Some don't.
- 5.42** Some private equity funds provide growth capital to those tech companies that were funded by VCs. So let's say it's a VC-created company. It's done OK. It's not one of the home runs. It's just doing OK. It's not going to go public, but it's still profitable, but it needs to grow. It needs more capital. The VCs are not interested anymore because it's not a potential home run. It's just like a double or a triple, maybe, and they're just not going to invest. A private equity fund will now give it growth capital to allow it grow, to allow it to become bigger, and then they can either sell it or maybe even take it public. But the valuation is just not going to be as explosive as some of these others.
- 5.43** So all of these are mechanisms by which you can invest in a company. The challenge, of course, here is that once you make the investment, it's very hard to get your money back. You can't just turn around and sell it. It's private. So while today, in particular, there are private markets where you can sell private stock, it's not easy. So the way to exit, the way to get your money back, the way to get a return, is to either sell the

company or list it on a public market like a stock market. It has to be, or sell your stake in some private auction or some private negotiated deal.

- 5.44** A lot more work, a lot more or a lot harder in order to get out. If I go to the stock market, I can go now. Market is already open. I could buy some stock. I can sell it later, anytime. The transaction cost is basically zero. I can do that all day. If I invest in a venture capital fund, go to sleep for 10 years, and then maybe I'll start getting my money back. Well, usually it's less than 10, but I don't do anything, and I can't get out. If I'm an individual and I'm invested in a private equity fund, I can't get my money out until the fund itself starts liquidating its investments and sending me money. So no liquidity, money is locked up.
- 5.45** Whereas in the stock market, I can come in and out anytime I want, pretty much. Certainly when the markets are open, but these days you can even trade when markets are closed. So a huge amount of liquidity. All right. So companies can sell a stake of ownership in themselves. You can start a company, and venture capital, the finance markets, will provide you with capital if you've got a good idea and have that potential for a home run. But at a certain point, at a certain point, to raise large amounts of capital, which are required today or have been required over the last 100 years, maybe in a sense less today than in the past, at some point, you need large amounts of capital.
- 5.46** And to get large amounts of capital, you're not going to get it from a VC; you're not going to get it from a private equity fund. Those are limited in terms of how much capital they can deploy for any one investment. The beauty of our system is you can list on the stock exchange. And what that means is, you're selling shares to anybody who wants to buy. And now the potential market is hundreds of millions of people, maybe more than that, because it's everybody in the world who can buy shares from you. And now the amount of capital you can raise, the amount of money you can raise to grow the company, to invest in a company, to do what you want, and also to cash out your profits, is astronomical.
- 5.47** You can do large raises. Again, if you think about Apple, when they first got the first few investments from people like Don Valentine and others—now, remember, this is 1970, so money was a little different back then. But we're talking about hundreds of thousands of dollars that they got. But when they listed and went public, we're talking about tens of millions of dollars that they got. That's the difference. So going public is a way to raise large sums of capital.
- 5.48** And what it does is it distributes ownership. Now, instead of a few people, like three venture capital funds maybe owning 60, 70, 80% of the company and a few employees owning the rest, now you've got millions of people owning the 60, 70% of the company. Maybe a few large investors own a big chunk, but generally it's widely distributed. And this is, of course, why you can raise so much money, because taking \$10 from a million people is \$10 million. \$10 million for any one individual to give is a lot of money, but

\$10 for a million people, not a big deal. So it just scales. It just scales in ways that no other form of raising capital allows for.

- 5.49** So one advantage is the scale of the capital. You can also keep coming back. That is, if the company does well and it wants to grow even further in the future, you can issue more shares, you can raise more capital, and you can continue to grow the business in that way. So the primary function of stock markets, the primary function of stock markets, is to allow corporations to raise large amounts of capital, large amounts of capital in one go. That is the reason they exist, if you will, the reason they exist. And it works.
- 5.50** 300 years ago, there were no large companies. I mean, the largest companies were state monopolies, like the East India Company that had monopoly over trade, with the colonies usually, or with the Empire. But in the 19th century, with the invent of capitalism and the industrial revolution, suddenly private businesses grew to scale. Competitive private businesses had the opportunity to grow. They reached points where the banking system and these just wealthy individuals, friends and family, all these other mechanisms by which you could raise money just didn't have enough capital to be able to provide them to facilitate the growth that they needed.
- 5.51** And while stock markets existed even before this on smaller scales, they really blew up. They really became big once vast quantities of capital were necessary. A lot of money was raised for railroads during the 19th century, both in England and in the United States, using the stock market. And why are people willing to provide this capital? Why are people willing to go in and buy these stocks? Well, again, it's a form of saving. It's money that I have, that I've not consumed. I don't plan to consume it now. I want to use it in the future. I'm planning to use it in the future.
- 5.52** Think about the fact that a lot of the big investors today are pension plans. What are pension plans? Pension plans are, they've accumulated the money that people have put aside for one day when they retire. And they've accumulated a lot of money for individuals, and they promise those individuals a return, so they have to invest the money to be able to make sure that they can live up to the expectations of people. But generally, we invest, we put money in saving accounts because we're not going to consume it now.
- 5.53** We're willing to delay our consumption, or indeed, we want to delay our consumption because maybe we're worried about one day not having a job or maybe one day retiring or maybe one day wanting to buy something like a big house or something that we can't afford to do today, so we save. So people are incentivized to invest in productive companies because they expect a return. They expect a return on that investment. And we'll get to what kind of return, what that means, and how you calculate that in a minute. But it provides them with the potential return, and also provides them with liquidity.

- 5.54** Just like with a bank that you can put money in a deposit account and withdraw it, in a sense, in a stock, you can buy a stock, you can sell it, and buy, sell, buy, sell constantly. Not recommended, but you can. There's transaction costs involved. But it's not like you're putting your money away like a venture capitalist does and forget about it for eight years. The nice thing about a stock market, or if you invest in mutual funds or you invest in ETFs or any other aggregator, you can at any time sell what you bought. You're not locked in, which is a huge advantage in terms of your long-term savings and your needs, both short-term and long-term.
- 5.55** So, a stock market provides the public, if you will, investors, with an opportunity to participate in economic growth, to participate in businesses that they would not otherwise have any access to. I can buy Apple stock, but if Apple was private, how would I even get there? And indeed, in a modern sense, if you invest in a portfolio, if you're diversified, if you buy mutual funds, you're investing in a lot of companies you don't even know about, you've never heard of, you know nothing about, and you don't really care. As long as they're growing, as long as their stock is doing OK, and even if it's not doing OK, because you've got a lot of them, some are doing OK, some are not doing OK, on an average, you do fine.
- 5.56** Again, we'll talk about diversification later. So stock markets provide the ability for ordinary people, if you will, to participate in the growth of industry, to participate in the success of business. We can all own a piece of Apple, which is unique in history. It's unique in history. This has not always been available. And it's a phenomena, primarily since World War II. Stock markets were very thinly traded. I mean, the 1920s, everybody was in because there was a— during bubbles, everybody gets involved and engaged. But generally, mass participation in the stock market for the common person, whether it's directly or through pension plans and insurance plans and stuff, is a phenomena of the last less than 100 years.
- 5.57** So what stock markets allow us to do is they allow us to transform fixed assets and future profits, profits that haven't really materialized yet, into liquidity, into cash right now. So what's the company selling when it goes public? What's the ownership stake that we're buying? What's it based on? It's based on the fixed asset that the company has. In the case of Apple, not much. Maybe a little warehouse, maybe some facility where they're putting stuff together. But there are not a lot of fixed assets in the case of Apple.
- 5.58** In the case of US Steel, when it went public in the early part of the 20th century, a lot of fixed assets, machines, furnaces, big buildings. In case of high tech, very little. But in both cases, the stock is really not based just on those fixed assets. That's a small portion of the valuation. What are you really buying when you buy a share when they go public? You're buying the future. You're buying the expectation of future profits. You're buying a piece of those future profits, the cash flow in the future. So we're monetizing, we're taking all those future profits, and we're monetizing them today.

- 5.59** And by monetizing them today, the company raises the money. It can make those future profits a reality because it makes the investment, and it builds up the company, and it makes those profits possible. If you really think about it, it's an amazing thing. You're taking an abstraction called future profits. Nothing's materialized yet. And you're making it real today. And it's that reality today that makes those profits possible. I mean, it's still risky. It's still not guaranteed, but now you have the capital to invest and make them happen. So they provide huge amounts of liquidity in this capital to get going.
- 5.60** But there's another really crucial aspect for the company aspect of going public that is really, really important, and that is that it provides a massive incentive for entrepreneurs to be successful and a massive incentive for venture capitalists to invest in good companies. Venture capitalists can make thousands of percent on their investment. Managers, founders can become billionaires at an IPO, initial public offering, certainly millionaires, but many of them billionaires. That's a massive incentive to work hard, to do the best that they can, to hire great people, to make sure the product is the best that it can be.
- 5.61** The venture capitalists have an enormous incentive to monitor the company, to provide it with good guidance, to bring in experts, because they're going to get rich if it's successful. So there's a built-in incentive here for everybody to do the best that they can do running this company, because if they do it well, the IPO, going public, is going to make them rich. It's a good thing, not a bad thing. So liquidity, capital, and incentives. That's the real value of going public, raising capital in the stock market.
- 5.62** We talked a little bit about these small investors being able to invest in large companies. You own a tiny bit of the company, a little piece of it. All of us suddenly, average people, can become capital providers. We're actually participating in capital provision. Of course, we do that when we put our money in a bank as well, because that becomes loans to businesses. So again, what the financial industry over and over and over again, you'll see, is doing is turning our, the amount of money we have, the wealth we have beyond what we consume, our excess, excess money, excess wealth, into productive investment. Banks do it. Stock market does it.
- 5.63** It turns us all into providers of capital. It allows us to, again, benefit from the success of these good companies, these amazing companies that otherwise we would have no way in. It's converting saving into capital and allowing all of us to prosper when and if the economy does well, our businesses do well. What else? Now, you could argue that all of that is true, at least a lot of what I've just said is true, for the IPO. When a company goes public, it gets all that money, and that's great. But what about all the trading that happens afterwards? Apple didn't just go public, and we all bought the shares, and that was it. But we actually buy and sell it constantly.
- 5.64** Now, when you buy a stock, does the money go to the company? No. Who does the money go to when you buy a stock? To the person who sold it. It goes to the person

who sold you the stock. I mean, that seems like a zero-sum game. It seems like the company doesn't get anything. So where's the productive value of us trading? What's the value of that on a regular, the fact that we trade? There's a number of productive purposes to this trading.

- 5.65** First, if you couldn't trade after the IPO, we wouldn't be able to participate in the IPO. We wouldn't provide the capital to begin with because then we'd be locked in. It would be private equity. I'm stuck. I can't get out. The only reason we're willing to participate in the IPO is because we know that we can sell the stock afterwards whenever we need the money, when it's convenient; we could keep it for a long time or not. The options are open. So that is a big reason why trading in the stock market is so valuable. It's just the ability to exit.
- 5.66** But what if everybody who did the IPO is out now? Why do we keep trading? What is the value here? It's really, really important to understand why people trade. Why are people trading in stocks? So what is it that we're doing when we're trading in stocks? Basically, why would I sell a stock? I mean, there are really two reasons. One is, I need the money. I don't really care. I just need the money right now. I've got other things I need to do, whatever, I need to sell.
- 5.67** The other is that I think the stock is worth less than what it's trading at. So I want out because I think it's overvalued. And then I might go in to a stock that I think is undervalued. At least I think this is what professional investors do. We'll talk about what all of us do or don't do. And in the sense that that is going on, all of our opinions about the value of this stock are being integrated into its price. Some of us think it's undervalued, and therefore we buy. Now, we're buying from somebody; why are they selling? They might be a liquidity seller, or they might think that it's overvalued. They might be disagreeing with me.
- 5.68** We're all going to disagree. We'll see in a minute, we'll see maybe at the beginning of next class, evaluation model and what it looks like. And we're not going to agree on what any particular stock is worth. But the aggregate of our opinions gets reflected in the price. And indeed, the opinion of the most influential among us will get reflected in the price more than the opinion of the less influential of us. Let's say a stock is trading at \$10 a share. I am sure that this stock is worth \$15. I've done my research and I've studied, and I've figured it out that this stock is worth \$15. What am I going to do at \$10?
- 5.69** I'm going to buy it, from people who don't think it's worth \$15. They think it's worth 10 or 9, or 8. They're going to sell it to me. The more I buy, what's going to happen to the price? It's going to go up because I'm buying it. People are observing that. They're bidding up. I need to convince more and more people to sell to me. The only way for me to do that, at 10, I've convinced everybody who believes the stock is worth 9. But if now everybody believes the stock is worth 10, and I want to buy it, then I have to buy it at 11. I have to give them a profit.

- 5.70** So I'm going to keep buying it until when? Until it's almost at 15. Because even at 14, if I buy it at 14 and I truly believe it's 15, I will make another buck. Why not? I'll take it. I'll buy it until it's almost 15. Now, if I'm really good at this, and if my opinion really is informed, and if I know what I'm doing, I'll make a lot of money. And by making a lot of money, I'll be able to do this over and over and over again. If I'm not good at what I'm doing, and what's going on here is, I'm completely bamboozled, I'm completely wrong, what's going to happen? I'm going to lose a lot of money, and that's going to make me less effective in the future. Even if I see a stock that's at 10, and I think it's 15, I just won't have the money to influence it.
- 5.71** Money in the stock market tends to go towards those who are successful and away from those who fail. And therefore, the more successful you are, the better at this you are, the more influence you have on stock prices. The more influence you have on stock prices, the more stock prices are reflecting the knowledge, the information, that the better investors have. So stock prices are crucial. And again, in a few minutes, we'll talk about how we determine them. But stock prices are crucial because stock prices provide us with really, really important information for everybody.
- 5.72** For example, they tell managers whether they're doing a good job or bad job. A manager might be so focused on what they're doing, they might miss something in the big picture. But if they start seeing their stock go down, they're going to have to stop, think, consider, "What am I missing?" Look, discover, change strategy. Hopefully, they're a good manager. And if not, the board will notice the stock is going down, hopefully, because they're stockholders, maybe. And they'll force the manager to reconsider. So the information is incredibly valuable for inside the company because what it's getting is feedback from the outside, saying, "You're doing something wrong."
- 5.73** Let's say you're in the early part of the 20th century and you run the best buggy company in the world. You make amazing buggies, really comfortable horse-drawn buggies. And you don't notice over in, I don't know, Detroit, that they're building automobiles. That's a different world. What's that going to do with you? But the market notices, and the market starts blowing your price. Now you have to wake up and notice and realize, "My business is dead." It might be dead tomorrow, might not be dead in five years, but in 10, 15, it's dead. What do I do?
- 5.74** That's really crucial. The earlier you know that, the more you can control the damage, the more you can shift, the more maybe you can go into the automobile industry, or maybe you sell at the peak to somebody who's not noticed this. But the market is giving you a signal. I bet you, with the rise of PCs, the very early rise of Apple, IBM stock started to decline. And believe me, IBM did not realize what was going on. And even though they tried to go into the business of PCs themselves, too late, not good enough. IBM stock by the early 1990s was crushed, crushed, which required them to make a change.

- 5.75 And they brought in a CEO who completely revamped IBM, completely changed it, put it on a track, but a track away from building mainframes because mainframes were dead. And IBM became more of a service provider, more of a consulting firm, more of business consulting, information consulting. And the whole culture in IBM changed because up until then, even into the mid-1980s, IBM were king of the hill. They thought they were untouchable, and yet they were dying. And they didn't know it, but the market knew it. So the market will send you a signal. The market will let you know whether you're doing well or badly.
- 5.76 So it's crucial for managers. It's crucial for boards of directors. And it's crucial for investors. Investors who might not be experts at obtaining information and valuing companies, but just generally, what sectors are doing well, what sectors are doing badly, what should I be in, what should I be out, how should I diversify, how should I invest? The market fluctuating and giving you a price that makes sense is crucial information for you as an investor. And on a bigger scale, it's crucial for investors like venture capitalists or banks or private equity funds.
- 5.77 "So the mainframe business is dead. OK, I shouldn't invest in mainframe startups because it's dead. Apple stock is going up. Oh, maybe I should invest in startups that are going to exploit that somehow." So the market is sending signals to the venture capital community and others that are suggestive of where to look for the next big thing. And the market can do that because there are people in the market trading who are constantly aggregating information, constantly thinking through, constantly modeling. "What do I think this firm is worth? And what do I project into the future that it's going to be worth in the future?"
- 5.78 And of course, what this does ultimately is that it helps allocate capital on an economy-wide scale. As shares of the buggy industry are going down, capital is withdrawn from the buggy industry and invested in the automobile industry. As shares of industrial companies are going down in the 1980s, capital is withdrawn from what's called the Rust Belt. And where does it go? Where does the capital go in the 1980s? From the Rust Belt, it goes to Silicon Valley. Silicon Valley cannot exist without, in a sense, the capital being freed from dying industries that are Rust Belt.
- 5.79 So we might mourn all the loss of jobs and all of that, even though unemployment is going down at the same time, all the loss of jobs and the loss of manufacturing in the Rust Belt in 1980s. But we missed that it all went to Silicon Valley, which was much higher growth, employees, many more people, much bigger salaries because it's a much more productive industry. And we missed the whole economic picture, but somebody got it. The people who are actually doing it, the people who are facilitating that transfer, that reallocation of capital. The stock markets, finance, generally allows us to reallocate capital.
- 5.80 As the stock price goes down, the cost of raising more money to build out, to expand, the buggy guy, he wants to build another buggy plant because he's the best. But his

cost of capital is going up because the lower the price, the more shares he has to issue to get the amount of money to build the plant. Or if he goes to the bank, the bank will say, "But your stock price is down." "Oh, no, that's temporary. Don't pay attention to that." And the banker goes, "No, I take that seriously. Let me take a look. Oh, there's an automobile over here, company over here. Maybe not a good time to expand." Again, the bank is not going to give you money to expand.

5.81 So even markets, financial markets that are not directly related to the stock market, are going to pay attention to the stock market, see what's going on, interpret it, and determine their allocation of capital. So you're going to make fewer investment in bad companies. And when you're going to see things on the rise, things are exciting, things are raising a lot of capital, that's an industry I'm interested in. I know AI right now, whatever, capital will flow into that area. Sometimes maybe too much capital, but that's part of the way it works. As I said, of every 10 investment, at least five go bust, that's going to be true. That's going to be true here as well.

5.82 All right. So prices in the stock market reflect information. They reflect crucial information. And that information is used by the rest of the market participants to determine allocation of capital across the economy. How are those prices determined? So we're going to look at that when we come back in the next class. Do we have any questions?

5.83 You talked a bit about startups, and I currently work at a startup in New York. I'm just curious, if or when we get to an IPO, what does that price that it comes out with depend on? What makes it go up down?

5.84 On the day of the IPO?

5.85 Yes.

5.86 Well, on the day of the IPO, it basically is going to come up or down. It will depend on whether the market, the participants in the market, believe that the pricing that was given to the IPO by the investment bank that you're working with is optimistic or pessimistic. Is it reflective of what the market really thinks the future is, or whether it's reflective of whether they're over-optimistic in no way? Now, most IPOs, it goes up. One of the reasons for that is, no investment bank wants to bring a company to market and for the stock to tank after it's priced it. Like, the institutional investors that have gone in in the pre-IPO and committed to buying a certain amount at that price are going to be pissed off. So they always leave a margin.

5.87 So usually a stock goes up in the IPO. How much it goes up will depend on people's sentiment, people's beliefs, people's ideas about what the future holds for that particular company. And of course, what happens in the first day is not going to be important to you because the reality is, you can't sell. You'll have a lockup. You won't be able to sell probably for six months. So what you really care about is what happens

in the first six months. And that, again, will depend on the market's evaluation of the company.

- 5.88** Now that it's public, it reports much more extensive financial information. People will analyze that. People will try to predict what the future holds. And over those six months, it will trade, and after six months, reflect if the market is efficient, we'll get to that, the best available information about your company at that point. And you can only sell then, so that's what's relevant. First day is relevant for certain speculators. And some people can sell on the first day, but most employees are stuck until a few months later. They've got a lockup.
- 5.89** When Donald J. Trump, of course, when he lowered, I believe it was the tax rate, this was the corporate tax cut or whatever, because a lot of companies have gone to Ireland and have done all kinds of shenanigans, the common sentiment was, well, these companies are coming back to the United States, and with this new cash, not only is the US better, but they will invest back into their companies. But in reality, we saw a lot of stock buybacks, which is a different kind of investment for a different crowd. Should the government get involved in this transaction unlimited or the free market reigns?
- 5.90** Let's take this one at a time. Let's do corporate taxes, and then we'll do stock buybacks. I don't think the two are necessarily related. And we'll do stock buybacks, even though I think I'm covering that later, but we'll do it. All right. So, corporate taxes. Corporate taxes are some of the dumbest taxes invented, right? Because what is a corporation? A corporation is us. It's the people who own the stock in the corporation. So theoretically, this is a way to tax us, again, on money that we've already been taxed on.
- 5.91** Because how did we get—we go to work, we make some money, we save up some money, and we pay taxes. We save up some money, and then we take this money that's after tax. We've already paid taxes on it, and we invest it in a company. And then the company makes a profit, and theoretically, now it pays taxes on money that was already taxed when we made it. So it's doubly taxed. And then, again, in theory, if the stock goes up and we make capital gains, we're taxed on it again on the capital gains taxes, on the piece of the capital gains. Or if they give us a dividend, we're taxed on a dividend. We're taxed many places.
- 5.92** But the reality is that we as investors don't really pay the corporate tax, or we don't pay most of the corporate tax. Every study that looks at this, basically, the corporate tax is paid by customers with higher prices. So we as investors, we have a certain expected return on a stock, and we calculate those expected returns on after-tax dollars. And the way we price everything is with after-tax dollars. We don't care about taxes in the sense of how we buy and sell our stocks. So we're not paying, really, the corporate tax, at least not in anything significant.
- 5.93** What the company does is, it takes the tax as another cost, and it basically, because investors are demanding a certain return on their money, companies will bid up the

cost of the stuff that they sell to cover much of the tax that they pay. So consumers pay it. And the second people who pay it, and this is probably even more significant, is to cover the cost of the tax, companies will build down the wages that they pay employees. So when corporate taxes go up, prices go up and wages go down.

- 5.94** When Donald Trump cut corporate taxes, what actually happened—it's not always clearly observable—what actually happened is, prices went down relative to where they would have been otherwise. And more importantly, wages went up. And you see that in the data. You see wages going up. They don't go up instantaneously. That never happens. But over time, indeed, wage growth increased after the corporate tax rate was cut. So corporate taxes are just a proxy, just a way to tax labor. It's another way to tax us and to tax consumption. When you cut corporate taxes, you're increasing wages and you're decreasing prices, which is great. The logic there ultimately should be, the corporate taxes should be zero.
- 5.95** Because the money has already been taxed, the money that got invested in the corporation. It was taxed when it was income. It just gets passed on to these other groups. It's a hidden tax on everybody else. That's not clear. It's not fair. We pretend that evil corporations pay it. Evil corporations pay it. Nobody cares if corporate taxes go up because this evil corporation is paying it. But it's not corporations paying it. It's actually all of us paying it. It's a very deceptive, non-productive tax. So the lower, the better.
- 5.96** And the reason to do it was not so much to bring the money back because there are all kinds of reasons why companies are in Ireland, one of them being a low corporate tax, but not the main reason. But it was mainly to get this effect and to grow the economy so Trump could say, "I succeeded in growing the economy." I don't think he cares about the economy. He cares about what he could say about the economy. So that's corporate taxes.
- 5.97** What was the other issue? Stock buybacks, God. Look, stock buybacks don't have a big effect on the market. They never have. People say companies buy stock to manipulate their stock, but they can't manipulate their stock. Because the reality is that, for every buyer, there is a seller. If I'm buying the stock, what's happening? The number of stocks is shrinking. So you say, "Oh, if the number of stocks is shrinking, the price must go up." But no, because your assets are going down. Why are your assets going down? Because your asset included the cash that's used to buy the stock.
- 5.98** And what's the offset? The assets go down, and therefore potential future earning go down, by exactly the same amount as the shrinking of the stock. It's an exact equation. So stock prices shouldn't move when you do a stock buyback. Now, why do companies do stock buybacks? They do stock buybacks because they have nothing better to do with the cash. And this is the sad thing about stock buybacks. It's not that companies are somehow manipulating their stock. The real sad thing about the stock buyback is, these companies don't have anything better to do with the stock.

- 5.99** That is, they believe that buying the stock, their own stock, is as good an investment, better an investment, than anything else they could do with the cash. If they could take the cash that they have on their books and invest it in something that generated a higher return than buying their own stock, then they would do it and their own stock would go up because now they're invested in higher return things. The reason to do a stock buyback is, I've got a lot of cash, and the reality is, I look around and the investments are not very attractive for me as Apple or for me as Google or whoever. So I'm going to give it back to my shareholders because it belongs to them. I'm going to do a stock buyback by giving them.
- 5.100** Now, why not give them a dividend? Why buy their stock? Because dividends are taxed as regular income and capital gains are taxed at a lower rate. So the government, by setting taxes the way they set them, has basically said, "Don't pay dividends." They have a higher tax rate. Do a stock buyback. Give them capital gains; that's better for them. So stock buybacks make no difference other than it's an efficient way to get money out of the hands of people who don't know what to do with it and into the hands of people who might be able to do something with it.
- 5.101** Because if the company knew what to do with it, they wouldn't buy their stock back. It's a sign of diminished investment opportunities, particularly—now, Apple produces so much cash that you can understand that it can't invest at all. There's just not enough good opportunities to invest it. So they do a lot of stock buybacks. But there are a lot of smaller companies. I celebrate stock buybacks because I think it's a way of increasing efficiency. It takes cash again from people who don't know what to do with it and gives it to people who might know what to do with it. All right, guys. Thank you.

Lecture 6: Stocks and Strategy

At a glance — Stocks and strategy.

- 6.1 Imagine a company that has done really well. It's really big. It's got huge amounts of assets. Indeed, it's done so well that it's expanded into a variety of different areas. It maybe produces widgets over here. It does financial services over there. It's in the food business over here, maybe makes some jet engines over there. And it's got, I don't know, 12 subsidiaries all doing different things. And it keeps growing in terms of assets. It's growing and growing and growing.
- 6.2 And the manager, the CEO of the company, is doing fantastic. I mean, he's flying around the country in this private jet. He stays in the nicest suites in the nicest hotels. All the managers of all these different subsidiaries are doing fantastically well. They're all making huge salaries because, as it turns out, maybe the board has decided to compensate them based on sales. So they're maximizing sales like crazy. They're growing like crazy. And the only problem with this business is that it's not making much profit. It's selling a lot, but profits are not very high.
- 6.3 And since the board is compensated, for some bizarre reason, compensating the managers based on maybe assets or sales or things like that, the board's happy, the CEO is happy, shareholders not so much. No profits or very low profits. Company is huge, but no profits. Now, you can imagine that, but to a large extent, this was indeed the state of American business during the 1960s and '70s. American businesses grew into massive conglomerates. Largest companies in America, largest companies in the world sometimes, were conglomerates. Conglomerate means many, many subsidiaries doing all kinds of things unrelated to one another.
- 6.4 Managers were compensated in a weird way based on sales or asset growth or things like that. And indeed, they were not profitable. They were not good companies. And it makes sense that they would not be good companies because it takes extraordinary skill and ability. It takes being really good to do one thing, to take one product, one industry, and really be good at that, world-class at that. And now you got 12. It's very, very rare. There are a few CEOs in history who have been good at it, but it's very, very rare that somebody can multitask at that level, can be really excellent at every one of those things.
- 6.5 But why would you do it? Why would you have lots of different subsidiaries? It lowers risk. If you're a CEO, you're never going to lose a lot of money. Why? Because if there's a recession and some industries are hurting, other industries are probably doing OK. So you balance out your losses and gains, and overall, you do fine no matter how. But of course, when things are going well, some of the industries might not be doing that well.
- 6.6 So if you're a manager and you're risk-averse, you don't want to take a lot of risk, you don't want to lose your job, and you're being compensated in this weird way through

sales and assets, what you want to do is grow assets and lower risk, which is exactly what American businessmen did in the 1960s and '70s. And they built these massive conglomerates. The consequence of that really is that American businesses got really crushed by competitors from Japan and Germany. American industry was not doing well going into the 1980s.

- 6.7 Just think automobiles, but you can think beyond that. I think the VCR was invented in the United States, but didn't last very long. TVs, all TVs used to be made in America. No TVs by the 1980s were made in America. And over and over and over again, you got businesses coming in from outside, beating the Americans in a variety of different industries. And a big reason for that was this conglomeration, lack of focus, and not paying attention to what really mattered, which is profits and shareholders. And indeed, the stock market didn't do well, partially because of inflation, but partially because the companies weren't that profitable. They weren't doing that well.
- 6.8 And stock prices reflected that, particularly of these conglomerate companies. Now, it's true that in the '50s and '60s, there was no international competition because we'd basically flattened Germany and flattened Japan. They were still building, and by the 1970s, they had built enough so that now they were competing. But their companies were a lot leaner and a lot more focused and a lot better than ours.
- 6.9 One of the things that a stock market provides us is with a mechanism to fix it. And this happened in the 1980s. We fixed it. There's a reason why the US economy did so well in the 1990s, is because we fixed it in the 1980s. And how did we fix it? Well, a lot of these companies had all these conglomerates. They were worth as a whole less than what they were worth as parts. Because as parts, focused, they could produce more profits than once you put them all together with a loss of focus by management. And that was reflected in stock prices.
- 6.10 So what happened in the 1980s is there was an opportunity here for people to come in, buy these big companies, and break them up. Make them efficient, sell off the pieces. A lot of times when somebody wanted to buy the company, the company said, "No, we don't want to sell." Why would they? CEO was pretty comfortable. That private jet was in the back. He was doing pretty well. Why would he want to lose his job? And that's where the term hostile takeover came. So I don't care whether you want to sell it. I'm going to force you to sell. And I'm going to force you to sell because you don't own that much stock.
- 6.11 One of the characteristics of American businesses in the '60s and '70s is that CEOs did not own much stock in the company at all. You don't own stock. I just need to buy enough stock so I can take you over and control the company, sell off the assets, make a lot of money for myself, for other shareholders. Each one of these companies is now going to be much more profitable and much more focused and contribute much more to the economy, but also contribute much more to the shareholders who land up owning it.

- 6.12** There are a number of challenges, of course, in doing this. Before 1968, you could literally go and buy up shares in a company, get 51% of the shares, walk into the CEO's office and say—how does Donald Trump do it? "You're fired." He does it better than I do. "You're fired." But managers didn't like that. Nobody likes to be fired. And managers are a pretty influential group of people, CEOs of large conglomerates, particularly in the 1960s. So they all got together and they lobbied Congress, and they got Congress to pass a law that basically says, "You can't do that anymore. You can't buy up 51% stealthily, in secret, and just take over a company."
- 6.13** Why? I don't know why. It's not fair, it's not right, but they probably made it about, it's not good for shareholders or something like that. Although there's nothing wrong with shareholders. But the argument would be, here's the argument. If I buy up the share stealthily and then I get 51% and fire the managers, then the people who I bought the shares initially, because as I buy them, the stock price will go up. But the people whose stock I bought initially are not going to benefit as much as the people who buy the shares at the end, right? When I buy that 51% of the shares, that last 1%, the price is going to be much higher than when I bought that first batch of shares.
- 6.14** And that's not fair, people will say. Because these guys didn't benefit from the fact that I was buying up the company. Congress bought it. The public bought it. Everybody bought it. In 1968, a massive legislation gets passed, and part of that legislation is to say, if you want to buy a company, hostile or friendly, if you want to buy a company, first of all, once you reach 10% of the company, you have to tell the world what your intentions are. Are you going to marry the girl or not? You can't get her pregnant and not. You got to make your intentions known.
- 6.15** So 10%, you have to file a document with the SEC saying, "I bought 10% of the company. I'm just here as an investor. No intentions." Or, "I bought 10% of the company. I intend to buy a lot more. I intend to own this company, to have a majority." And in that case, then you're required to place a public tender. You're required to tell the world, "I'm buying the company. I'm going to buy 10%, 20%, whatever it is, and this is the price I'm willing to pay for it." So you can see now that more people benefit from this, but it's also more expensive for me to buy the company. And I have to let the world know.
- 6.16** And now the manager can go, "Oh, my god, he's coming after my company. I can defend myself. I can go to Congress and try to stop him." And the 1980s were basically this big battle between existing managers who were running these conglomerates and young entrepreneurs, maybe not so young sometimes, trying to take over these companies and break them apart and make them more efficient. And the existing managers came up with all kinds of legal and other ways, things called poison pills and all kinds of financial shenanigans to try to keep their companies.
- 6.17** And the people who are trying to take over the company came up with all kinds of ways to get around it and ultimately take over the companies. And this is a huge battle. And

the movie Wall Street depicts some of that. If you really want to get a sense of what this looks like, I highly recommend, there's a fantastic movie in terms of finance, probably the best movie in terms of finance that I know of. It's a movie called Other People's Money. It's got Danny DeVito and Gregory Peck, of all people. Danny DeVito plays the financier. Gregory Peck plays the businessman who's trying to protect his business. It's a little, if you're sensitive to vulgar language, beware.

- 6.18** But other than that, the speeches Danny DeVito gives about finance are some of the best you'll hear anyway, in terms of just the sheer truth of what he's saying in terms of how it all works and what it's all there for. So I highly recommend that. It's an example of a hostile takeover, trying to take over a company that's basically dying, dead, where the assets are worth more sold, melted, if you will, than they are running because running a company is actually losing value. And at the end, Gregory Peck gives a speech, Danny DeVito gives a speech. Brilliant, just brilliant. And so I highly recommend watching that movie.
- 6.19** Anyway, this battle went on through the 1980s. A lot of movies were made about this, not just Wall Street and Other People's Money. Many other movies were made about it because it was a real cultural phenomenon. If you're familiar with the name Michael Milken, this is the era of Michael Milken. Michael Milken basically was a banker who provided a lot of the funding for the people who wanted to buy the companies and break them up. So he provided what were called the junk bonds or high-yield bonds. These were very highly leveraged, very risky bonds. But there was the funding that gave the buyer the cash to buy out the company.
- 6.20** And Milken is responsible through that for restructuring a vast proportion of American industry. Also, he used these instruments, just to milk him for a minute, he also used these instruments, junk bonds, to finance a lot of very risky ventures at the time. They couldn't get venture capital. They wanted huge amounts of money, and the regular bond market wouldn't give them because they were too risky. And he used these high-yield bonds, bonds that paid very high returns. He used these high-yield bonds to make these investments.
- 6.21** And these investments changed the world in ways that, certainly, I don't think he envisioned at the time. I don't think anybody envisioned, but it was MCI putting the first fiber optics into the ground. I mean, almost nobody knew what fiber optics would do, what impact they would have, but we wouldn't have an Internet today without them, right? This was the first competitor to AT&T's monopoly. They put fiber optics and they offered long-distance phone calls, which was a big deal back then. You guys are too young to remember when long-distance phone calls cost a lot of money. And of course, that's the backbone of the Internet, ultimately, the fiber optics that are in the ground.
- 6.22** He funded the first cell phone company, McGraw Cellular, which built cellular towers and sold mobile phones, those big brick phones that used to be cellular phones. But

that is the infrastructure on which cell phones ultimately got built. Indeed, the entire telecommunication industry, as we know it today, was funded in the 1980s by, basically, Michael Milken's arranging financing for all of that through this instrument called junk bonds.

- 6.23** The way we repaid him for restructuring American business and creating the infrastructure for the modern internet, modern telecommunication economy, is by trying him for 99 counts of all kinds of stuff and hounding him and his family until he pleaded guilty for four or five of them. Pretty minor things, all of them pretty minor. Usually, you'd get a slap on the wrist and off you go. In this case, the judge sentenced him to 10 years in jail and basically said in the sentencing that he was making an example of him, so financiers would never do what he did again. Change the economy, make life better, make us all—anyway, there's some good books about this.
- 6.24** But in my view, he's a real hero of American finance, one of the great financiers of all time, and wrongly convicted. It's one of the reasons, I don't know if we want this on video, but it's one of the reasons that I will hate Rudy Giuliani forever. It was Giuliani's reign of terror on Wall Street during the 1980s that brought down, ultimately brought down Michael Milken and sent him to jail, destroyed also his investment bank firm called Drexel. Drexel also had to fold because the government went after it. But basically all of that.
- 6.25** And their sin, at the end of the day, their sin was facilitating this breakup of conglomerates, this reallocation of resources from the Rust Belt to Silicon Valley, this completely restructuring the US economy, the building of the infrastructure to make the Internet economy possible. That was the sin. And that pissed off a lot of people. Look, a lot of people lost because of that. But the reality is that almost all of us gained from it enormously.
- 6.26** And one guy who's still around, who's still active in financial markets, that was, made his name back then, was Carl Icahn. Carl Icahn is still around. We might talk about him when we talk about meme stocks because he shorted them recently. But Icahn was one of the takeover artists. They called him takeover artists. He would go in, take over companies, break them up, make a lot of money for himself, but for shareholders, and again, improve the efficiency and productivity of those companies in that economy.
- 6.27** Note that this is what a stock market makes possible. The reason I went into all of this is, this is one of the advantages of a stock market. It makes it possible to change control. It makes it possible to get rid of lousy managers. If the stock price goes down and I look at it and say, "Huh. Stock prices are 20. I actually think if managers were good, if they actually did the right thing, if they were productive, if they were focused, this could be a \$40 stock. I'll buy it." By whatever means, gain 51%, kick out the managers, replace them with somebody who knows what they're doing, and hopefully it goes up to 40, and I've made some money.

- 6.28** So the ability to change control, replace bad managers with new managers, with better managers, is one of the most important functions of a stock market. And one of the great tragedies, I think, of American regulation is that over, since 1968, but certainly since the 1980s, we've made that harder and harder and harder to do. So many states, this is state law, so many states now have passed laws that basically make hostile takeovers basically impossible or illegal or too expensive. And that's sad, because one of the reasons, in my view, for example, that Japan, which has been in economic malaise since the early 1990s, it had the same problem America had in the '70s.
- 6.29** It had these conglomerates that were not productive, not efficient. And there was a headline in Forbes in the mid in the late 1990s, and the headline was, "What Japan needs is Michael Milken." And it's absolutely true. What Japan needed was somebody to go in and break up the conglomerates, make them more efficient, make them more focused. And since nobody did that, nobody could do that because of the regulatory environment in Japan and the culture, which is very antagonistic to something like hostile takeovers, they never got their act together. They never restructured their businesses.
- 6.30** We had the benefit of having those people around and having a regulatory environment that allowed us in the '80s. It doesn't exist today. So do we have a lot of inefficient big companies today that are generating a lot of cash that could be broken up, that could have better managers? Probably, because the tool for getting rid of them is gone. That ability to do, I mean, you still have takeovers, but they're nowhere near what they used to be. And you don't get individuals taking over big companies. In those days, somebody like Carl Icahn, who was young, he was probably in his 30s, be my guess, 40s maybe. And he could raise money, raise enough money to take over a massive conglomerate. And he was a nobody.
- 6.31** I mean, somebody trusted him, Mike Milken, to do the job, was willing to lend him the money. But today, you couldn't imagine anybody doing that because of the regulatory environment in which we live. So one of the things that we've lost over the last 40 years is that flexibility in capital markets, that flexibility in the stock market, to be able to initiate, to create changes in control. It's still true that another company could buy this company, but hostile takeovers these days are pretty rare. They happen once in a while, but they're pretty rare.
- 6.32** And I don't think it's because we have such an efficient market that we don't need them. I think it's because it's way too expensive because of the regulatory environment. Again, regulations making things worse, not better. Finally, there's also a flip side to this. And the flip side is that good managers, if they're compensated with stock, can do very, very well, in a sense, are rewarded for how good they are. So one of the innovations that happened during the 1980s, and it's still around today, is that managers are compensated with stock.

- 6.33** Stock options in the past, they're out of favor these days, unfortunately. But stock, stock options, ways in which the managers, not just the CEO, but managers generally in the company, can benefit from the upside. You work at a startup, so startups, of course, give stock. Everybody's motivated now to get that IPO and to make a lot of money. But give stock to managers in all companies. Give them that same incentive to work to maximize shareholder wealth. Isn't that what the business is there for? It's to maximize shareholder wealth, not to mix the CEO's income, not to reduce risk. It's to maximize shareholder's wealth.
- 6.34** What's the best way to guarantee that managers will maximize shareholder wealth? Make them shareholders. What's the best way to guarantee that boards of directors work to maximize shareholder wealth? Comprise boards of directors of shareholders. Again, not favored today. Today, there's a big push to put on boards of directors, outsiders. Outside directors. What are outsiders by today's definition? Outsiders by today's definition are people who know nothing about the business, who own no shares, therefore have no incentive to maximize shareholder wealth. That's an outsider.
- 6.35** So politicians are good outsiders. Lawyers are good outsiders. Academics are great outsiders. They know nothing about the business and they own no shares. Warren Buffett sitting on your board of directors? Not so good because he owns a lot of shares. Now, in my view, that's the best director possible. Why? Because his incentives are aligned with the other shareholders. And what is a board there for? It's supposed to represent shareholders. So what you want the board to be are shareholders.
- 6.36** Now, they can be outsiders in a sense that they don't get income from the company. They're not beholden to the CEO. They're not his golfing buddies. That's good. But you want them to be shareholders. You want them to be invested in the company. You want them to believe in the company. And you want them to be incentivized to do what's good for the people they represent, which are shareholders. Sadly, again, for a variety of regulatory and other reasons, but also cultural reasons, that is out. That is out. You're supposed to have boards of directors that are driven by all kinds of goals, whether they're DEI goals or whether they're just this idea of outsiders or somehow being good.
- 6.37** Disinterested people. We love disinterested people, to represent our interests. That's bizarre, right? I want people representing my interests who are interested in me, have the same incentive as I do. If I'm a shareholder, I want them to be shareholders. Indeed, I would want my board of directors to literally take money out of their pockets and buy shares. If you want to sit on the board, I want a significant portion, a decent portion, let's say, of your net worth in this company. That would make a motivated board of directors because they know, if the company fails, they suffer. If the company succeeds, they benefit enormously. That's a great board of people.

- 6.38 Very few companies do that. Very few companies actually have that. But that's a great way, again; another benefit of the stock market is, you have this tool to provide these incentives to managers and to board members, to make sure that they are rewarded for doing good. And doing good in this case is maximizing share price. All right. So the primary function of a stock market, as we said, is to raise capital for companies, whether it's in the initial public offering or later on, when they want to grow, they raise capital in that way.
- 6.39 But all these secondary functions provide us with an opportunity to save and participate, provide for these changes in control, provide information; all these others, of course, make that first possible. And make, generally, the market and the economy far more productive, far more efficient, and significantly better. All right, so let's see how all this happens, and to really see how all this happens, we really need to talk about how to think about stock prices. And really, this is not just about stock prices. We should think about how to value any financial asset.
- 6.40 What we're going to talk about is really the way to value stocks, but it's also the way to value bonds, and it's also the way to value real estate. It's really the way to value anything. Now, we're not going to get into the technicalities and the deep equations here. I'm just going to put up one simple equation because that's all you really need at this level, at least. So the price today, what's going to determine that price? What do I get from owning the stock? Well, I get some cash flow. That cash flow could be dividends, but it could really be, if the company doesn't give me the dividend, but instead reinvests it, then it's reinvesting to produce a greater return.
- 6.41 So what I really care about is not the dividends. What I really care about is how much cash does the company have at the end, in a sense, of every year, every quota, or whatever it is? The more cash it has, the more opportunities it has to invest, or the more opportunities it has to give me the cash back, whether through a dividend or capital gains. So this is true. If you think about it, if I buy a house and I rent it out, what is the value of this rent? Well, it's how much I get every month over the many years that I'm going to own this house, discount it back, based on some interest rate, which we'll talk about, discount it back to today's price.
- 6.42 And then I get a price on that. And if I spent more on the house, plus the value of the house when I sell it, I want to discount that back. And hopefully, what I paid for the house today is less than what this cumulative, all these cash flows and the value of the house in the future, is. Then I've made good money. And you should include, by the way, in the cash flows, subtract from the rent, all the maintenance, if you have a mortgage, the mortgage payment. So your net, your net cash. So this is true of any financial asset. Price is going to be the sum, from 1 to some n , over whatever period of time.
- 6.43 Of all the cash flows, and cash flows here mean net cash flows. This is after all expenses, after everything is paid out, divided by 1 plus a rate of return—which is what

rate of return I expect on this investment; we'll get to that in a minute—to the power of n . I take, as we talked about, money tomorrow, money a year from now, is not the same as money today. I have to discount it back. I have to take into account the fact that I could have done something else with the money. I could have used it, I could have spent it, I could have invested in something else—my opportunity cost.

- 6.44** So this r reflects a real rate of return, a time up for money. We'll call this i . It's the interest rate. It's the time value of money. It has to reflect inflation. And it has to reflect—what am I missing? Risk. It has to reflect risk. Now, it's not exactly that, but we'll get to some other considerations in a minute. But here's the, I think, important point. If you buy a bond, let's say, government bond, you're buying something that's going to pay you 5% every year for 10 years. And at the end, you get the principal back. It's a \$1,000 bond. You get 10% every year, and then you get the \$1,000 or 5% every year, and then you get the \$1,000 at the end.
- 6.45** What of these variables do we know for sure? We know the cash flows. This we know. It's \$100. It's a 5% of the \$1,000 every single year, and then it's \$1,000 at the end of the 10 years. So the cash flows, we know for sure. What don't we know? We'll put it this way. What might we disagree about? We might disagree about this all. Now, what's the risk of a government bond? Zero. It's relatively easy because risk is taken out. But we might not agree about what inflation is going to be over the next 10 years. We might disagree about that.
- 6.46** So even when it comes to a bond, which has no risk, where we know exactly what the payments are going to be, we might disagree because our r , our required rate of return, what we require to get on this investment, might be different because we might have different expectations with regard to inflation. Indeed, our i might be a little different. Our time preferences might be a little different. We usually can get i . This is the real rate of return from the marketplace. But the reality is, it might not be enough for us as individuals. The market might determine i as x . Usually, real rate of return is somewhere between 1 and 2%.
- 6.47** By the way, in the early 20-teens, it was negative, because the Fed held interest rates so low that this was actually negative, which nobody really thinks that this was negative. But bonds were priced as if this was negative, which is bizarre. Another distortion and perversion that you get from central bankers. So you can see that even when it comes—even when we know what the cash flows are, there's going to be some disagreements or there's going to be trading. I'm going to trade these bonds.
- 6.48** And we're not all going to agree on the price because some of us have different expectations about inflation. Some of us indeed have different expectations just about our time value of money. Not going to be the same. Now, let's take a stock. Do we know the cash flows? No. We're each going to come up with a different expectation with regards to cash flows. We're not going to agree on what the cash flows are going

to be. I mean, is the Macintosh going to do well in the future or isn't it? I'm sitting in 1984 when that ad comes out.

- 6.49** I don't know if you've ever seen the first original ad for the Macintosh. Revolutionary advertising, just that ad changed advertising, I think. It was stunning, right? And if you haven't seen it, check it out on YouTube. A 1984 ad for the Mac, and of course, it's 1984, which is important because it's Orwell, so you can see the connection when you watch the ad. We don't know how many units it's going to sell, but we can all make estimates about that and plug them into the model, and we're not going to agree. It's going to be unusual for all of us to agree on what those estimates are going to be.
- 6.50** If you're going to do it, we'll probably put different probabilities and different outcomes. We'll weigh them differently. So these cash flows are going to be very—it's going to be a wider rate. Are we going to agree on the r ? Well, even if we can agree on the i , and even if we're going to agree on inflation, are we going to agree on the degree of risk that this is going to involve? Probably not. So this price is— we're all going to come up with a different p . And depending on how confident we are in our projections, we might be willing to buy if the price is below what the market—if the price is above, our price, is above what the market thinks it is, or we might sell if it comes out below.
- 6.51** And you can see how this generates trading. And as I said before, the more confident somebody might be, the more his views are going to be reflected in the price; the more he's going to move the price and make it reflective of his views. People who are particularly successful at this are going to do better than people who are not very good at this. So this is a very simplistic, very basic model of finance, but it's really how things should be priced. Whether they are or not is a different question, but this is how everything should indeed be priced.
- 6.52** This is very different—and I think, for the most part, this is what drives changes. If cash flows suddenly fall off the cliff, if suddenly we get very pessimistic about the future of the company, what happens to price? It goes down. We know that. That's the relationship. If risk suddenly increases, they are the competitors. Cash flows haven't been touched yet, but they're competitors. The risk that cash flows might go down goes up. This goes up. It all goes up, price goes down, and of course, everything in reverse as well.
- 6.53** So we know that this impacts it in these kind of relationships. The higher the expected return, the lower the price is going to be. I'm willing to pay a lower price. The higher the cash flow is, the higher the price is going to be. So we can see the relationships, and those relationships exist in the market, and we can see them play out on a pretty regular basis—not always, but on a pretty regular basis. On the other hand, this is what John Keynes, in the General Theory of Employment, Interest, and Money, in 1936, wrote about the stock market. Just again, to give you a sense of how people think. This is considered by many, not by me, but by many one of the great economists of the 20th century.

- 6.54** "When the capital development of a country becomes a byproduct of the activities of a casino, the job is likely to be ill-done. The measure of success attained by Wall Street cannot be claimed as one of the outstanding triumphs of laissez-faire capitalism." So for Keynes, the stock market is a casino. The price in the market has nothing to do with any of this. And of course, if it's a casino, then everything we've talked about so far is moot, because it's not a good gauge of information. It doesn't actually allow for control changes because the price doesn't reflect anything. It doesn't compensate good managers and penalize bad managers. None of the things that we argued work if it's a casino.
- 6.55** But for the most part, it's not. We'll get to the aspects of it, maybe, where it is a kind of a casino. Now, I'll just say that this expected return, this r , once you start thinking about your investment not in terms of just investing in this one stock—once you're thinking about your investment in the context of a portfolio, in a context of all the investments you're making, all the other stocks you're in, all the other bonds, real estate, whatever you're in, then the way to think about this becomes not in terms of these components. But the way to think about this is, what is the expected return of this in comparison to everything else that I own?
- 6.56** And a lot of times you'll hear the term beta. You know beta? The beta of a stock. And the beta of the stock is basically, how sensitive is this stock to movements in other stocks of the market, generally, often? If a stock is very sensitive, then we would say it's riskier, and therefore, my expected return should be higher. If the beta is lower, that is, let's say it's 0.5, so for every movement in the market, it only moves half that movement, then we'd say the risk is lower and r is lower. Now, this is a particular interpretation of risk. We'll talk a little bit about that later.
- 6.57** But it's important to note that the way we think about all this changes when we start thinking about a portfolio, and we're less concerned about a particular company, and we're more concerned about the relationship between the different asset prices. Most investors, 99% of investors, don't ever look at cash flows. I mean, how many of you have bought stocks based on this? You don't. Most of you invest in a mutual fund, and the mutual fund is diversified, and they diversify, and most of them are not doing this. Somebody is. And that's how the information gets into the stock price, because somebody is doing that work.
- 6.58** We often hear criticism of speculators, speculators in the marketplace, people who are buying and selling. Well, people who are speculators are buying and selling based on the information. They're the ones who actually embody the stock with this information; it's not us. It's not even the mutual fund, certainly not the ETF. But it's active investors, investors who actually do the research, investors who actually have an opinion, investors who actually care about a particular stock—are the ones that embed the stock price with the information, with knowledge about what's going on. The rest of us are passive.

- 6.59** So speculators are heroes of the stock market, not the bad guys of the stock market. Not only are they providing us with liquidity, so for when we want to sell, somebody is buying, but they're also the ones that are constantly buying and selling based on their information, their knowledge, what they believe about this stuff. And we're talking about professional speculators, not the amateurs, which we'll get to in a little bit. You count on the professionals to make the market efficient, to the extent that it is efficient. So there are opposing views on the values represented in this equation. Embed prices with information. Embed prices with information.
- 6.60** And prices are very sensitive to information. I mean, study after study after study shows this. I did some of those studies a long, long time ago. You can see that, as soon as something changes in the world out there that affects either the cash flow or the risk, prices change. And they change sometimes very, very fast. Now, who's doing that? Most of us are not doing that. We're just sitting on our mutual funds. And it's even the mutual fund probably not doing it because most mutual funds are just diversified. They don't actually follow any particular stock. It's somebody doing it. And those somebodies are usually, we might call them speculators.
- 6.61** They might be hedge funds, they might be investment banks, they might be somebody who is actively engaged in trading stock. So prices are very, very important, as we said, for the allocation of capital and everything else. And in that sense, traders, speculators, people who are actually engaged in figuring this out, are crucially important economically, crucially important to the marketplace. OK, one other fallacy that I think exists out there in the world about stocks—there's this notion that stocks reflect short term and not the long term. And there's a sudden truth to that, because the reality is that the further away this cash flow is, the less reflected in the price it is.
- 6.62** Because we're discounting it over many more periods. That just makes sense. \$100 a year from now is worth less than \$100 today, but \$100 ten years from now is worth less than the one year and much less than today. So the cash flow, when I sell the house way down there, is worth a lot less today because I'm discounting it over all those years. But there's another sense in which all those future cash flows are counted in the price today. Not as much, the further out you go, but they still matter. They still count. So all stock prices reflect long-term cash flows.
- 6.63** Certainly not quarterly. You can't value this based on just one quarter. It wouldn't mean anything. And if everybody was valuing everything just based on quarterly, some smart investor would see that as a great opportunity to look long-term and buy and sell accordingly, and make money off of the people who think he's short-term. And he makes so much money that he would drive their prices to his evaluations of a long term. Because the long term actually happens. The long term, at some point, there's realization. It's not just a theory. The long term is not just theoretical, it's real.
- 6.64** So it's a fallacy that we hear over and over and over again, that the stock market drives people to be short-term, only think about next quarter, only think about short-term

effects. And you can see this, by the way, just simply by the very fact that, I don't know, we get IPOs constantly or you get venture capital investment constantly in companies like biotech. How long does it take a biotech company to make money, if it ever makes money? Decades. It can be easily decades, more than a decade.

- 6.65** Take Moderna. Remember Moderna? Probably remember it because it was a producer of the vaccine. And I know this is controversial. Put aside the vaccine nonsense. Moderna came up with this great idea of using mRNA technology in order to deliver medication, induce the immune system of the body to heal itself in essence. And for years, they'd raise money. I think they went public based on the expectation that they would be able to use mRNA technology to kill cancer. And they tried. And over 10 years, they tried. They never made a dollar. Not a dollar. Money kept flowing because the promise was great. And if you could cure cancer, you're going to be very, very rich. But they never made any money until Covid hit.
- 6.66** And suddenly, the technology had an application, and it was a relatively simple application; it wasn't that hard. So they could devise it very quickly, produce it very quickly, get it out there, and make a fortune. And suddenly, the shareholders of Moderna made a lot of money. And the beauty of it is, is, by making a lot of money, it means they have a lot of capital. And Moderna has been spending that capital over the last three years, four years, in massive investments in vaccines against cancer. So the likelihood now that we will actually get a treatment for Moderna against cancer is much higher because of Covid.
- 6.67** Because so much capital now was internally generated through profits that can be invested in actual cancer treatment. And there's a bunch of stage 2 and stage 3 trials going on using Moderna. Moderna, and BioNTech is the other company that did another vaccine. So the Covid vaccine has this other effect of spurring innovation in biotech beyond itself. But the idea is here that Moderna for years and years and years had no income, and yet it had some value in the marketplace because there was always that potential.
- 6.68** And I think that illustrates how long-term investors are. They're not short-term. They're actually quite long-term. Much longer, I think, than government bureaucrats who get very impatient very quickly with these kind of investments. So no, stock markets are long-term. Investors are long-term. There are times when they become shorter-term. When do you become shorter-term? If this goes up, then you're valuing cash flows now much more than cash flows in the future. The lower your expected return is, the more you value in the future versus the present.
- 6.69** So in times of great uncertainty, when there's risk, let's say economic risk, not just risk to this company, but economic risk, we become more short-term. When inflation is higher, if this goes up, we become more short-term. If there's uncertainty about regulations, that increases risk. We become more short-term. So you can incentivize us. That is, there are external incentives to be long-term or short-term. One attempt that

government has got for investors to be more long-term rather than short-term is they're taxed differently. If you sell a stock within a year of buying it, you pay a much higher rate on capital gains than if you sell it after a year and one day, which is a massive market distortion.

- 6.70** Because as we said, speculators who are buying and selling constantly, embed prices with information. Well, they're not trading as much because they're being taxed. They'd rather trade less and be taxed less. So it's massively distorting of the market for somebody to decide what short term is and what long term is; and we don't like short term and we like long term. But speculators are short-term investors who are thinking long-term. They have to. That's how that's embedded in the cash flow. It's not just tomorrow's cash flow. It's all cash flows. And they wouldn't be able to make money speculating if they didn't think about the long term, if they didn't take that into account.
- 6.71** So any attempt to manipulate people into being more long-term or more short-term than some philosopher-king or some bureaucrat believes we should be is distortive and destructive to the actual activity of markets. There's this question out there of, are markets, are stock markets, efficient? And what do we mean by efficient? What does efficiency mean? Since stock prices reflect information, information about cash flows and risk, then an efficient market is a market in which prices reflect all available information.
- 6.72** And that means that unless you have new information, unless you know something nobody else knows, you can never really make excessive returns on the market. You can never make more than what the market is already expecting. There's nothing on the table. There's nothing left. The price, this price, the market is already reflecting all the information available. And if you don't know anything the market doesn't know, you're not going to do better than this p. This is the price. You buy it and you'll get the expected return, but you're not going to get anything more than that. There's no profit above and beyond the market return.
- 6.73** There's a joke that's told. A lot of this idea of efficient markets comes out of the University of Chicago, big on efficient markets. All the big theorists in finance have come out—many of them have come out of Chicago and promoted this idea of efficient market. And the joke is that a student is walking along with his finance professor in the street. They're talking. And the student says, "Oh, look, there's a \$100 bill on the pavement." And the professor says, "There can't be. If there was, somebody would have picked it up already."
- 6.74** It's efficient, right? There's just no \$100 bills on the pavement. They don't exist. Now, of course, what this is missing is, what? But somebody has to pick it up. Somebody has to trade on the information that is available. And since information is a constantly moving target, somebody has to constantly make money by being ahead of everybody else and by incorporating the best information into the price. Somebody had to pick up the \$50

bill. And as I said, the speculators, the hedge funds, the various types of professional investors who do this for a living, day in, day out, they're the ones picking up the \$50, \$100 bills.

- 6.75** And it's very, very, very unlikely that any one of us is going to find \$100 bill on the pavement because those guys are constantly scanning the pavements to pick them up. And what this suggests is, don't be a day trader. Don't pretend like you know more than people making seven figures trading stocks, or have spent their lives analyzing companies and knowing everything about them. And you can, as an amateur, just figure out patterns or figure out information. By the time you think it through, it's already done. Somebody's already picked up that \$100 bill.
- 6.76** And for the most part, that's what it means. That's what it means to be efficient. And for the most part, over time, markets are efficient, but somebody has to do the work to make them efficient. If the work is not done to make them efficient, they will not be automatically efficient. Somebody has to be trading based on the information. And indeed, that is, for the most part, what happens. But there can be occasions where this p completely is divorced from cash flow and risk. And really, we saw this in maybe the most dramatic fashion during the meme stock phenomena of, what was it, 2021, January, or 2022?
- 6.77** Anyway, what happened during this point is, today, for a variety of reasons, both regulatory and technology, it's almost free to trade. You can go on and trade with no transaction cost, and anybody can go trade. In the past, trading was expensive or trading was difficult. You had to call your broker or whatever. But we've made it so it's easy. Everybody can do it, and they can do it like that. If you have an account at Robin Hood, Robin Hood is an online broker, you can basically trade at no cost. They get paid elsewhere. They get paid from the market makers who they funnel all the transactions through. But you, as a buyer or seller, don't pay any cost.
- 6.78** During this period, January 2021, I think, somebody on Reddit basically decided that GameStop, that store that sells video games, was undervalued. Now, GameStop had been declining for a long time because while GameStop had once been a fairly good stock because they had—gaming is huge, right? And they would rent out games, sell games in their stores. At some point, the market realized, efficient markets, information, right, that a lot of the games were moving online. They were moving to the cloud. You didn't need to go buy a game. You could just download a game. You just play it in the cloud, play it online.
- 6.79** Indeed, the number of these, number of purchases and number of renting was going down, and the stock price reflected that, reflected the best available information. The idea was that GameStop somehow could morph itself into an online gaming thing and compete with the big platforms. But that was hugely risky, hugely improbable. Anyway, somebody on Reddit decided that this stock was undervalued and was valued at a lot more than what it was.

- 6.80** They also decided, for whatever reason, that AMC, the movie theater business—now, again, movie theater business, it's an OK business. For a while, it was very highly valued. But then came Netflix and Prime and all the others. And that business became less profitable because people stopped going to the movie theater. They'd rather stay home and watch a movie on Netflix. So the long-term prospects of AMC are not great. They're not zero because, once in a while, it's cool to go out and to see a movie on a big screen, particularly if it's a big spectacle. But generally, people are staying home more and watching on Netflix or watching on their phone, which is even more ridiculous, right? Watching movies on their phone.
- 6.81** So the same people who said GameStop was undervalued said AMC was undervalued, then Bed Bath and Beyond was undervalued, and a bunch of other old-line companies. They said, "Oh, no, they've underestimated how much upside there is in these companies." And within a few days, these stocks just went like that. Like, a bunch of young people—there's a movie out, I think, recently about this phenomena, which I refuse to go see because it makes them out to be the good guys. But a bunch of these kids went out there and just bought shares on Robin Hood and kept hyping it on Reddit, so more people would come in and buy shares, and it just went through the roof.
- 6.82** And the valuations were absurd. And the market in these shares was completely inefficient. It did not reflect the best available information. It reflected the dumbest available information. And it even got a certain philosophical excitement because what they were doing was, they were basically going after Wall Street. They were proving the Wall Street model didn't work. They were proving these companies didn't know what they were doing because they valued GameStop at 10 and GameStop went up to 400. And that was the real value.
- 6.83** And then there were short-sellers. There was a number of companies that were short-selling, particularly one that had shorted these stocks. Bet that the stocks would continue to go down, which makes complete sense because these are dead, I mean, Bed Bath and Beyond since then has gone bankrupt. It's gone to zero. So they expected all these companies ultimately to go to zero, and my guess is they all would have. They were going to crush the short-sellers, which they did. One of those short-sellers went out of business.
- 6.84** By the way, when you crush a short-seller, who loses the money? Well, the owners of the hedge fund lose some money, the short-selling hedge fund. But it's pension plans, insurance companies. It's your parents' pension that's really taking a hit. Not that you should care, but that's the reality. Anyway, very inefficient. And indeed, once it got to that high, it started coming down, but it took a while to come down, because it's much easier to push a stock up than to push a stock down. It's much easier to buy a stock than to short the stock.
- 6.85** Everybody know what? Well, let me tell you what a short is because people on the video probably don't know, or some people don't. Shorting is when you believe a stock

will go down. What you can do is you can go to a broker and say, "I want to borrow the stock." So you borrow it from them. Let's say it's trading at \$20. And you borrow the stock from them and you sell it. You get \$20 you put in your pocket. And then what you're hoping is the stock goes down, let's say to \$10. You buy it back for \$10. So now you've only made \$10 in your pocket. And you give it back to the person you borrowed it from. So you're up 10 bucks.

- 6.86** Why is this so—so people short stocks when they think they're going to go down; they short them. The most you could make in a short is the stock goes to zero, you made the 20 bucks. Can't make more than 20 bucks. How much can you lose when you short a stock? An infinite amount. If you short a stock, and the stock price goes up, there's no limit to how high it can go. Now, when you buy it back, you can lose a lot more than that original 20 bucks. You can lose everything. It's exactly what happened to these short-sellers. At some point, you have to pay back.
- 6.87** There's a lot of things that make short-selling very difficult. One is the risk. Much more risk than buying a stock. Second, there's a bunch of regulations that tell you under what conditions you can short and what conditions you can't short that make it more complicated. There are margin calls. There's leverage involved here, and the banks can demand that you, in a sense, buy back the stock when you don't want to, in conditions that are not very good for you. And it's just expensive and difficult.
- 6.88** Generally, stocks and markets are much more likely to be overpriced than underpriced. Underpriced is easy to fix. People buy, push the price up. Overpriced, difficult to fix. People have to short; short's expensive. Shorting is difficult, much more difficult to actually do. When the meme stocks went up like they did, crazy. It just happened again a few weeks ago. We're now in 2024, and it happened again. This time it corrected very, very quickly. But at the time, it didn't correct quickly, it corrected slowly.
- 6.89** Partially because everybody saw the guys who were shorting got killed, they don't want to get caught again. And partially because, funnily enough, if the stock price goes up a lot, then what happens? These companies then issue new shares, and they raise capital with these incredible valuations, and then they can use that capital to survive a little longer, which is what happened with AMC and I think GameStop. So, to the extent that stocks can be manipulated by groups of people all rushing in and changing the price, they're obviously not efficient.
- 6.90** And there are probably lots of other circumstances in which people are trading, investors are trading, markets are trading in ways that do not reflect the best available information and the best available prices, particularly in smaller stocks, particularly in less followed stocks, where people are doing less research and people are less focused on because they can't deploy enough capital to make a difference if you're a professional. So there certainly are parts of the financial market that are not efficient, parts of the stock market that are not efficient.

- 6.91** By the way, we talked about Carl Icahn. One of the things Carl Icahn did, it's his own capital that he had. He actually shorted the meme stocks when they went like this and made a lot of money as they came down, a lot of money as they came down. But it's risky. It's risky. Imagine shorting a stock at 20, like we gave the example before, and it going up to 250. You've just lost \$200 on a \$20 investment, on a \$20 short. That is painful. I can tell you because I've done it. It's happened to me, and it's painful.
- 6.92** So these meme stock phenomena in the, in the old days, in the '90s and early 2000s, there were bulletin boards where people would spread rumors about stocks and people would jump in and buy and sell based on those rumors. During the dot-com bubble years, where a lot of internet stocks were hyped, they became irrational in the sense of their valuations. The market corrected for that in 2000. But these phenomena do happen. You can indeed get bubbles under certain circumstances. You can get these irrational phenomena where stocks just go completely out of whack.
- 6.93** Of course, one of the ways in which most of us as investors take care of that is being widely diversified, which reduces our risk of being exposed to any one of these crazy phenomena, up or down, and basically not trying to beat the market, but just trying to benefit from the fact that, for the most part, in the stock market at least, markets have gone up systematically over many, many, many decades. If you hold stocks for the long run, you tend to make money. No guarantee about the future, but as long as the economy is growing, as long as the company is making money, then stock prices will go up, to the extent that they reflect that.
- 6.94** At the end of the day, the more efficient a market is, the better capital is allocated in the economy. The more the price reflects the best available information, the more efficient capital is allocated in the economy. Meme stocks are horrible. Not only because some people lose money who shouldn't, but they're horrible because they distort the allocation of capital in the economy.
- 6.95** GameStop and AMC should not have raised capital. Capital should not be available to them. The only reason it was, because of this artificial, crazy, emotionalistic group of people doing this over quite an extensive period of time. A lot of people, by the way, lost a lot of money. The meme stock kids, a lot of them lost a lot of money. You'll never hear the stories about this. But when the stock came down, they lose a lot of money. Questions?
- 6.96** Throughout history, there have been a lot of bubbles, and it's been proven time and time again that short-sighted thinking doesn't really work. And yet it seems people keep falling into the trap. So generally speaking, what do you think causes bubbles and what attracts people to that kind of short-sighted investing?
- 6.97** I think the main cause of bubbles is the government; it's monetary policy. If you look at bubbles, almost every single one of them, maybe every one of them, they are always preceded by artificially low interest rates. They're always preceded by a period of what

people call easy money, which is easy to get your hands on loans and money, and people are using leverage to buy more and more stocks.

- 6.98** In 1996, I think it was, Alan Greenspan said he thought there was an irrational exuberance in the market. But he did nothing about it. Indeed, interest rates went down quite a bit during the later part of the 1990s. Now, he thought 1996 was irrational exuberance. I don't know what he thought about '98 and '99 when the Nasdaq went up, what, 90% one year or something ridiculous like that. I mean, it was just insane. But he was keeping interest rates very, very low. This I was very low. The inflation expectations were nonexistent. And the perception was that nothing was risky anymore.
- 6.99** And indeed, Alan Greenspan encouraged that perception. There was something called the Greenspan put. The Greenspan put was this idea that if ever the stock market went down, Alan Greenspan would save it. It would provide liquidity in order to force it back up. And this happened in 1987. He was just chairman of the Fed, just had been chairman of the Fed for a few months. And on Black Monday, I think it was Black Monday, the stock market went down 25%, 25 % in one day. Alan Greenspan's response was, "Flood the market with money." Interest rate's down, flood the market with money.
- 6.100** And of course, the market recovered. Took it a while. I think this was in October. And by the end of the year, it had mostly recovered. It's not clear why it would go down 25%. Maybe it had to do with the Treasury Secretary going on television, saying he wanted to make the dollar cheaper. Maybe it had to do with other things as well. There was new information. If the dollar is going to be cheaper, that is important information about interest and inflation and the value of purchasing power. But the increase was basically funded by Greenspan.
- 6.101** If you look at the dot-com bubble, it was before really low interest rates. If you look at the bubble in real estate prices in the 2000s, that led to the great financial crisis of 2008, Greenspan engineered a massive decline in interest rates because of 9/11 and because of the dot-com bubble crash in 2001 and 2002. Basically for that, much of that period, I think, two and a half years, interest rates were negative, real interest rates were negative. Money funneled into the housing for a variety of reasons, went to housing and not into the stock market. But low interest rates always, seem to always precede bubbles.
- 6.102** One wonders about the world, where did low interest rates in the 20-teens, where are the bubbles? There are a variety of reasons why they might have not burst yet, or we might not be having bubbles because there's, some of that money that came in was taken out through other measures, through bank reserves, which increased during that period. So I think the main cause is cheap money, easy money, combined with some change that's happening, some incentive to get people revved up.

- 6.103** So there was clearly a bubble in railroad stocks because it was a new technology, and everybody was really, really excited, and everybody rushed in, and they drove prices up and it quickly corrected down. It makes sense. And there was a lot of investment in that. And the same happened, the same happens every time there's a new industry, there's a lot of investment. I think that, so you combine the cheap money in the 1990s with the Internet. A lot of excitement about the Internet. Everybody wanted a piece of it. Everybody wanted to be in. And in the 2000, incentives to buy homes that come from the government, combined with cheap money, drove home prices up.
- 6.104** So it's always this combination of something exciting with cheap money. That creates what we call bubbles. And bubbles, this accelerated increase in asset prices, followed by a decline. It's very hard to spot a bubble while it's happening. It's very easy to spot it after the fact, but it's very hard to spot it while it's happening, because while it's happening, it could be just things are really good. How do you differentiate between that and a bubble?
- 6.105** It occurs to me that market efficiency is one of the goals. The insiders in a company, insider trading, is one way to increase market efficiency. What are your opinions on insider trading or some form of insider trading?
- 6.106** No, that's a great point. Absolutely. The reality is that the more information is in prices, the more information is embedded in prices, the better. And who has the most information about a company? Insiders. So to the extent that we can get the information that insiders into the stock price, capital allocation will be better. My view of insider trading as a consequence is that the government should not be prosecuting it. It should be a contractual issue. Let's say you're a company and you don't want, for whatever reason, insiders to trade. Then when you hire them, put it in the contract and allow trade on inside information or trade however you want them to do.
- 6.107** So exchanges might say, "If you want to list on our exchange, we allow inside information or we don't allow inside information. And let's compete." Let's see what happens if you have competition. So I think this should be something that is left to the market, not something dictated by government. At the end of the day, you want a lot of the information that today is considered insider to be reflected in prices. And I don't think there are any losers from that. If you don't have information and you trade with somebody who has information, so what? You're not losing. You don't have the information. That's a fact of reality.
- 6.108** According to our standard of fairness, which everybody is the same, it's not, quote, fair, but that standard of fairness is ridiculous and absurd. Note, too, that inside information is a completely ambiguous term. Sometimes it's clear-cut. A CEO knows the company is sometimes clear-cut. But other times, I don't know, you notice a piece of paper, and the piece of paper happens to be some legal document that says something, and you don't have any fiduciary duty, responsibility to shareholders or anybody else in the

company, and you trade on whatever you read in that piece of paper that maybe is in the trash somewhere. Is that a crime?

- 6.109** Well, the courts disagree, and sometimes it is, sometimes it isn't, depending on the circumstances. Insider trading is a very, very, very ambiguous term. The meaning of it changes over time, depending on who is interpreting it at any given point in time. I would like to see that whole thing disappear. I think markets would be more efficient, and we can figure out, based on market parameters, what is the best way to deal with the fact that some people have more information than others.
- 6.110** I mean, the whole point of markets is to have better information. The whole incentive is to have better information about cash flows and have better information about risk. We should be incentivizing people to get information as long as they're doing it legally. You can't break into somebody's office and steal information. That's not a problem of inside information. It's a problem of stealing. You can't hack their computers to figure out what's going on. The crime there is the hacking, not the information.
- 6.111** So as long as they do it within the bounds of the law, you want to encourage research. You want to encourage thinking. And sometimes people do the research and thinking, and then they say, "Oh, but you have more information than the rest of people." That's wrong. It's an egalitarian standard that is completely corrupt and wrong. People have more information than others. That's what makes markets work. We disagree. We disagree based on information, based on knowledge, based on ability. And trying to force us to be equal is distorted and perverted.
- 6.112** Yes, markets would be a lot more efficient if we allowed for inside information. And I think if we freed up the markets from these laws, the market would find the right equilibrium between not doing things that are really stupid, like a CEO should not be able to short your stock and then drive the company bankrupt and make money. That's clearly wrong. But no, everybody thinks that's wrong. You don't need insider trading laws to know that that's wrong. That's a violation of fiduciary responsibility.
- 6.113** But why shouldn't he buy his stock if he knows things are going well? Why shouldn't he be rewarded for the fact that things are going well? Now he's not allowed to. So, the way, the way stock purchases of insiders happen, they've only got one day, like, in a quarter to do it. It has to be away from earnings announcements, anything like that. So it has to be with blindfolds on. But that's exactly, denies the market, the information that they obviously have. All right? Thank you.

Lecture 7: Risk and Return

At a glance — Risk and return.

- 7.1 Underlying everything we've been talking about, all these financial assets and markets and how they work and the pricing model that we talked about earlier, underlying all of this is this concept of risk. The future is unknowable. The future is uncertain. We don't know what it holds. And risk really is the recognition, the identification of the fact that we don't know what the outcome is going to be. Now, we don't really worry about outcomes that are going to be better than expected. Risk really represents those outcomes that are worse than expected.
- 7.2 So risk is a recognition of the downside, the recognition of the fact that things don't always work out the way we would like them to work out, whether that is true in every aspect of our lives, and that can be true certainly in the world of finance. But what's interesting is that finance as a field has not only developed tools to deal with risk associated with finance itself, with investments, but also developed tools to deal with risks in life. So there are lots of potential losses that we all could incur in life.
- 7.3 There is the, we could die. Now, we don't lose anything when we die. I know this is a shock because you're dead, so you don't know you've lost anything. You're just dead, right? But other people around you who might depend on you, your children, your wife, are losing, certainly losing financially. And as a consequence, things like life insurance were developed in order to be able to protect the risk of death, the risk of loss, to those who stay alive, not to the one who's dead. Can't fix that.
- 7.4 You get fire insurance because maybe things will burn down, something will burn down, and that's a loss. And you have to recoup, you have to rebuild. We have earthquake insurance. We have all types of insurance. We'll talk more about insurance in a minute, that protect us from these catastrophic losses, these one-time shocks to the system. And so that is one of the ways in which we protect from loss in life more broadly.
- 7.5 Financial industry also has a wide variety of ways in which to basically protect us from loss or to reduce our losses, to reduce the pain from loss in the financial world itself, from stocks going down, market collapsing, and of course, events happening outside of our control. So risk is the risk of downturn, of a downturn, of something negative happening. Now, it's interesting that the way we measure risk, mostly, not exclusively —there's a variety of different measures of risk—but in finance, the main way in which we measure risk for financial products is with volatility, with this idea of variance from expectation.
- 7.6 What's interesting about that, but there's no easy solution to it, but you should recognize it because when people talk about the risk in finance, this is what they're often talking about, is that it's symmetrical. That is, variability around an expected

value could be negative or it could be positive. A lot of times when people in finance are talking about risk, a big chunk of that is about doing better than expected, which shouldn't really be called risk. But because we use a term that involves variability, volatility, that's up and down. It's not just up.

- 7.7** As we saw from the equation before, risk is something we want to be compensated for. If we take risk, we want to be rewarded for taking that risk. Otherwise, why take it? So if I have two alternative investments, and let's say they both have the same expected return, but one is less likely to have that expected return, and one is more likely to have that expected return, which one am I going to invest in? The one that's more likely, the one that has less risk.
- 7.8** Now, of course, what happens if everybody invests in the one that's more likely and nobody invests in the one that's less likely? Then the price of the one that's—the expected return on the one that's less likely will rise. The price will go down until it's higher than the one that's more likely, because only then will people start buying it. I need to be compensated for the risk that I'm taking on. Otherwise, I'll invest in something else with less risk if the expected return is the same. The way we get compensated for taking on more risk is that our expected return, and note, our expected return, is higher.
- 7.9** That doesn't mean the actual return will be higher. Expectations don't always materialize. On average, over time, you'd expect your expectations to be realized. Otherwise, you're going to change your expectations, hopefully. You're going to change the whole algorithm by which you create those expectations because you're wrong so often. But on any given point in time, any given day, your expectations might not be realized, and therefore, you might earn less or you might even lose and not get your expected return.
- 7.10** The more risk, the more the expected return that you anticipate. The lower the risk, I don't know, like, you put money in a bank, it's insured by deposit insurance. You can take it out anytime you want. What's the risk of that money disappearing? Or what's your expected return? Therefore, because the risk is basically zero, what's your expected return on that money? Pretty much zero, maybe even negative. It's very low because there's no risk. And you're getting a certain service from the bank. We talked about that. You're getting a certain service by them keeping your money and giving you a new debit card and all of that.
- 7.11** The safer the investment, checking account being the safest, let's say, cash being the safest. What's the return on cash? Really negative if you count inflation. That's why you're not encouraged to put money in the mattress. Because money in the mattress, the return is—it's safe, but the return is negative. In a checking account, maybe you'll get a small interest return, maybe it will compensate you for inflation, maybe, but it's going to be very small. A bond, if it's a government bond, you're pretty much

guaranteed to get the promised cash flow, but you don't know what inflation will be, and you don't know if you might need the money early.

- 7.12** You don't know what the price of the bond will be exactly. So you can't say the risk is zero, but it's low. A corporate bond, now it's a corporation promising to pay you over the next 10 years a certain interest. The cash flows are known, but are they certain? No. Now there's a risk. Company could go bankrupt. Maybe if it's a triple A Apple, the likelihood that Apple goes bankrupt in the next year, in 10 years, pretty low, maybe even lower than the US government going bankrupt in the next 10 years. Not sure. Pretty low. So risk is low on a bond from Apple.
- 7.13** Because remember, you're going to get the money from the bond as long as they don't go bankrupt. Even if profits tumble, even if they don't do well, the bond will still pay you as long as they have enough money to pay you, which means as long as they're not bankrupt. There are what's called high-yield bonds. Why would they be high-yield? Why would they give you a high return? Because they're risky. So the reason they're high-yield is because they're risky. Low yield, low risk, high yield, high risk.
- 7.14** So when you see, for example, "Oh, I can buy this bond or this company, and it's paying 15%. This is fantastic. I can't get 15% anywhere else," there's a reason. There's a reason. During the S&L crisis, S&L wanted you to deposit money with them because they were— remember, we talked about this—they were buying lottery tickets. They were taking huge risks. They needed capital in order to make the risks. And so, what would happen is, in the Wall Street Journal every day, you would get a little column where they would report on the S&Ls paying the highest rates on checking or saving accounts, highest rates.
- 7.15** And these were often double-digit rates. You could get 12, 15% on a simple checking account during the 1980s. And why did people send them the money? Guaranteed deposit insurance, right? So I take no risk, zero risk, right? As long as it's less than, at the time was \$100,000, as long as it's less than \$100,000, zero risk. And if I had more than \$1,000, then what would I do? I'd take all five banks listed in the Wall Street Journal. I'd send them each \$99,000, and I was covered completely. And I would get that nice 15% return on the entire amount, right?
- 7.16** How many of those banks do you think went bankrupt? All of them, basically. All of them. Because what were they doing with their money? They were taking on high risks. They all went bankrupt. FDIC paid me back. I was cool. Everybody was happy except the FDIC and taxpayers who had to bail the whole thing out. So when you see a bank, or when you see a financial institution, or when you see a bond, or when you see somebody promising you a very high return, red lights should be going off in all directions. Why? How can they do this? What's the risk?
- 7.17** And maybe the answer is, yeah, they're doing something maybe stupid, but the money is guaranteed, so I'm going to get my money back. So yeah, I'll take that return. I'll take

that deal. But most of the time, it's not guaranteed. Most of the times, you're not protected. Most of the times, if somebody is offering you a deal that looks too good to be true, in life generally, it's not a good deal. Something, there's risk there that you have not acknowledged or haven't recognized or something. So, high-yield bonds, yeah, that sounds great. Yeah, but that means high risk. Anything high-yield is going to be ultimately high risk.

- 7.18** Stocks yield typically more than bonds. Why? Because they're riskier. The reality is that bonds get paid first. If we go bankrupt, the high-yield bond will get the money before the stockholders do. The stockholders get the residual, whatever's left. And if the residual is negative, they get zero. Or if it's zero, they get zero. Bondholders will get, maybe you won't get a buck for every buck you invested, but you might get 80 cents or 75 cents. You'll get something. Almost always, you will get something. Stockholders get nothing. So just that risk of bankruptcy makes stock more—generally, of the same company, stocks will always be more risky than the bond.
- 7.19** Just because the worst-case scenario is bankruptcy, and stocks suffer the most in bankruptcy. It goes to zero. So just remember risk and return, that relationship. There's no such thing as a free lunch. If the return is high, the risk is matched. Unless you're in a meme stock, and then all is fine, don't worry, be happy, and just do what the guys on Reddit tell you, and you'll be fine.
- 7.20** Remember that somebody bought that stock at the top. A bunch of those bored kids during Covid sitting at home who just got a stimulus check, right? They just got a stimulus check, and they plowed it into GameStop, and some of them bought it at the top. Some of them sold, but some of them bought it at the top, and then lost it all or lost a big chunk of it. So beware of not being the sucker left right at the top there to take the hit. Stay away from meme stocks, generally.
- 7.21** All right. So relation between risk and return, we're all good? One of the ways in which we mitigate risk, and this is really in the stock market, but generally in investing, it comes out of work that was done in the 1950s, and a lot of work has been done since to refine the models and to improve them. And they're always going to be refined and improved because you can't model anything really involving human beings mathematically, not perfectly. You can estimate, you can give an approximation, but the reality is that, as meme stocks prove, people have free will, and free will leads people sometimes to do good things, sometimes to do bad things, sometimes to do crazy things. But it's not modelable. You can't put it in an equation.
- 7.22** But a lot of work was done from the 1950s on, on this idea of, how do you construct an ideal stock portfolio? How do you construct an ideal stock portfolio so that you get the best return given the risk? So always return given a risk. So yes, I could have higher expected returns, but the risk is higher. But how do I find the optimal trade-off between return and risk? So I'm making the most return given the risk. There's a

famous paper in 1952 by a guy named Markowitz, who revolutionized the field and started people thinking about this.

- 7.23** There are a lot of different models out there that have come out of it ultimately. But the main insight, the main insight, is that if you invest in a portfolio of stocks, that the return on that portfolio of stocks will be the average return on those stocks. But the volatility will not be the average; it will be less than the average. That is, the volatility, in a sense, cancels each other out. And that means that we can find a combination of securities, depending on how that correlated one with the other, that will optimize this idea of return given a level of risk.
- 7.24** We could choose the level of risk, in a sense, dictate what kind of portfolio we should earn to maximize the expected return, always expected return, given the level of risk that we have chosen. So for example, if you're young and you've got a long time to deal with volatility, then you're going to be, you should be, more willing to take on risk. OK, so I'm willing to take on more risk: what would be the optimal portfolio? Because I don't want to take unnecessary risk. I don't want to take on risk that I won't get rewarded for. I want to take on the—I want to create a portfolio where I get the optimal return given that risk.
- 7.25** So you can construct portfolios that are ideal for individuals based on their risk preferences, which are often based on age, but also based on wealth and based on a lot of other factors. And this is this big insight that came out of portfolio theory and is, again, still being developed and still being thought about. There's mathematics and a lot of math behind it. If you combine securities, and you want to combine securities, then one of the things you want to think about is how you're combining. What you ideally want is to combine securities that don't move together. If I have two securities that move exactly together, then it's just one security. It's exactly the same.
- 7.26** But if I have securities that move in different directions or don't have any relationship to one another, they move independently. They could be correlated negatively. They could be correlated positively, but not exactly the same, and they could be uncorrelated, with a correlation of zero. They're completely independent. What you want is to construct a portfolio with correlations that are as small as possible, where they do not move together. So that if something hits a particular industry, maybe those stocks go down, but there's another stock over here that might benefit from that and actually go up.
- 7.27** And you want to be able to cancel out as much of the downside as you can through diversification. So, generally, diversification lowers risk while preserving returns, while preserving average expected returns. And as an investment strategy, I think the right approach, generally, is diversification, is diversification. Now, in the theoretical literature, they come up with these perfect portfolios, efficient frontiers. They map it all out exactly. But there are a lot of problems with the math, with the modeling, that are just not real.

- 7.28** It doesn't mean that the underlying intuition, the underlying idea, is wrong. But the actual math in terms of somebody telling you, "OK, this is it. I got that perfect," there is no such thing. There are all kinds of things that they assume that are just questionable. For example, the math assumes that all distributions are normal distributions. All distributions of stock returns are normal distributions. The reality is that stock returns are not typically normal. They typically are fat tails. That is, pretty extreme events happen more often than you'd expect.
- 7.29** We don't really know how to add and subtract and do math on non-normal distributions. We only really know how to do math on normal distributions. It's just statistics, probability. Again, the math that's being done makes an assumption that we know is not true. We assume that all people in the marketplace are rational. One could make the argument that the rational ones are the ones that matter because they're the ones who determine prices. But again, as we've seen in meme stocks, bubbles, and other occurrences, there can be enough irrationality in a marketplace at any given point in time to take us out of this efficient market or rational assumption.
- 7.30** Also, the assumption of rationality in economics, rationality in economics means a little differently than I think it means in epistemology, in philosophy. In economics, rationality means pretty much that you know everything. So you know everything and you make the right choice always. We don't know everything. We don't always make the right choice. So the economic rationality is just not true. It's an assumption that just is not reflective of reality. Indeed, we can all be rational, but it doesn't mean we know everything. We're all rational within the context of what we know, hopefully.
- 7.31** But even that is not good enough for the economic models. The economic models assume a lot more than that, and therefore they're flawed. That, of course, leads to an assumption that markets are 100% efficient, that there's no ability to pick up the \$100 bill. Nobody's actually picking them up. It's just never there because somehow it got picked up without anybody actually doing it. But somebody is picking up the \$100 bills, so markets are not 100% efficient, not at any given point in time, not in every different security.
- 7.32** And then, of course, there's the question if variance is really risk. Variance is, again, symmetrical, up and down. And what we really, really care about is the down, not the up. So this whole line of reasoning, while valuable, gives us real insights, gives us a real idea of what we should be looking for and how we should be investing for the most part as non-professional investors. We should be diversifying. It's actually pretty rare to find managers who will manage our money who can do better than the market. It's very rare. I'm not saying there aren't any. There are. But it's rare to find.
- 7.33** Let's say most of us invest with mutual funds or with ETFs. And some of those mutual funds are run by what's called active managers; active managers are managers who actually trade on a regular basis, based on their assumption that they have information that others don't, or they know more than others, or they can predict the economy

better than everybody else. Very few of those active managers actually beat the market. You're usually, not always, there are exceptions, but you're usually better off buying the market. It's very cheap to buy the market. ETFs, index funds, very, very cheap, very low transaction costs.

- 7.34** You're not paying a manager for his brilliance. You're paying an algorithm to just buy and sell as the market changes. And as a consequence, you can buy it very cheaply and you just own the market. And owning the market is about as good of a strategy for most investors and just passive. Just go to sleep, don't think about it, and just let it ride until you need the money. Buying even the market is a little tricky in the sense that you have to think about, the market is not just the S&P 500. S&P 500 is only big stocks. The market includes small stocks.
- 7.35** But if you really think about the market, the market is also real estate. The market is also private companies. The market is a lot of different things. And the market is not just the US. The market is global. The market is emerging markets who often outperform stable, older markets like the United States. So one of the things to really think about is, if you're going to diversify, diversify; take it seriously. So you want to have S&P 500, but you also want to have small stocks, and you also want to have international stocks, and you want to have emerging markets, and you want to have real estate, and you can buy real estate like a stock. There's things called REITs that you can go out, buy a portfolio of real estate.
- 7.36** Is that diversified? You want to make sure that it's diversified, at least across the country, probably not globally. That is a harder thing. But you want to be able to expose as many asset classes as you can, so that you're really exposed to, you're really canceling everything out. You're getting the most that you can, given that you're not an active investor, given that you're not going and digging into companies and you're not figuring all this out. You're letting the professionals do that.
- 7.37** We talked a little bit about beta, a big one of the things that comes out of Markowitz's work is the importance of this correlation between stocks and the market, however you define that market; that we call beta. If everything has a beta of 1, everything is correlated, it doesn't help you in diversification. Again, you want uncorrelated risk. Beta of zero is really nice. If I can find securities that have a beta of zero, not correlated with the S&P 500, let's say, or not correlated with the rest of the market, that is ideal.
- 7.38** We talked about short-selling, but why would anybody, other than if you had a real sense that a stock was overpriced for some reason, like the meme stocks, you might want to short, but you better know what you're doing. Maybe Carl Icahn does. I think most of us don't. But you want to short; but why otherwise would you want to short stocks? Why do participants out there in the marketplace short stocks? Well, often, shorting a stock is going to diminish the volatility of the rest of your portfolio. It's, in a sense, insurance against the market going down.

- 7.39** So let's say, so one of the kind of financial institutions out there is called the hedge fund. Hedge funds are investors. They used to be in the old days relatively unregulated, so they could do a lot of different things. Hedge funds, mutual funds are very regulated. So for example, if you're the manager of a mutual fund, you cannot be compensated based on your performance. Kind of funny. But you're not going to get a percentage of the profit you make. You get a salary. You maybe get a bonus about performance, the value by managers. You can't get 10% of all the profits you make for your investors. That doesn't exist with a hedge fund.
- 7.40** A hedge fund, typically a hedge fund is, the structure of payments that the manager gets is 2 and 20. 2% is a fee. It's 2% of the assets, 2% of the investment you get every year just to cover expenses. And then 20% of all the profits you get. Very lucrative. A lot of billionaires in the hedge fund business. You accumulate a significant amount of assets. It's very easy to become a billionaire off of 2 and 20 if you do well, assuming you do well. And if you do well, you get more assets.
- 7.41** One of the phenomena that happens in hedge funds, not surprising, is as you get bigger, what happens to returns? They go down. It's much harder to make a lot of money. It's much harder to beat the market when you're big than when you're small. If you're small, you can do very limited strategies into small stocks where the market is not very efficient. As you grow big, you have to deploy large amounts of money. The only way to deploy those large amounts of money is in parts of the market that are pretty efficient, and you can't really beat the market in a sense.
- 7.42** But one of the reasons—so again, hedge funds have a different compensation scheme. Mutual funds, for example, are not allowed to short stocks. Hedge funds are allowed to short stocks. And indeed, one of the reasons they're called or used to be called—they're still called hedge funds, but now a lot of them don't hedge. In the old days, hedge funds were called hedge funds because they hedged. So what they would do, for example, is, let's take a particular industry. Let's take biotech. And they would invest in those companies in biotech that they thought would outperform, that had really good prospects.
- 7.43** They would short those stocks in biotech that they thought were overvalued. And they wouldn't make enormous returns if biotech went up through the roof. They made returns based on the difference between what they bought long and what they bought short. But the volatility was reduced dramatically. Biotech is very volatile. Companies go in and out of favor and stocks will take off for a year or six months and then they'll crash. But when they crash, you make money on the short to compensate for the losses on the long. And when they go up, you make money on the long, which is compensated on losses on the shorts.
- 7.44** So your returns are much smoother, much more normal. They still can be very high, but much less volatile than the returns. So shorting is done, most of the shorting is done by hedge funds. Some funds specialize just in shorting. They find rotten companies and

short them, or bad companies that short them. There's some companies that specialize in that. They'll find a company where they think there might be fraud or they think they're just bad managers. They're doing a really awful job. They'll short the stock and then they'll publish a report telling the rest of the world why this company sucks.

- 7.45 And if the rest of the world believes them, they'll crash the company and they'll make a lot of money. But notice they're doing the world a favor. They're providing us with new information. They're making the market more efficient by doing so because they're giving us information. If it's true, if they're not just rumor-mongering, then the stock was overvalued. Now it's crashed back to where it actually should be, given its true nature, given the information that we have about it. So shorting is often used to mitigate risk of a portfolio, but don't recommend it for individuals.
- 7.46 As I said before, you can lose a lot of money, very, very quickly. Quite shocking. But it is something that hedge funds—that's what, for the most part, hedging means. You can do the same thing by using options. We'll talk about options in a little bit, but you can do a similar thing in terms of protecting your losses by using options. It turns out, for the most part, shorting is cheaper than buying options. But buying options is another option in which to cover your downside risk.
- 7.47 All right, before we get to options, let's talk a little bit about insurance because the history of insurance is really interesting. So you can go back to the Greeks and you can find insurance even back then. Insurance on maritime, on ships going out and never coming back, on storms, even types of life insurance in ancient Rome. But in all these cases, insurance was basically gambling. They had no real knowledge of the risks involved. The probability, probability is a modern theory. There was no theory of probability, back in the Greeks.
- 7.48 Indeed, it's interesting. Like a lot of things you will find that the Greeks make a lot of innovations. They're advancing in math. They're doing a lot of interesting things in math. They don't have the base 10 numerical systems. It's very difficult for them to really take off, particularly with probabilities. But they do a lot of stuff. They're also gamblers, but they never actually connect math with gambling and never actually develop a theory of probability.
- 7.49 And then once Rome basically disappears, for about 1,000 years, nothing happens. It's generally the case. From between about 3 to 400 until about the 15th century, nothing much happens. No astronomical observations are made. Zero. Except in the Arab world, where the Arabs go through their golden era in around 900 to about 1200, maybe 1300, if you count Spain. In the West, nothing happens. It's just dead. There's no progress. There's no discovery. I mean, there are little things that people do, but nothing on major scale.
- 7.50 And then with the Renaissance, there's an explosion, explosion of discovery. You all know, Copernicus, Kepler, Galileo in astronomy. Suddenly, everybody's making

observations. Something changes in the Western mind when they rediscover the Greeks, which is what the Renaissance is all about, the rediscovery of Greeks. But for a long time, it's just dead in terms of knowledge. Well, the same is true of math. There's gambling going on. People continue to gamble, but you'd be surprised how little they understand what they're doing.

- 7.51** I mean, even throwing a dice. They have no concept of the probabilities of what number will actually come up. I mean, maybe some dice throwers have a better intuition about it, but there's no theory. There's no math. I mean, that math really is developed in the 1600s. So, while there is maritime insurance and life insurance and fire insurance and all these things, it's all basically gambling. It's all basically blind luck in terms of what the insurance actually means. Today, if you think about insurance today, life insurance requires very sophisticated actuarial models.
- 7.52** Actuaries make a lot of money because it's not easy to be an actuary. The math is pretty intense. The statistical modeling is pretty intense. So it's hard. Not a lot of people want to be actuaries because the math is so hard. I'm sure AI is going to take their jobs soon. But the reality is, actuarial sciences, both for life insurance, for weather insurance, for all the different types of insurance, is not easy. To make money in insurance is hard. But we know what the math is. Imagine doing it without that math, completely blind. It was basically just another form, as I said, of gambling.
- 7.53** But starting in the 1660s, something happens. This is the Enlightenment. This is when reason becomes, people focus on reason, people focus on rationality, people focus on discovering and understanding the world. Science is, in a sense, reborn. There's a rebirth in science. There's incredible innovation scientifically. This is the era of Newton. This is a period in which, from 1660 to 1800, where there is super excitement about the world, about progress, about knowledge, about reason, about the human mind. And this affects insurance.
- 7.54** Pascal and Fermat—Pascal, you might have heard of—Pascal, Fermat were both mathematicians. They come up with probability theory, the first time we've got a theory of probability. They have this problem that they're faced with. There's a game of chance. Everything in terms of progress in probability and statistics, all of it was in an attempt to try to explain gambling. Gambling was at the bottom of all of this. Everybody's a gambler. Anyway, Pascal and Fermat, there was a problem.
- 7.55** Let's say, you're playing a game of chance. There's a certain—this guy's won a little bit more, this one's won a little bit less. You stop right now and you have to project. You have to project who's going to be the winner and who's going to be the loser, how to divide the spoils. Different mathematical problems, not really, not with the knowledge we have today, but at the time, quite complicated. They developed a whole theory around this and explained it.

- 7.56** Up until this era, 1660s, again, nobody knew how long people lived, not on a life expectancy basis, not on an average. Nobody had ever run the numbers. Nobody had ever looked at it. Nobody really seemed to really care. But a guy named John Graunt actually got all the—now, a lot of it was not even recorded further back, but by then there were records of these things. So he got all the data from all the different districts in London, and he got it all together and he made charts of who died, who has died from what, at what age, who was born, how many people were born in any particular year. Nobody knew this. Nobody knew what the population of London really was.
- 7.57** But he put it all together and actually created these tables. And then Halley, famous from Halley's Comet, an astronomer, he then turned this into not just mortality tables that now could be used by life insurance companies. What's the likelihood that a 20-year-old is going to live to be 30? Now, today, it's almost 100%, but not in London at the time. Life expectancy was much shorter. Life expectancy was probably 29. Now, that's at birth. But at 20, life expectancy is much more than 29. But nobody knew that. Nobody understood that until Halley actually puts tables together of telling you, at any given point in time, what is your expected life?
- 7.58** He comes up with the idea of confidence intervals, statistical significance, how to estimate a probability of something just off of sampling. Like, today, we do these surveys, we do polls, we sample 2,000 people, and we know what the American public is thinking. It's pretty crazy. But the math is solid. And the math is based on stuff that—sorry, this is not Halley, this is now Bernoulli. This is Bernoulli in 1705, all based on Bernoulli's ideas. The whole idea of sampling confidence intervals and being able to tell us what the significance of something is based on a sample.
- 7.59** So you get these real developments of probability theory, life expectancy tables, the whole idea of certainty and uncertainty and how to measure it is now with these confidence intervals and statistical significance. Again, in the 18th century, you get this idea of normal distributions. Things that repeat lots and lots of times in nature tend to be normal distributions. Tend to be a certain average result, and things get to be distributed normally, what we call the bell curve.
- 7.60** Then another Bernoulli, Bernoulli turned out that this Bernoulli family was like a family of geniuses. There were five uncles, nephews, whatever, all made major contributions, just in this era, to mathematics and other sciences. They were all super brilliant. It's also an interesting fact that when you look at families like that, a huge number of them are really, really smart, the reality is that, after a while, generation after generation gets dumber. They revert to mean. This is mean reversion. This is the bell distribution. Dumber families tend to get smarter. Human intelligence tends to gravitate towards a mean.
- 7.61** Anyway, Daniel Bernoulli came up with this idea. Different Bernoulli came up with the idea that value is related to utility. We're not going to value \$100 exactly the same. If I'm a billionaire, \$100, I'm not even picking it up off the street. If I'm really, really poor,

I'm running to pick that \$100 up. We value \$100 differently based on who we are, where we are in life, what else we've got to do right now, what are our priorities. Context matters. Context of your life.

- 7.62** Then Bayes. You probably heard about Bayesian statistics. Bayes comes up with the idea of inference and the idea that, if you multiply the probability of an event times the payout, that gives you its value. And Bayes develops a whole branch of statistics, a whole theory of statistics, on top of that. Anyway, mathematics, at the core of insurance is mathematics. And therefore, insurance can't develop until the math is there. But once the math is there, insurance just explodes.
- 7.63** So, starting in the 17th century, you get the first real life insurance companies that are actually looking at these mortality tables, that are actually doing primitive forms of actuarial science, that are actually trying to match people and change premiums depending on how old you are and depending on your circumstances. You actually get maritime insurance, and Lloyd's explodes and becomes big, and other maritime insurers become very, very big as they try to use science, they use statistics, to try to figure out, what is the probability of the boat coming back?
- 7.64** It's gone off to the colonies. What's the probability of it back? What does history suggest? What do we know from the past? How risky is this? What's going on? Is there war right now? Well, that increases the risk; charge higher premium. So all of this is going on during this period. You get fire insurance becomes big. London has this massive fire that basically destroys a big chunk of London in the 17th century, and people learn from that. There's this huge downside. I want to buy protection. I want to be able to protect myself from this downside. It's a life-changing event to have all your property destroyed by fire. I'm willing to pay upfront to get, ultimately, to get a peace of mind.
- 7.65** The ideal is that you pay upfront, you pay, you pay, you pay, you pay, and you never get the money back. That's the ideal, right? That doesn't quite work in life insurance because you know you're going to die at some point. But even life insurance, most life insurance policies don't go until old age. You buy a term life insurance, it might cover you for 30 years. After 30 years, it's gone. And what you really want is never to get that money back. What you really want is to be alive when the 30 years are over. So you've wasted all that money.
- 7.66** You see this a lot in the press when people buy insurance and they don't use it. "That was a misallocation of resources." Not taking account that the reason they bought it was in order to protect themselves if it happened. But if it doesn't happen, it's even better. But you see this a lot when they criticize the financial markets for buying insurance that is never used. That's the idea. It's ideal that it never be used. You don't want the fire, you don't want the earthquake, you don't want to die. The reason you want life insurance for 30 years is, you want life insurance while you have a family,

while your kids depend on you. Once a kid is an adult, if you've done a decent job, then they're not relying on you.

- 7.67** If you've saved money, then your wife is not relying on you, or you're not relying on your wife, whoever happens to bring in the money, right? Or you're not relying on the life, you're going to be dead. But whoever your dependent is, is not relying on it, right? So our point of insurance is never to use it, but it's a cost. And it's a worthwhile cost because it reduces our risk. It reduces the downside of if something really bad actually happens. People don't have a good sense of what insurance is. So they get upset when it seems wasted, that you're not getting stuff back.
- 7.68** For example, I'll just give you a quick example. There are two forms of life insurance. One is universal life and one is term life. Term life is, you pay, and if you're not dead at the end, your premiums are gone. That's it. You lose it all. Universal life is, you pay in. If you die at any point, you get the insurance. But if you actually live through the thing, you get the money back with interest. So universal life is a combination of term life with a saving account. Term life is just term life. People prefer universal life because they want to get that sense that I can get my money back.
- 7.69** But the fact is, universal life makes absolutely no sense. All you have to do is buy term life and then get a savings account. Much cheaper, much more profitable, much easier to manage the risks involved. You're in control, not the insurance company. But people don't like this idea that, what if I don't die and I never get my money back? It's gone. That's the point. Also, one other point about insurance. How does insurance fit into this model of savings and investment in productive capacity, in production? Well, it fits perfectly. You're sending premiums to your insurance company. That's a form of saving.
- 7.70** It's a different form of saving because it's saving—you get it only when the catastrophe happens, only if it happens, but it's still a form of saving. You're reducing uncertainty in the future. And what does the insurance company do with all that money? Just sits there in the vault, holds it? No, it invests it. It buys, puts it in venture capital; it buys stocks or it buys bonds or it buys assets that are put to productive use. So again, even insurance is a way of taking savings and turning them into productive activity, which is what all of finance ultimately is about, the purpose of finance.
- 7.71** All right, finally, when it comes to risk, let's talk a little bit. We're not going to talk a lot about this. Let's talk a little bit about futures and options markets, which, again, are often discussed in the media and often out there. And these are ways of reducing risk. And really, futures markets, which are the basis for much of the derivative market—these are all called derivatives—futures market developed really out of agriculture. So I'm growing corn, and I'm completely dependent on the weather. Good weather, have a bumper crop. Bad weather, bad crop.
- 7.72** But that's not it, right? Because I don't know if I get a bumper crop, and everybody gets a bumper crop, then there's a lot of supply, and the price will go down. So what I'm

really concerned about is, I've got the weather, whether my crop is going to be good or not, but I've also got the issue of price. I don't know what price I can sell this crop in a year because I don't know what supply or what demand. Demand could change. Some health guru could say, "My corn is no good anymore," and nobody buys corn anymore. I want to find a way to lock in the price so that I don't have uncertainty about it.

7.73 Now, why would somebody want to lock in a price for me in advance? Why would somebody say, "I'll buy all your corn at this price, at 20 bucks a bushel or whatever it is, 20 bucks a bushel, no matter what's going on in the marketplace?" Why would somebody do that? Well, let's say I'm a manufacturer on the other side, and I use corn to do corn syrup. I don't know, that's what Coca-Cola does, right? Let's have Coca-Cola. And I want to guarantee my price of corn syrup. I don't know if you've looked at a can of Coca-Cola. It's all corn syrup. All our soft drinks basically have corn syrup in it for a variety of regulatory reasons. We don't use sugar in America. We use corn syrup, all driven by regulations and tariffs and taxes.

7.74 So Coca-Cola says, "Look, I'm not sure what the price of corn is going to be in the future. I'd like to know exactly, how about we negotiate a price? I'll buy all the corn you produce at 20 bucks a bushel. That way, I know what my cost of production is going to be in the future. You know what you're going to get on the corn." Everybody's happy. It could be that the other side is just a speculator. Somebody says, "I'll bet against you. I think the price of corn is going to be 30, so I'm willing to give it to you 20." You're worried the price might be 10, so you're willing to take the 20. It doesn't really matter who is on the other side. The farmer is now protected.

7.75 Hopefully, at \$20 a bushel, he can make money. Now, if it turns out that it's 30, he'll regret it in a sense, but it's the same as buying insurance. Buying insurance, you don't die, you regret it? No. All the futures contract is, is insurance. There's nothing to regret. And Coca-Cola might guarantee the \$20, and it might turn out the price of a corn is 10. Farmer is really, really happy. Coca-Cola is not happy. Are they going to regret it? No, it's exactly the point.

7.76 And again, you see in the press over and over again, when somebody takes a loss on a future— that is, the price is different than what they secured, it's less than what they secured or more than what they secured, depending on which side of the transaction they're on— the press will say, "Oh, wow, that's terrible. They lost money." No, they bought insurance. It's exactly what it is. So agriculture futures, futures contracts, are standardized by exchanges. And so you can go into the exchange. If you're Coca-Cola, you don't have to find a farmer; you can just buy a contract on the exchange that guarantees the price. If you're a farmer, you can go to exchange. The exchange will match buyers and sellers, right?

7.77 And speculators fill in the gaps. Speculators can take the farmer's position. If Coca-Cola wants to secure more contracts than the farmers are willing to do, a speculator could step in. And it's not that literally the corn goes from the farmer to Coca-Cola. It's that

they settle on in cash, and then they go to the actual market and buy whatever they need or sell whatever they need. So futures contracts are typically settled. If you speculate on, what is it—if you speculate on corn futures, you're not actually going to get bushels of corn in your backyard, if you landed up buying; you'll settle in cash, and the farmer will sell at the market price in the marketplace.

- 7.78** These contracts have existed, again, for a very long time, but they became really, really prevalent in the 1970s or since the 1970s. Actually, in farming, they've been prevalent for a long time. But from the 1970s, you could get futures in currencies. You get futures in interest rates. You can get futures from 1982 on. You can get futures in stock prices. So I can bet, in a sense, on what the stock price will be in the future. I can buy or sell a futures contract on one of those things.
- 7.79** Mostly, though, people use options. Options are cleaner because options don't actually guarantee that a transaction will happen. So what's called the call option, a call option, is the right—but not the obligation, you don't have to do it—it's the right to buy a particular thing, a stock or anything, an agreed-upon quantity of that stock and agreed-upon price from the seller. So I have a right, if we take the bushels of corn, I have the right to buy five bushels of corn at a price of \$20 a bushel. If the price in the market is 15, do I exercise the right? I have the right to buy it at 20, but not the obligation.
- 7.80** If in the market it's only 15, I don't exercise it. Whatever I spend to buy the call option is gone, but I buy the corn at 15. If I bought it at 20, that would be really dumb. If it's 25, I can buy it at 20 and sell it for 25. I can make money. So options work that way. You can also buy a put option. A put option is a bet on the price going down rather than the price going up. So I'll only exercise the option if the price goes down. So options are what mostly are used with regard to stock options, stocks, and it's how a lot of people hedge. Instead of shorting, they buy put options. So they make money if the stocks go down.
- 7.81** You can buy put options on indexes. You buy put options on S&P 500. So if the S&P 500 goes down, you make money. If it goes up, all you lose is what you paid for the option. You don't have unlimited losses like in a short where the loss is unlimited. Here, your losses are only—now, the person writing the option, you can write options. That is, I sell you the option, I can lose a lot. That's why you have to pay me upfront. But for example, if I sell you a put option, I need to be able to provide you with that stock at that higher price.
- 7.82** So options are tricky and they're pretty dangerous, and you could lose a lot of money on the writing side. On the other side, you can't lose a lot of money, but you can lose what you paid for them, and very few people are good at it. Again, they're used by hedge funds and sophisticated investors to hedge their risks, to balance out long positions with a put option or short positions with a call option. So call options, you make money when it goes up. That's short, you make money when you go down. So you want to offset the two.

- 7.83** Finally, you have swaps. Again, you'll hear about them a lot in the media. These are situations where banks or companies, they've got bonds, they've got interest rate commitments, but they, one, maybe have a variable rate interest, like a variable rate mortgage where the rate moves with interest rates. The other one might have a fixed rate mortgage, so the rate is flat. Maybe this guy wants fixed rate and this guy wants variable rate; even though they have the opposite, they could swap the interest rate, giving this guy a fixed and this guy variable. And these can be very, very complex transactions. And generally, these derivatives can get crazy in terms of how complex and how difficult they can become.
- 7.84** You can today buy catastrophe bonds. So bonds that will only pay you if a catastrophe happens. Like, I'll buy a bond on earthquakes, and if an earthquake happens, I get paid. It doesn't happen, my money is gone. So today, you can do pretty much anything. In terms of structuring different cash flows under different situations with different scenarios, some of them to hedge against all kinds of risks, and some of them just speculative, because I think X is going to happen or not happen, not hedging anything known.
- 7.85** So risk mitigation is a massive industry. It covers the insurance, it covers a lot of the option trading, it covers the futures markets. These are massive trades that involve both real-world things that happen, catastrophes, weather events, things like that, but also that cover the financial markets and how do I protect myself, given the type of investment that I'm making? Now, again, how does this risk mitigation help with the allocation of capital, which is what finance is all about? Well, it makes it easier to allocate capital because now I can protect myself. I'm much more likely to invest in risky ventures because I can buy protection.
- 7.86** So a lot of things get funded that wouldn't otherwise get funded. And on the savings side, I'm willing to put my savings at risk because, again, I can protect them. I can buy protection on them. So the increased saving, the increased investment, which is what finance does; and by controlling the risk, I mean, think about life insurance. If I didn't have life insurance, couldn't buy life insurance, you would save more in very low-list risk assets. Because you'd want to accumulate a nest egg that if something happened to you, would be guaranteed. You would not invest your savings in something risky.
- 7.87** But if you have life insurance, and even though you have kids dependent on you, now you can invest your saving in a risky asset because you know if you die, the life insurance kicks in and takes care of them. So it allows you much more flexibility in terms of how you manage your assets, how you manage your investment, how you manage your savings, because you now have the tools to mitigate risk or at least reduce risk. So the aspect of finance that involved in lowering risk or in controlling risk or managing risk is crucial to every other aspect of finance, which is involved in savings and in allocation of capital to its most productive use. All right. Thank you, guys.

Lecture 8: The Future of Finance

At a glance — The future of finance.

- 8.1** All right, our final class. We've been talking about what finance does, and really what finance is doing is allocating capital in the economy by providing a return on saving and by managing risk. Now, this is a crucial function. The world in which we exist today could not exist without it. Indeed, you cannot imagine any level of advanced civilization without finance. You can actually look around the world at the various countries and how well they're doing and how civilized they are, and finance is directly correlated. It's not causal. It's not because they don't have finance, they're not civilized. It's because they're not civilized, they don't have finance.
- 8.2** Finance depends on the rule of law. Finance depends on property rights. Finance depends on contract law. Ability to sign contracts and keep to them. A stock is a contract. A bond is a contract. A life insurance contract is a contract. If I can't trust the authorities in terms of abiding by contracts, financial markets would disappear. They rely on trust, trust in a rule of law, trust in property rights. Countries that tend to be free, countries that tend to have property rights, countries that tend to have protection of contracts, have robust financial industries.
- 8.3** It's a reflection of that freedom. It's a reflection of that arrangement, if you will, that political arrangement. So if you go and you think about the financial hubs of the world, you think of a place like Hong Kong, at least before the Chinese took it over during Covid. It was an unbelievably robust financial industry, and that's because they had the rule of law, they protected contracts, and they had property rights. Or London, or New York, United States. These are countries that are robust in terms of contracts, property rights, et cetera.
- 8.4** And finance is an organic phenomena. It's something that grows out of the market. It was not planned. Nobody sat down and figured all this out. Stock markets started in coffee shops, people trading pieces of paper. You could argue that the only thing planned was money, and even money—the way banks, for example, became the issuers of money, the way money became pieces of paper rather than gold, all that was an evolution in the marketplace.
- 8.5** Finance has evolved, in a sense, spontaneously, but really through the work of entrepreneurs, and often mathematicians, scientists, people who are figuring out how these things function. Without the math, there would be no insurance, certainly no options or futures. We actually can price options. It's complicated to price an option. What happens if this happens? You get one outcome. This happens, you get a different outcome, and you have to be able to price all the different variations and the probability that each outcome happens.

- 8.6** Not easy, but it's doable because we have the mathematical tools to do it, because science, engineering, and entrepreneurship create whole industries, and they create these opportunities within the realm of finance. So finance is this organic thing. But it bumps up, and we see this constantly; it bumps up against a reality where finance is not trusted. Financiers are not trusted. They're not believed. They're viewed as zero-sum. They're viewed as crooks. They're viewed as bad guys. They're viewed as wasteful. Like, buy insurance, what a waste. You never used it.
- 8.7** And as a consequence of the fact that they're viewed this way—they're greedy, they're self-interested, self-interest equals bad, that is in our culture—and as a consequence of that, what happens? Well, we want to protect ourselves against them. They might do really bad things. They might do horrible stuff. They might lie, cheat, steal, violate their contractual obligations. They might run away with all our money. They might lie to us, commit fraud. So even though we have laws against fraud, we have laws against stealing, we still don't feel protected. So we regulate. We control.
- 8.8** What is regulation? Regulation is an attempt to catch a crook before he commits the crime. Or to constrain the crook, so he cannot commit the crime, more accurately, right? It's to make it impossible for the crook to commit the crime by telling him there's certain things he cannot do. But how do we know he's a crook until he committed the crime? The implicit assumption in regulation is that all financiers are crooks, all businessmen are crooks. And we need to restrain them because otherwise, they'll commit the crime.
- 8.9** Finance, I think, because it is so suspect by the culture, because at some point in one of the lectures we talked about, there's a pure profit. It's money off of money, making money for money that is so distrusted— that our culture feels like it has to control and regulate every single aspect of finance. And finance is only increasing in the amount of regulation. Once in a while, they deregulate. They did a little bit in the '70s. They did some in the '90s. But overall, more and more and more and more controls.
- 8.10** And the more controls you place, the less entrepreneurship, the less innovation, the less fewer new ideas, because new ideas, by definition, are going to break out of that regulatory environment, and the regulations are supposed to try to hold you in, hem you in. So we have a financial system that every aspect of it is regulated. Banks, we've talked a lot about all the different regulations of banks, but investment banks are regulated, and the stock market is regulated, and how we trade is regulated, and whether you can short a particular stock on a particular day is regulated.
- 8.11** The option markets, the futures market, have their own regulatory agencies with their own rules. It's an endless stream of government controls, government controls that hem you in and constrain what you can do. And that is a drag, a drag on the efficiency of markets. It's a drag on the ability of markets to respond to changes in the environment, to changes in the world, to changes in the need for capital. It's a drag on

the ability of financial markets to respond to new risks. And it's a drag on the ability of markets to respond to what is going on in the world, particularly if things go bad.

- 8.12** And yet the regulations are not the worst part of it, in terms of finance. Regulations are often—investors can figure out ways around them, or they can take them into account and price them in, in a sense, and do stuff in spite of it. Somehow we find ways to survive with the regulations. And again, a lot of—I mentioned this the other day—a lot of energy, a lot of thought, a lot of sophisticated math goes into trying to find ways around regulations.
- 8.13** I'll give you the example of the money market accounts which were created to get around Regulation Q, which didn't allow banks to give interest. But you can go on and on. The Euro market, there's a Euro dollar market, or there used to be, less today because the regulations went away at some point. But there used to be, American companies could not pay interest, or American banks could not pay interest on corporate checking accounts. So European banks created accounts in Europe that were denominated in dollars. Every night, the US bank would send the money to Europe. It would sit overnight in Europe, get an interest, and come back, all kinds of stuff like that.
- 8.14** And then companies would start issuing bonds in Europe in dollars because, again, it got around certain regulations and certain controls. This happens every day, all the time. Every time there's a new regulation, we find ways around it. It's just a waste, but it's also constraining. It's also, things like deposit insurance clearly lead to things like moral hazard, which lead to disasters, lead to real losses. But as I said, it's more than the regulation. The reality is that the product of finance, the thing that finance deals with, is not only regulated, it is completely 100% in control of a government entity.
- 8.15** That is money. Money is not private. Money is not a market in today. It is a market phenomena. It came out of the market. It was designed by a market, created by a market, and even all the innovations all the way down to paper money were products of the market. But at every step of the way throughout history, governments have tried to monopolize, tried to control. Where there was originally government coins, OK, we can do gold, but the gold has to have the king's picture on it, and that's the only legal tender, is the stuff that has the picture of the king on it. And therefore we get a cut.
- 8.16** It's a way for us, and it's a way for us to control and maybe take some of it and keep it. And once you get rid of gold, once you get rid of any tether to something real, only the government can do this. There has to be a monopoly over it because then anybody could issue any kind of paper anywhere and there's nothing connected. So you need one centralized authority, a central bank, that is responsible for the issuance of money. But of course, how do they determine how much money? How do they determine, and they do determine this, what the interest rate should be?
- 8.17** And how much of the decisions they make with regard to interest rates and to the amount of money that's put out there are determined based on economic models—

which I'm not sure how good they are to begin with—but how much of it's based on economic models? And how much of it is politics, just plain politics? There's this phenomena that the Fed never increases interest rates in an election year. It usually decreases to help the incumbent because the incumbent is their boss, ultimately. How much of that is economics? I think we know, right? And how much of it is politics?

- 8.18** So we've got this, we've got money, which is the product that finance uses. It's what is being allocated, which is under political control or under the control of people who think they have economic models that explain how to do this. But the economic models are all suspect, and you can see that. You can see that by the fact that, I mean, this is shocking to me. During Covid, the Federal Reserve basically gave us \$5 billion, sorry, \$5 trillion, right? Handed it out to us. Different usses, but handed it out to us.
- 8.19** And then it was surprised there was inflation. I mean, what kind of economic model are they running? This should not have been a surprise. Certainly not to people who all they do is think about this stuff. This is their job. And yet it really did seem to surprise people. There's a sense in which, like, the Bank of Japan has been for 30 years trying to create inflation in Japan, doing what they think would cause inflation in Japan, and they've been unsuccessful.
- 8.20** So how good are the models? How good is the economic thinking behind this when they can't do something simple, supposedly? Like predict when inflation is coming or prevent inflation or create inflation or whatever it is that they want to do. You have to start being suspicious about the Federal Reserve and central bankers when they tell you their goal is 2% inflation and you go, "Why 2%? Why not 1, 1.5, 3? Why 2? Where did that come from?" Out of nowhere. It's just a nice low number.
- 8.21** And they can't do it. They've been trying to get to 2% inflation. Through the 20-teens, they couldn't get there. It was always underneath. And now they're trying to get to 2% and it's above. And these decisions have real consequences. Inflation is incredibly damaging to the net worth of Americans, the purchasing power, ability to buy stuff, ability to use our money to live. It makes us poorer, all else held constant, makes us poorer. And it also distorts the allocation of capital because inflation in and of itself is not an easy number, right? What does inflation mean?
- 8.22** They use CPI, consumer price index, which is a basket of goods. They try to adjust a little bit for quality of goods. So for example, the price of computers, really, I buy computers every few years. They're always the same, same price, flat. But the computer I'm getting is much, much better. In a sense of per unit of computing, prices are going down dramatically. They try to adjust for that a little bit, but they don't know how to. How do you adjust for it?
- 8.23** What do you do about televisions? I don't know how you adjust for television. You guys don't remember. I used to watch television when I was a kid on a black-and-white box that was this thick and this big. And we thought, cool, black, and cool. And then color

came and it was mind-blowing. And now, what would I do without my 85" super-thin television with 4K, 8K, whatever, resolution? I mean, it's just unimaginable, the difference in quality. How do you measure that?

- 8.24** And by the way, the 85" today cost less than my black-and-white box did back then. How do you measure that? What does inflation even mean if you're looking at prices? Whose prices? My basket of goods, the things I buy on a weekly basis, I'm sure, is very different than yours. All of ours are different from one another. So what does inflation mean, if my inflation and your inflation are different? And even the Fed says, "Well, there's CPI, but then there's CPI without the volatile food and gas prices," as if we don't buy food and gas. And then there's PCE, which is the producer price index, or maybe PPI. And then there's other bunch of inflation indexes.
- 8.25** And it's mind-boggling even for somebody in finance to figure out what exactly they're tracking, what they care about, what matters. Is it going up? Is it going down? Because again, once you get involved in central planning, it's really impossible to tell, what is inflation? I mean, there was a traditional definition of inflation. It's just an increase in the money supply. But they don't want that because the money supply is constantly increasing. They don't want people to feel like there's a lot of inflation constantly. There's a sense in which what they want is to capture the pain points, and the pain point for most people is prices going up.
- 8.26** But if money is increasing all the time and prices are not going up, then where's the money going? Into asset prices, so the money enters in the banking system, the banking system now has more money than it did before. It allocates that money into certain investments. Those investment prices go up. So maybe you have asset inflation. You don't have consumer inflation. But the Fed doesn't count that. And again, misallocation of capital because if prices in some area go up, not because they're more profitable, but because money happens to be chasing it for some other reason, well, that is going to distort the way capital is allocated across the markets.
- 8.27** But it gets, it can become really, really bad. If the Federal Reserve misses on its interest rates, if it sets interest rates too low, makes lending super easy and cheap, guess what people do? If you offer me a loan at 2%, I'm going to max out because I can make more than 2% in the market. So give me money at 2%, I'll take it. I'll buy a home, cool mortgage. I did. Most of us did. When mortgage rates were like three, three to four, we all bought homes. Why wouldn't we? Cheap money. God, I'm so happy I have that 3-point whatever mortgage that I have now, when mortgage rates are 6%.
- 8.28** And I bought a home that, at 6%, I couldn't afford. So if I had to reprice that home today, couldn't afford it, because I can't pay the 6%. Can pay 3, can't pay 6. So you keep interest rates really, really low, people borrow money. They might borrow money to buy stocks. Give it to me at 2%, my expected return on stocks is 8 over the long run. So why wouldn't I make that trade? I borrow all I can and buy as many stocks as I can, if I

can get it at 2%. Not sure I can at 2. Maybe it'll cost me 3, maybe it'll cost me 4, but my expected return is 8. So I can afford to pay a lot of interest if that return is real.

- 8.29** Now, there's risk associated with it, but that's easy to forget when interest rates are very low, the economy is doing well, and the government is telling you, "Things are fantastic. Go for it, people." 1920s, unsurprisingly, interest rates were very low. Very low. Guess what people did? They borrowed a lot of money. Some of their money, they borrowed it to buy stocks. Certainly in '28 and '29, they did it to buy stocks. Margins for stocks, that is the lending for stock purchases, really high. Stock market crash. Suddenly, they can't pay the debt that they have. Not only did they lose their money in the stock market, but they also borrowed money, and they lose on that.
- 8.30** No question, I think, most economists believe that the Federal Reserve and its management, or mismanagement of money, interest rates, before the Great Depression and during the Great Depression made the Great Depression what it was and made it worse than it needed to be and made it longer than it needed to be. Now, you can also blame Hoover and his policies like doubling income tax or starting a trade war in the middle of a depression or a lot of other things that he did. But then FDR, all the things he did, we actually didn't exit the Great Depression, not in real terms, until after World War II.
- 8.31** That's a real danger. That's a real struggle. And of course, what happened during that Great Depression, hundreds of banks went bankrupt. Hundreds of banks had to be shut down. Our gold was confiscated because they went off the gold standard. They didn't want you to own gold. They wanted you to own paper. And of course, the economy collapsed, but who do they blame it on? Finance. Stock market, speculation, investment banks, Wall Street. And what was the response? 1933, 1934, two of the biggest regulatory bills ever. Glass-Steagall and the Investment Act, both of these acts regulated banks, investment banks, all of finance, because they blamed finance for the Great Depression. I gave you a quote from FDR a few classes back showing that.
- 8.32** A great financial crisis just recently, recently in my mind, at least, maybe not in yours, but in my mind, it's recently. It was in 2008. Not an accident that it comes right after a period of really low interest rates, negative real interest rates, negative real rates. And all the money went into housing. Why did it all go to housing? Because the government wanted us all to own homes. George Bush had a goal, 70% home ownership. And they asked Freddie and Fannie, these government agencies, to buy as many mortgages as they could at whatever price it took in order to repackage them, in order to subsidize our homes. So the government was basically subsidizing, and they still do.
- 8.33** You get to deduct the interest rate on your mortgage. You don't get to deduct other interests, but only on your mortgage. So even if you pay 3.5% on your mortgage rate, you're actually paying a lot less than that because you're deducting at least up to the first million in your mortgage. You can deduct the interest on that. So your actual rate is close to under 3 in an after-tax sense. And that's the sense in which that's the

government subsidizing it, or at least incentivizing you to funnel your money into a home. Whether you want to or not, whether maybe you prefer to rent, but more economical to buy, at least in the 2000s, maybe not today.

- 8.34** And the consequence of all that was a really brutal recession, which resulted in hundreds of banks going bankrupt, as it always does. Stock market being crashed, a lot of us losing a lot of money. And of course, a massive bailout, massive bailout, which ultimately didn't cost taxpayers. Kind of cool because the banks paid all the money back. So the government actually didn't lose anything. Taxpayers didn't actually lose anything, but that's even worse because now the government thinks that they're good investors. The bailouts cost less, but of course, they're costly.
- 8.35** Part of the cost is, you don't let the really bad guys disappear. Part of what recessions are for is to get rid of incompetence. During Covid, God, during Covid, in 2020-21, the government bailed out everybody. It wrote checks to everybody, everybody. I mean, there's not a segment of the financial industry that did not get some kind of bailout. It was truly stunning. Every credit facility, every tranche of bond, they got involved in everything. And the response, of course, to 2008 financial crisis was Dodd-Frank, another massive regulatory bill; hasn't really helped because we still have fragile banks.
- 8.36** And I guess the—oh, and just last one is what's called the Greenspan Put. Stock markets go down, the Fed will flood the market with cash so that the market doesn't go down too much and it rebounds again. Had that in '87, but it also had that, every time the stock market goes down a lot, the Fed Reserve does something about it. Preventing, again, the natural forces of the market cleansing out stupid decision-making. Luckily, they don't do that when meme stocks go down, but maybe one day.
- 8.37** The point of all this is really to say, our financial industry is very fragile. Very fragile. And not because we don't have enough regulations. Because we have too much regulation, we have regulation at all. It's the regulation. It's the bailouts. It's deposit insurance. It's low interest rates when the government wants to spur economic growth, artificially low interest rates. It's negative interest rates in some cases. It's those things that make this system super fragile, super sensitive to collapse.
- 8.38** We don't get regular recessions, anyway. We get crashes. We haven't had a recession since 2008. Hasn't been a recession since 2008, which is a long time. Well, Covid. But Covid went like this, down and straight up, because they gave us all money, which we spent. We got inflation. We didn't get a recession. Now the Fed is raising interest rates. Everybody expected a recession. I expected a recession. It didn't happen. They keep everything afloat. It's not natural. It's not real. I think all it does is creates all the stress factors, all the energy around, when the recession does happen, it's going to be a big one. It won't be just a little one.
- 8.39** Because there's so many distortions in the pricing mechanism. There's so many distortions in what business survives and which don't. Which businesses are going

bankrupt, to which are not. So many distortions that at some point all of that has to be corrected. It could just be corrected slowly and mildly over years, which means just slow economic growth and a lot of stagnation. That's possible. That's one path. The other path is big correction, big downturn.

- 8.40** But given the fragility of the system and given the incompetence, really incompetence, are the people who are in charge of the financial system, in the sense of the regulators and the central bankers. It's hard to see an upside. There are various scenarios of downsides, but it's very hard to see an upside. Upsides, when things go really, really well, it's a consequence of economic freedom. It's a consequence of leaving markets alone. We're far from that. Nobody in our political spectrum today is advocating for more economic freedom. Everybody in the economic spectrum today is advocating for more controls.
- 8.41** It's hard to be optimistic about finance as we move into the future, given the amount of controls and given the fact that they control money. And money is so crucial to everything that we do, everything that we do. And it's so difficult in the world in which we live today, even to get a grasp of money and what it is, because it's really not, we don't use this. This you can add up, you can see how much you got. But now money is just bits traveling at the speed of light all over the place. It's just entries in ledgers all over the place. Who has what?
- 8.42** And it's very difficult to keep track, very difficult to understand what's going on, and very difficult to project what would the world look like if it wasn't so controlled, if it wasn't so centrally planned? So I think the system is super fragile. And fragility is not good. And there's no macro hedge that you can do here. Buy gold, I guess. But it's not a productive hedge. It doesn't add anything to the world. So we have to live with fragility, with this idea that banks could go bankrupt. I mean, Silicon Valley Bank was out of nowhere. Nobody predicted it. Nobody saw it. Should have. Should have. Nobody did. How many more Silicon Valley Banks are there? How much would it take?
- 8.43** As I said, any bank run on any bank will drive the bank bankrupt. Any bank, because of the level of leverage. But that level of leverage is not a market phenomenon; it's a regulatory phenomenon. But that's just banks, again, across the entire spectrum. Every one of these institutions has similar kind of regulations. And now that we're bailing everybody out, moral hazard that we talked about is everywhere. So generally, one thing you can guarantee in the world is that people are taking more risks than they should. Finance companies are taking more risks than they should. It's almost guaranteed, given the moral hazard, given the bailouts.
- 8.44** And this is bad because a lot can go wrong. And it's also bad because we're living during a period where a lot could be going right. There's really a revolution going on right now in finance, in our lives more generally. It's been going on for decades now, but it's starting to affect finance in a significant way, and that is, call it the Internet, call it

computing power. It's going to change the way these entities function. It's going to change the way finance is done.

- 8.45** I mean, if we allow for this to happen, and I'm not sure we will, but if we allow for this to happen, do you really think that, in 10, 15 years, to get approved for a mortgage, you'll have to sit in front of a mortgage banker, fill out a bunch of forms, and wait for their decision to come down? I mean, really? Won't you go onto an AI app, tell it to go search your financial records, and talk to the AI app of whoever's providing the mortgage? It could be anybody providing the mortgage. It could be any institution. And those AI apps basically negotiate.
- 8.46** I mean, an AI algorithm is probably better at deciding whether you should get a mortgage or not than the guy at the bank. Probably. I mean, maybe not right now, but think about how much it's changed just over the last few years and project that at 10, 20 years into the future. Banking is not going to be the way it was, not if technology is allowed to change it. I mean, as it is, some of us still remember checks, but when's the last check you wrote? I don't remember when the last check I wrote.
- 8.47** And Venmo? You guys Venmo? I want to pay you money. I just send you money over Venmo. What do I need? I need some place where I can store my cash, and I need some place where you're storing your cash, and boom, there it goes. It could be a bank. It could be any kind of institution. It could be any depository where I store my cash, electronically. Insured, not insured. Everybody's offering insured deposits. You can do brokers insuring, providing those kind of accounts. Your investment banker is; it's not just banks anymore. You can transact all kinds of entities.
- 8.48** And of course, banks are doing less and less and less lending. Private firms are doing more and more lending. What do we need banks for exactly? Banks really exist. And certainly, are we going to have 5,000 banks? In the future, 5,000? Who needs 5,000 banks? I don't think we've ever needed 5,000 banks. Even when we had 20, we didn't need five. In those days, maybe you could say we needed 400, given the size of the country. Today, 20, 40, maybe. These banks are going to disappear. They could disappear through mergers and acquisitions, through M&A. That is the preferred way, although it's happening very slowly, much slower than it should.
- 8.49** Again, because the regulators have to approve every single one of these transactions. It takes them at least six months to give a thumbs-up on a deal that's already been negotiated between the banks, and there's agreement, and the boards have agreed, and shareholders are about to agree, and the regulators take six months to approve the deals, at least. These days, it's even nine. And generally, the Biden administration doesn't like M&A, so they've even slowed it down more than it used to be. But it'll take a long time to get from 5,000 to 50, just an M&A.
- 8.50** Maybe another financial crisis where a few hundred banks go bankrupt. Don't know. But something's got to give. 5,000 banks cannot make money in this economy, not

when I have Venmo, not when I have PayPal, and I can store money in PayPal. It's not like, I can put money in Venmo. I can keep the money in Venmo. It's not a bank, but I can keep money in Venmo and then just send it to people and people can send it to me and I can keep it there. I can do the same in PayPal. I think Zelle is only connected to a bank or something. But anyway, and this is just the beginning, I can take loans now, not from a credit card, not from a bank, but from a "buy now, pay later" service provider, which the government wants to regulate heavily right now.

- 8.51** The CFPB is coming out with new regulations on buy now, pay later fintech companies. So technology is completely uprooting finance. It's completely changing the rules of the game. And it's only the beginning. And AI is going to have a radical impact on what is going on in finance. Finance will, again, not be the same in 10 years as it is now because AI can do a lot of things that people are doing now or not doing now. A lot of the investment algorithms are probably going to be run by AIs in the future, rather than just a statistical match, a statistical regression like they are today.
- 8.52** Pension plans are huge right now because a lot of companies still have pension plans, but pension plans have really gone away. What do we save mostly today? 401(k)s. So rather than defined benefits, we now do defined contributions. That's going to shift the way money is allocated in the next few decades. Now, the government still uses defined benefits, and that might stay forever, so public pension plans might be big in the future. Insurance is now becoming more and more fragmented. You can insure more things. It's becoming much more discrete.
- 8.53** But again, AI is going to change all that, and it's going to change our level of accuracy in terms of forecasting, and it's going to change our ability to price these things. With all this technology and all this fintech, are regulations going to evolve to take that into account? Or is all this fintech just a way of getting around regulations? One of the things that I worry about is CFPB, the Consumer Financial Protection Board, wants to regulate buy now, pay later. That's quick relative to the past. In the past, they would have had a few more years before they regulated. It worries me. We have so many regulatory agencies all competing on who can regulate whom faster.
- 8.54** And is that going to suppress innovation? What exactly is going to happen? How is this all going to evolve into the future? But there's no question, the future is fintech. The future is fintech, and the regulators can either catch up or they can kill it. They certainly can kill it. But then it's a real problem in the US if it gets killed over here. But the future is technology. The future is not—finance will become just a tech, just an app. Again, why do I need a bank? The bank does nothing for me that an app can't do. It should be able to take my money and allocate it. I should be able to give it the criteria, "Buy me a bunch of stocks, buy me a bunch of bonds, buy me a bunch of this," and it's done.
- 8.55** How does regulation evolve to take this into account? How do we deal with the fintech revolution that's happening when the industry is so fragile? Maybe this is a solution. Maybe this gets around the fragility of much of finance. Then finally, of course, this

relates to fintech and this relates to technology. What about crypto? Crypto is a real innovation. It's a different way of doing things. Right now, there's a bill in Congress to regulate it. It passed the House. It's probably a decent bill because the alternative to passing a bill to regulate crypto is to just let the existing regulatory agencies just make up regulations as they go along, which is what the SEC has been doing.

- 8.56** For example, there was a period of time where some startups in Silicon Valley were not raising capital from venture capitalists. They were issuing something called tokens. You could get this token, which you bought from the company. They took the cash they had available, and that's what they used to grow the company. They paid the employees with these tokens. These tokens were tradable, but these tokens also provided, let's say, some special access to the product that you were going to get, or some membership or some exclusive thing that was happening.
- 8.57** So it was a way to raise capital with something that's exchangeable, but not in a traditional VC model, not from VCs. Anybody could buy this token. And the SEC just went apoplectic. "You're raising capital without going through us, without doing it the traditional way, the conventional way, without having the contracts that we always had. You're issuing securities, tokens, without abiding by security law." And they banned it. Now, there's still tokens out there being used in a variety of different ways, but not, in a sense, to raise capital. What a shame.
- 8.58** This could have been a new innovative way to raise capital. Maybe it works, maybe it doesn't. But we will never know because we didn't run the experiment. Why not run the experiment? People are voluntarily doing this. Nobody's forcing anybody to participate. Why is it the government's business if I want to buy a token as a security? You can imagine all kinds of uses of crypto that I don't know if they'll work. I don't have a strong opinion. But I know that a lot of really, really, really smart people, really smart people, people I respect and admire in Silicon Valley, are really excited about this technology.
- 8.59** They're investing huge amounts of money. They're putting their money where their mouth is. If you'd come to me with the idea of eBay, eBay, you put up stuff and people bid on it and you sell it, if you'd come to me with that idea before it was actually created, I would have said, "You're nuts. Nobody will do that. And you can't make money on it. It's just a bulletin board. It's just like advertising in your newspaper, just online. What's the gimmick?" And yet eBay made a fortune. I would have passed on it as a venture capitalist. So what do I know? The real venture capitalists are putting a huge amount of money in crypto.
- 8.60** And a lot of it is because they think, I think they think, that crypto is a way of getting around a lot of the fragility and a lot of the risk and a lot of the regulation that is involved in finance. This is to create a new parallel finance industry that is less regulated on the side. Crypto is a great way to move money around the world. It's not cheap, but it's a great way to move it if you don't want the government to know what

you're doing. Now, if you live in—you know there are 160 different currencies in the world? Most of them suck. There are like four or five good currencies in the world.

8.61 And the dollar counts as good, even though we, living with the dollar, don't like it that much because it's inflated. Relative to other places, it's pretty good. This is why Argentina wants to replace the peso with the dollar. I want to replace the dollar with something better. But they'll settle for the dollar because they know what the peso is like. They know what real bad currency is. Well, if you're in Argentina, if you're in Egypt, or if you're in, I don't know, Congo, and you're dealing with your local currency, or in Zimbabwe, which I used to carry around with me, I used to have a bill. It was 100 trillion Zimbabwe dollars, 100 trillion. And you could buy maybe a loaf of bread with that. I have a few of them at home. They're not worth anything.

8.62 So if you're in Zimbabwe and you want to preserve your capital, you don't want to hold Zimbabwe dollars. And it might be the case, usually is the case, that you're not allowed to hold dollars, capital controls. People are not allowed to have dollars or Euros or Japanese Yen or any currency that's worth anything. Just stuck with the Zimbabwe, well, why not? Why not move money somewhere else in the world using bitcoin? Or why not buy what's called a stablecoin? Stablecoin is a crypto which is linked to the dollar or linked to gold or linked to something, and it fluctuates in value. So it's not like bitcoin that goes all over the place. It's never going to go up a lot, but it's going to stay pretty constant, which is pretty good if you're living in a country that has 100% inflation.

8.63 You can buy a stablecoin and keep it in a wallet and not report it to the government and not have them be able to track it down. And you don't lose the value as long as, of course, the stablecoin can be exchanged back into whatever currency you're using. So there's a problem here and a solution happening. It's a way to hedge currency risk. Stablecoins are a way to hedge currency risk, not so much for the dollar, not for Americans, but for people all over the world. It's a huge value.

8.64 Bitcoin is a kind of a ledger. It's still grappling with what bitcoin and ether and all of these others really are. They want to be money. They want to replace the dollar. They want to actually be a medium of exchange. And maybe they will be and maybe they won't. I'm skeptical. Primarily, I'm skeptical because I don't think the government will allow it. Again, I don't know what would happen in a real market. Let's compete. Let's see. Maybe gold returns as money. But I have this vague suspicion, and I can't prove it, but I have this vague suspicion that in the world of electronics, in the world in which we live, gold is probably not the best way to do money. Not sure, but maybe.

8.65 I'd like to see competing. I'd like to see bitcoin compete with the dollar standard and see what happens. I'm not sure even bitcoin is the right one. Let's see competition between cryptos and see which money wins. But none of that can happen. None of that is going to happen because somebody has a chokehold on what money is. Somebody has a chokehold in the distribution of it. Somebody has a chokehold on

interest rates and to be part of the financial system. Ideally, we get rid of centralized money, whatever money is, and we have decentralized money. If you have decentralized money, private money, competitive money, then whoever wins will win in a market economy that will produce the best result.

- 8.66** We're living in, I think, fascinating times in terms of finance. Incredible sophistication when it comes to hedging, when it comes to stock investing, when it comes to modeling risk, when it comes to all this thing. Unbelievable sophistication, even when it comes to figuring out the risk of a bank, huge amount of sophistication, huge computer models, people figuring out, people have economic models and financial models. It's a very sophisticated world on the one hand, but very fragile. It doesn't take much to shake up the system. It doesn't take much to cause bankruptcy among the financial firms.
- 8.67** They serve a hugely important role in allocating capital. In spite of all the controls, regulations, interest rates, and everything, they do a pretty good job. Looking out, tech industry, it's got the capital it needed, and it keeps getting the capital it needs. I'm not sure in the most efficient way, I'm not sure in the most productive way, but it's getting it. So at least we get innovation, we get progress. Companies still go bankrupt. That's good. We don't bail out everybody; that's good. We need bankruptcy. It's healthy to get rid of inefficiency and unproductive activities.
- 8.68** But the system generally is fragile and sensitive to mistakes by a centralized authority. And you can almost guarantee that mistakes will be made. While finance evolved as a decentralized system, it is constantly being centralized. And yet at the same time, we have all this technology, and this technology is a decentralizing force. Venmo is decentralizing. Crypto is decentralizing. We've got this competition between these two—centralizing by government, decentralizing by technology. I think the future of finance is going to depend on how this evolves. The future of finance is going to depend on, in a sense, what force is stronger, what force dominates.
- 8.69** Is it the centralization through government, which means, I think, slow economic growth, less efficient allocation of capital, and a lot of crises on the way, a lot of volatility? Or is it this decentralizing force of technology, which I think can lead to efficient allocation of capital and, ultimately, significant positive economic growth? That's where we are today. I don't know where this is all going to land up. I don't know what the future actually holds and which one of these forces wins out. I'm not particularly optimistic these days because I don't see our politicians moving in a decentralizing privatization direction. I see them consolidating power, if anything.
- 8.70** But things can change, and they can change pretty quickly. And again, technology keeps pushing the frontiers, pushing the frontiers of finance, really as it has since the beginning of time. We talked about this. We talked about the history. And at every point, the technology, the technology of finance, pushes and pushes and pushes and ultimately makes the world the better place.

- 8.71** I saw a Bloomberg interview with Jamie Dimon the other day, and he said something that really disappointed me. "If the government wants to make mortgages public, just make them public." I really like Jamie Dimon, so that was unfortunate to hear. But it seems like businessmen don't even want freedom anymore.
- 8.72** Well, I would have to hear the actual interview with Jamie Dimon because it sounds to me like what he's saying, and I don't know, but it sounds to me like what he's saying, "Look, we're subsidizing mortgages, we're controlling mortgages. We're, in a sense, nationalizing it because we package them up by Freddie and Fannie. All the mortgages get sold to Freddie and Fannie, or half of them do. Just go ahead and nationalize them, if you're really dabbling so much." I don't think he's saying that approvingly, necessarily. He's saying, "Instead of making it hidden, just be explicit about it and just do it. And maybe then we can have a discussion about whether it's a good thing or a bad thing."
- 8.73** By hiding it, we don't have that discussion. But look, generally, it's absolutely true. Jamie Dimon is not exactly a laissez-faire capitalism guy. He's not a free market guy. He never has been. I like him, too. But I like him only because, if I compare him to, I don't know, Biden and Trump, he seems like a saint. And he seems, first of all, he's sane. He still got his mind. And he understands a little bit of economics, which neither Biden or Trump have any knowledge of. So I like him by contrast, but not because he's some brilliant—
- 8.74** And generally, business people are just like everybody else. They're very good at what they do. But that doesn't mean that they have a better philosophy or better politics or a better view of economics than everybody else does. They're influenced by the same forces, same intellectual forces, the same ideological forces, as everybody else is. There are just as many leftists among businessmen, or maybe not just as many, but there are lots of businessmen who are leftists. There are lots of businessmen who are nationalists. Just because you're good at business doesn't mean you're pro-freedom. And it never has been like this.
- 8.75** I mean, Henry Ford was a fascist, semi-Nazi sympathizer. Who knows what J. P. Morgan's politics were, or Rockefeller's politics. I don't know, but they weren't exactly — none of them was political in a sense of advocating for real freedom. And this crop of CEOs, so many of them were attracted by woke. So it's not clear-cut in terms of politics and business. Never has been, unfortunately.
- 8.76** You mentioned 2008 crisis. A lot of it had to do with Freddie and Fannie Mae, which we just talked about. I guess in 2008, I believe they went bankrupt. At the time, there was talk of getting rid of both of them, and are we in the same position that we were in 2008, in terms of how much of the mortgage market they're controlling and is flowing through them again? And are we on the cusp of another, another 2008, possibly?

- 8.77** In terms of Freddie and Fannie, we're in a much worse situation than 2008. More mortgages are flowing through them than did in 2008. In terms of whether we're going to have another 2008, the answer is probably no for other reasons. So let's be clear about what happened. In 2008, both of them went under government receivership. Now, they're both government entities, really. The government guarantees their debt. So they're regulated by a government entity, just the two of them. So they're very much government entities, but they were taken over by the government because they were basically bankrupt in 2008.
- 8.78** And at the time, politicians, both on the left and on the right, said, "OK, we need to either break them up into 10 different companies instead of two." Ideally, we would privatize them. Some people argued—a significant number, not just a few, I mean, there was real energy behind, "We need to do something about Freddie and Fannie. We need to break them up. We need to privatize them. We can't go back to the way it was." Sixteen years later, in September, it will be 16 years, Freddie and Fannie are still under government receivership. There's not a single voice out there talking about privatizing them or breaking them up.
- 8.79** We've gone through a Democratic administration, a Republican administration, Democratic administration, doesn't matter. House Democrats, House Republicans, Senate Democrats, Senate Republicans, it doesn't matter. Nobody has the appetite to dismantle Freddie and Fannie. They're still there. They control about 50, they buy about 50% of all the mortgages. They repackage them, they sell them. The reason I don't think we're heading towards a housing crisis is that I think that the prices of homes right now are not a result of a financial bubble. Prices are very high, but they're not the result of a financial bubble.
- 8.80** Prices are high today because we haven't built any homes. Prices are high today, primarily, not only, there's an element of finance there, but because after 2008, we stopped building. First of all, most of the people who work in construction were illegal immigrants who left America. They literally crossed the border back into Mexico. People don't talk about this, but between 2008 and 2018 or something, illegal immigration was negative. It was leaving the country, partially because there were no jobs. So we don't have the workers to build homes.
- 8.81** And then the amount of regulations on home building has just gone through the roof. So you can't build the homes, particularly where they're desired, where people want to live. So today we have, I think the last number I saw was a shortage of 3.2 million homes. So there's a huge amount of demand, partially because we have immigrants, partially because we have people who want to upgrade or people who want to move or new couples being formed, new families being formed. And we don't have 3.2 new houses. And that's why prices are going up. They're being bid up because of the shortage of housing.

- 8.82** There is a part of it that I'm sure is financial, that has to do with all the financial shenanigans and the fact that interest rates were so low for so long. But I don't think that's a dominant thing right now in housing. The real question is, where's the bubble now? And I don't know. I don't have a good estimate, and I don't know what is going to cause the next financial crisis. I don't know. If I did, I would be very rich. I could make a lot of money.
- 8.83** So, internationally, the American worker is seen a bit as spoiled. You've seen the teamsters coming back, this constant fight between the workers and those who employ them. However, wages have not kept up with productivity. Productivity for the American worker is way higher. Without going into a wage spiral, is there a way that we can increase wages, or does this matter in the large scheme of things?
- 8.84** Productivity and wages are completely aligned. There's no wage productivity gap. I know there's a graph going around that shows productivity going like this and wages going like that. I think the technical term for that is BS. It's a bogus graph. There's a number of different ways, the way you measure earnings is all wrong there. They use a different inflator to adjust for inflation. They use a different inflator on productivity and on earnings. It's just wrong. If you actually look at proper graphs that have adjusted properly for all factors, wages and productivity completely aligned, the challenge we have— and wages have not stagnated for the last 40 years, as people pretend and people tell us about, just hasn't happened.
- 8.85** It's just not true. They don't measure wages right. They don't measure households right, they don't measure—they don't think about the kind of jobs people have. So there's a lot of, huge amount of problems in all those statistics. And most of what you hear from both left and right about wages and about "the last 40 years have been a disaster" is wrong; it's just wrong. Plain, straight false. All the stuff, Piketty, 12, 14 years ago, was publishing about inequality, just wrong, just not true. Papers are coming out now saying, "No, he's measuring all this stuff wrong." Just the numbers are wrong. It's not even a philosophical disagreement, which I have as well; but the numbers are just wrong.
- 8.86** So you have to be very, very careful, particularly when the left and the right agree on something. Probably wrong. Because there's something going on if they agree. But there's just a lot of mythology out there. And unfortunately, the mythology has a life of its own. And what happens is, people start believing it, people who are supposedly affected by it, not really affected by it, but supposedly affected by it. And they start believing that they're stagnating, even though they haven't.
- 8.87** I love to use my iPhone, but this didn't exist 40 years ago. And this doesn't cost that much. Almost everybody has them, or a Samsung or something. It didn't exist 40 years ago. How do you measure this? How do you measure the value of this, to a worker, to anybody, in terms of quality of life, standard of living, consumption? It's worth \$1,000

because that's what they pay for it, but it's worth a lot more than that. Changed our life by a lot more than \$1,000. That's why we buy it.

- 8.88** You can see this over and over again. We don't know how to measure quality and standard of living. I'll give you one final example because I use this for my inequality talks when I give inequality talks. You read J. K. Rowling? Harry Potter? You guys read Harry Potter? If you haven't read Harry Potter, you really should. Harry Potter is great. Anyway, I'm really pissed off at J. K. Rowling. I really hate J. K. Rowling. And the reason I hate J. K. Rowling is because I've spent thousands of dollars on Harry Potter books and movies and rides for my kids. Thousands of dollars. I had to read the books as well. Thousands of dollars have gone down.
- 8.89** So I became poorer by thousands of dollars, and J. K. Rowling became a billionaire. I mean, it's ridiculous. But that's the zero-sum economics that people do. But why did I spend thousands of dollars on J. K. Rowling's books? Because I enjoyed them, because they make my life better. Because I had a blast going on rides with my kids and going to the movies with my kids or reading a book with my kids. There's a concept for you. The value that I got from J. K. Rowling far exceeds the thousands of dollars I spent on it, but inequality increased. She got richer, and I got poorer.
- 8.90** As an economist, all I can measure is that my bank account went down by X number of thousands of dollars. Where's the spiritual value I gained from reading the books? Where's the measure of that? How do I capture that? I can't. People are so much better off today. They have no clue because they forgot what it was like. People think they want to go back to 1970s. Just try it, particularly in a hot climate when, in the 1970s, a lot of places didn't have AC, and today we have everywhere, pretty much. But life in the '70s sucked, certainly as compared to life today. We live in the best time ever, even for the worker.
- 8.91** But we keep telling them their life sucks, they start believing it. And it's not just this values that you can't put your finger on. Actually when you do the math, actually when you look at the numbers, they're better off. Now, are some people worse off? Sure. But mostly the people who are worse off are people who, am I allowed to say, deserve to be? People who don't work, the people who don't produce, the people who live off of welfare and stuff like that.
- 8.92** But no, I mean, all the numbers—put in productivity, wage gap, I can't remember, explanation, alternative, and you will find a number of articles that show—I think the best place, I think there's a guy at the American Enterprise Institute who's done some excellent work on this, AEI, American Enterprise Institute. All right, guys. Thank you.